

Eventual categoricity for abstract elementary classes

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Abstract

We prove Shelah’s eventual categoricity conjecture for abstract elementary classes in ZFC. For each infinite bound on the Löwenheim–Skolem number there is a uniform threshold such that categoricity in any one cardinal at or above that threshold implies categoricity in every cardinal at or above the same threshold.

Contents

1	Introduction	3
1.1	The statement	3
1.2	History and scope	4
1.3	Proof overview	5
1.4	Conventions	7
2	Set codes and models indexed by orders	7
2.1	A finite-arity partition argument	10
2.2	A fixed presentation on every order	11
3	Reversing maps on bounded tuples	13
3.1	Finite-support orders and their cuts	13
3.2	The orbit record of a short sequence	15
3.3	Bounded names for model orbits	17
3.4	Preserving every short record in a small subpresentation	18
3.5	The elimination argument	19
4	Stable diagrams and amalgamation at closed levels	21
4.1	Stabilization of strong-map classes	21
4.2	Natural diagrams and interchange of blocks	23
4.3	Bounded tests over a well-ordered presentation	25
4.4	Amalgamation and type counts	27
5	Working schemes and uniform comparison of diagrams	28
5.1	Systems and the working package	29
5.2	Refining the eligible levels	30
5.3	Directed unions and rich orders	32
5.4	The initial package and named data	33
5.5	Two routes to working diagrams	35
5.6	A set-theoretic code for diagram equality	36

6	The one-dimensional calculus	39
6.1	Testing, placement, and amalgamation	40
6.2	Approximations and stationary free extensions	44
6.3	Products and local character	47
6.4	Universal strips and approximate retractions	48
7	Isolation and dominated extensions	51
7.1	Relative isolation	51
7.2	Primeness and descent of isolation	53
7.3	Existence with a prescribed bound on tuple length	55
7.4	All bounded lengths at one level	57
7.5	Independent lists and coherent witnesses	58
8	Adjoining a marked increment	60
8.1	A template before chain closure	62
8.2	A testing bound beside an old small pair	65
8.3	Chain closure with all bounded tuple lengths	66
9	Finite iteration and exact boundary maps	69
9.1	Forgetting components and recovering types	69
9.2	The finite systems and their faces	71
9.3	Completing a boundary into an extension	73
9.4	A bounded support for boundary tests	76
10	A minimal type and its geometry	78
10.1	Finite configurations over the chosen model	80
10.2	Pointed levels and change of base	82
10.3	Eliminating proper extensions with unchanged geometry	83
11	Geometric systems and classification by bases	86
11.1	Shapes, relative bases, and arrows	86
11.2	Order presentations and their rank	89
11.3	Obtaining the working package from categoricity	91
11.4	Fresh coordinates and the two induction assertions	92
12	Exact isomorphisms of geometric systems	97
13	Uniform definitions and common closure levels	103
13.1	Set definitions at a given cardinal	104
13.2	A fixed register of bound requests	109
13.3	The common majorant and its limit refinements	113
14	Transfer to all larger cardinals	115

1 Introduction

Categoricity asks whether cardinality determines a model up to isomorphism. For abstract elementary classes, the eventual categoricity conjecture predicts that uniqueness in one sufficiently large cardinal forces uniqueness in every cardinal above a uniform threshold. We prove this prediction in ZFC, with the threshold depending only on a bound for the Löwenheim–Skolem number.

1.1 The statement

We specify the axioms, including the union clause used throughout the proof. All structures have sets as their universes, and all languages are set-sized and finitary. We write $|M|$ for the universe of a structure and $\|M\|$ for its cardinality.

Definition 1.1 (Abstract elementary class). An *abstract elementary class* (AEC) is a pair $K = (\mathcal{C}, \leq_K)$, where $\mathcal{C}$ is a class of structures in a fixed language τ and $\leq_K$ is a partial order satisfying the following conditions.

- (1) The class is closed under isomorphism. If $M \leq_K N$, then M is a τ -substructure of N . If $f : N \cong N'$ and $M \leq_K N$, then $f[M] \leq_K N'$.
- (2) *Coherence*: if M_0 is a τ -substructure of M_1 , and $M_0 \leq_K M_2$ and $M_1 \leq_K M_2$, then $M_0 \leq_K M_1$.
- (3) For every nonzero ordinal δ and increasing chain $(M_i : i < \delta)$ under $\leq_K$, its union U belongs to $\mathcal{C}$ and $M_i \leq_K U$ for all $i < \delta$. If also $M_i \leq_K N$ for every i , then $U \leq_K N$.
- (4) There is an infinite cardinal $\ell \geq |\tau| + \aleph_0$ such that, for every $M \in \mathcal{C}$ and $A \subseteq |M|$, there is $N \leq_K M$ with $A \subseteq |N|$ and $\|N\| \leq |A| + \ell$.

The least possible ℓ is denoted $\text{LS}(K)$. A *strong map* is an isomorphism onto a strong substructure of its codomain. The class is *categorical in κ* if it has a model of cardinality κ and all its models of that cardinality are isomorphic.

The chain axioms equivalently give unions and the common-upper-bound clause for nonempty set-indexed directed systems of strong inclusions. We also use the corresponding direct limits of coherent strong maps, obtained by renaming their injective images. No intersection axiom is included.

Theorem 1.2 (Uniform eventual categoricity). *For every infinite cardinal λ there is a cardinal $\mu(\lambda) \geq \lambda$ such that the following holds for every AEC K with $\text{LS}(K) \leq \lambda$:*

$$(\exists \kappa \geq \mu(\lambda) \quad K \text{ is categorical in } \kappa) \implies (\forall \kappa' \geq \mu(\lambda) \quad K \text{ is categorical in } \kappa').$$

The same threshold is used in the premise and the conclusion.

In particular, Theorem 1.2 resolves Shelah’s eventual categoricity conjecture positively. No amalgamation, joint embedding, tameness, no-maximal-models, or arbitrarily-large-models hypothesis is assumed. The starting categoricity cardinal may be a successor, a regular limit, or a singular cardinal. The theorem asserts the existence of a threshold, without prescribing a numerical bound for it.

On the resulting categorical tail, every model strongly embeds into every model of at least its cardinality. Indeed, the Löwenheim–Skolem axiom supplies a strong submodel of the smaller

cardinality, and categoricity identifies it with the given model. Taking a larger target gives proper strong extensions and a common target for any two models. Thus the tail has joint embedding and no maximal models.

It is useful to separate the uniform quantifier from the main structural argument.

Theorem 1.3 (The qualitative form). *An AEC categorical in unboundedly many cardinals is categorical on a tail of the cardinals.*

Here “unbounded” means unbounded in the class of all cardinals, and a tail contains every cardinal above some fixed cardinal. The set-coding argument in Proposition 2.3 proves the equivalence of the two theorems. Once that reduction is made, arbitrarily large models are available. The order construction in Theorem 2.6 then supplies actual models at every sufficiently large cardinal; subsequent categoricity arguments concern both existence and uniqueness.

The uniform theorem applies in particular to sentences of $L_{\omega_1, \omega}$ in countable languages, where countable conjunctions and disjunctions are allowed but quantifier strings are finite. For such a sentence φ , choose a countable fragment containing it and order its models by elementarity for that fragment. This is an AEC with Löwenheim–Skolem number $\aleph_0$ [BV17, Example 2.0.2]. The models and their isomorphisms are unchanged, so the same $\mu(\aleph_0)$ works for all these sentences. The prescribed $\beth_{\omega_1}$ threshold in the classical infinitary formulation remains a separate question [BV15, Conjecture 1.1].

1.2 History and scope

Morley’s categoricity theorem gives the countable first-order prototype: a countable theory categorical in one uncountable cardinal is categorical in every uncountable cardinal [Mor65]. Its structural approach connects uniqueness of models with stability and dimension. The geometry of strongly minimal sets, developed by Baldwin and Lachlan [BL71, Sections 1–2], provides a central antecedent for classification by bases. In an AEC, elementary compactness and the first-order description of types are not part of the axioms. The corresponding amalgamation, independence, and geometry therefore require separate constructions.

Shelah’s eventual categoricity conjecture asks for an eventual dichotomy between categoricity and noncategoricity; see [She09, Conjecture N.4.2] and [BV15]. The equivalent formulation using one threshold depending on an LS bound is discussed in [Vas17a, Remark 1.3]. Our set-coding argument makes that reduction explicit: it treats bounded categoricity spectra as well as classes categorical at unboundedly many cardinals. An explicit numerical threshold is a further question, separate from the existential threshold in Theorem 1.2; compare [She09, Conjecture N.4.3(1)(b)]. Under CH, the companion [Ope26, Theorem 1.1] constructs an AEC K with $\text{LS}(K) = \aleph_0$ that is categorical on a higher tail but not at $\beth_{(2^{\aleph_0})^+} = \beth_{\omega_2}$, ruling out this prescribed endpoint for arbitrary AECs.

Substantial special cases have been established. Makkai and Shelah proved transfer from sufficiently large successor categoricity for $L_{\kappa, \omega}$ -theories when κ is strongly compact [MS90, Conclusion 5.1]. Grossberg and VanDieren obtained upward categoricity transfer from one sufficiently large successor cardinal for tame AECs with amalgamation, joint embedding, and arbitrarily large models [GV06, Theorem 5.2]. Boney subsequently derived tameness from strongly compact cardinals and, combining it with the Grossberg–VanDieren transfer, established eventual categoricity for arbitrary AECs with a successor starting cardinal under a proper class of strongly compact cardinals [Bon14]. These results explain the importance of removing restrictions on the starting cardinal as well as structural hypotheses on the class.

Universal classes admit a ZFC transfer without assuming amalgamation [Vas17b, Theorem 7.3]. Tameness, amalgamation, and primes give another transfer [Vas18, Theorem 3.8]. Shelah and Vasey prove a ZFC categoricity theorem for excellent AECs and obtain broader results using strongly compact cardinals, or the weak generalized continuum hypothesis (WGCH), that is, $2^\kappa < 2^{\kappa^+}$ for every infinite cardinal κ , together with amalgamation [SV24, Theorem 14.2 and Corollaries 14.7, 14.10]. Under WGCH, amalgamation, and arbitrarily large models, Vasey also gives a classification of categoricity spectra [Vas19, Corollary 9.7]. These hypotheses supply substantial structure. Here the amalgamation, bounded-tuple calculus, and geometry needed for the working schemes are constructed at selected levels from the AEC axioms and unbounded categoricity.

The proof uses several established methods in forms that are proved locally. Order presentations originate in the work of Ehrenfeucht and Mostowski [EM56]; their AEC formulation and functorial maps are developed in [She09, IV.0.8–IV.0.9 and IV.0.12]. We retain a finite-support presentation throughout the proof. Its extraction uses classical partition-calculus methods [ER56], while the later isolation argument uses Fodor’s pressing-down method [Fod56]. Stationary independence and local character are central to Shelah’s good frames [She09, II.2.1]. Sections 6 and 7 establish the bounded-tuple calculus, relative isolation, and dominated extensions required for the systems used here.

A version-sensitive issue matters for the order argument. Boney and Vasey [BV15, Fact 1.4 and the discussion of Claim 1.5, pp. 3–4] identify a gap in the unrestricted syntactic-characterization claim of Shelah’s Chapter IV and withdraw an earlier attempted repair. Their valid variants with exponentiation or cofinality assumptions do not cover all the starting cardinals of Theorem 1.2. We prove the required reversal of strong maps directly in Theorem 3.11.

Finite systems and dimension-raising arguments have a long history in nonelementary categoricity. Shelah’s existence and uniqueness methods [She83, Sections 3–5] are antecedents of the passage from one additional dimension to a larger cardinal. The modern treatment in [SV24, Definition 8.16 and Lemma 11.6] distinguishes amalgamated completion from strong uniqueness onto a prescribed target. Both properties are needed below and are proved separately. Our final cardinal induction uses exact boundary maps for systems whose basis pieces may have different cardinalities.

Espíndola has announced eventual categoricity for arbitrary AECs and further accessible-category results [Esp22], and subsequently an explicit-threshold categoricity theorem [Esp23, Theorem 4.1]. These are separate public claims. The argument here constructs its working-scheme calculus, isolation, marked and geometric boundary lifting, and common-level closure directly. None of these interfaces is imported from those preprints.

1.3 Proof overview

By the set-coding reduction in Section 2, it suffices to fix one AEC K categorical in unboundedly many cardinals and prove categoricity on a tail. The proof first builds structure at selected cardinals, called working levels, and only at the end transfers it to every larger cardinal. A tuple is bounded at a level μ when its length is strictly less than μ .

From orders to comparison of tuples. An order presentation E assigns a model to a linear order: each element is represented by a label and finitely many indices, and an order embedding induces a strong map by substituting indices. Section 3 uses specially chosen orders to control the automorphism orbits of short tuples. A small subpresentation can be aligned with a larger presentation so that all its short tuples have matching orbit records. An elimination chain then

proves that a strong map can be reversed on any prescribed short tuple. The small model supplies the alignment; the larger model accommodates the enumeration of the small model's tuples. This avoids an exponentiation hypothesis at the categoricity cardinal.

In Section 4, the resulting equivalence classes of tuples under strong maps stabilize as the categoricity cardinal increases. These are the comparison diagrams used later. Finite supports give two further tools: a bound on the parameter sets needed to distinguish comparisons, and simultaneous placement of coherent bounded prescriptions. Together they yield amalgamation and homogeneity for bounded tuples at suitable countably cofinal closure levels.

A calculus that can be applied to systems. The same tools will be needed for finite systems of models, not only for individual models. Section 5 records their precise hypotheses as a working scheme: a finite system shape, specified arrows, one finite-support presentation, and compatible tuple comparisons. Sections 6 and 7 derive the calculus from this package. A type over a base is determined by identification of tuples in an amalgam over that base. Its distinguished free extensions are computed by long approximation sequences already in the base. The isolation construction gives extensions controlled by a specified mark: the full type of each additional bounded tuple, together with the mark, is determined by the mark's type and a bounded fragment of the base. Such extensions have the primeness and uniqueness needed for iteration.

Marked systems detect growth of a geometry. Section 8 shows that adjoining a finite mark and an extension controlled in this way again gives a working scheme. Iterating produces a model at each subset of a finite axis set, with commuting strong maps. Its boundary consists of the proper-subset vertices and their maps. Section 9 first completes a compatible boundary map into a common extension, then extends a boundary isomorphism onto the prescribed target. The return-tree argument keeps full maps coherent along individual branches but retains only boundary maps in the common accumulator. This distinction lets the orbit bound control joint comparisons across different faces, including their actual equalities.

Section 10 chooses a minimal singleton type, one with a unique nonalgebraic extension over each relevant base. Its realizations form the points of a pregeometry, with independence defined by finite products of the type. We retain a distinguished strong copy of its base model M_0 , with a fixed diagram for its enumeration, in the later models; these are the pointed models used below. Exact lifting for all finite marked systems at one common level rules out a proper strong extension with no new points. A downward construction then extends this no-growth conclusion to arbitrary larger cardinalities. Thus the geometry detects every proper extension, and its basis cardinality equals model cardinality above the resulting threshold.

From geometric systems to a categorical tail. Sections 11 and 12 organize bases into finitely many pieces assigned to vertices. Tiled order presentations give systems with these pieces; a paired induction establishes their working packages and classification by prescribed basis maps. A second boundary argument gives exact lifting for every finite geometric shape at suitable levels.

The two finite-dimensional applications require separate common-level choices. Section 13 supplies both: fixed set codes for all finite constructions and their bound requests produce a level where every finite depth is available. The geometric application is made after the marked application has supplied the no-growth threshold. Finally, Section 14 starts from a common geometric level and inducts on the maximum cardinality of the basis pieces, simultaneously for all finite shapes. One additional axis records the previously constructed isomorphism at each extension step; coherent unions handle limits. The zero-axis case gives pointed categoricity. The

Löwenheim–Skolem axiom and categoricity at one fixed smaller cardinal place the chosen base pattern in every sufficiently large model. Forgetting that base gives the qualitative theorem, and the set-code reduction gives the uniform theorem.

Figure 1 summarizes the proof and the two uses of common-level closure.

1.4 Conventions

All size variables are cardinals unless explicitly called ordinals. An assertion about bounded tuples at a level μ concerns tuples of length strictly below μ . Eligible working levels have countable cofinality and are strong limits closed under the displayed bounds. Whenever smaller adequate levels are needed cofinally below a working level, that requirement is stated and included in the closure bookkeeping. Automorphisms of an index order always mean automorphisms of its pure linear order. Charts, auxiliary hulls, and basis choices are not added to the structures whose isomorphisms are being compared.

2 Set codes and models indexed by orders

We first separate the uniformity over AECs from the transfer argument for one AEC. All directed families below are nonempty and set-indexed. We begin by deriving the directed forms of the chain axioms in Definition 1.1.

Lemma 2.1 (Directed unions and coherent direct limits). *If a nonempty set of models is directed under strong inclusion, its union belongs to K and every member is strong in the union. If every member is strong in a common model N , its union is strong in N . More generally, a coherent diagram of strong maps indexed by a nonempty directed partial order has a direct limit in K , with strong canonical maps. Every compatible family of strong maps from the diagram into a model induces a strong map from the direct limit.*

Proof. The union is a well-defined τ -structure: finitely many arguments are contained in one family member, and any two choices of such a member agree after passing to a common strong upper member. Here finitariness is used for functions and relations alike. We prove the union and common-upper-bound assertions together by induction on the cardinality of the family. A finite directed family has a member containing every other member, so its union is that member. A countable directed family has a cofinal increasing sequence: enumerate the family and successively choose a member above the previous choice and the next enumerated member. The chain axioms apply to this sequence, including their common-upper-bound clause.

Suppose the family is indexed by a set I of uncountable cardinality κ , and the two assertions hold for smaller families. Choose an operation assigning to each pair of indices an index of a strong upper bound. Express I as the union of an increasing continuous sequence $(I_\alpha)_{\alpha < \text{cf}(\kappa)}$ of nonempty subsets of size less than κ , each closed under that operation. To obtain this sequence, enumerate I by κ , use a cofinal sequence of ordinals below κ to specify the required initial segments, close each stage under the binary operation by countably many iterations, and take unions at limit stages. A union of fewer than $\text{cf}(\kappa)$ sets of size less than κ still has size less than κ ; countable closure preserves the size of each infinite stage. This construction also works when κ is a successor cardinal, using κ many stages.

Let U_α be the union of the family on I_α . The induction hypothesis gives $U_\alpha \in K$. For $\alpha < \beta$, every member indexed by I_α is strong in U_β . The inductive common-upper-bound assertion therefore gives $U_\alpha \leq_K U_\beta$. The chain axioms now give the required union over I . If the original

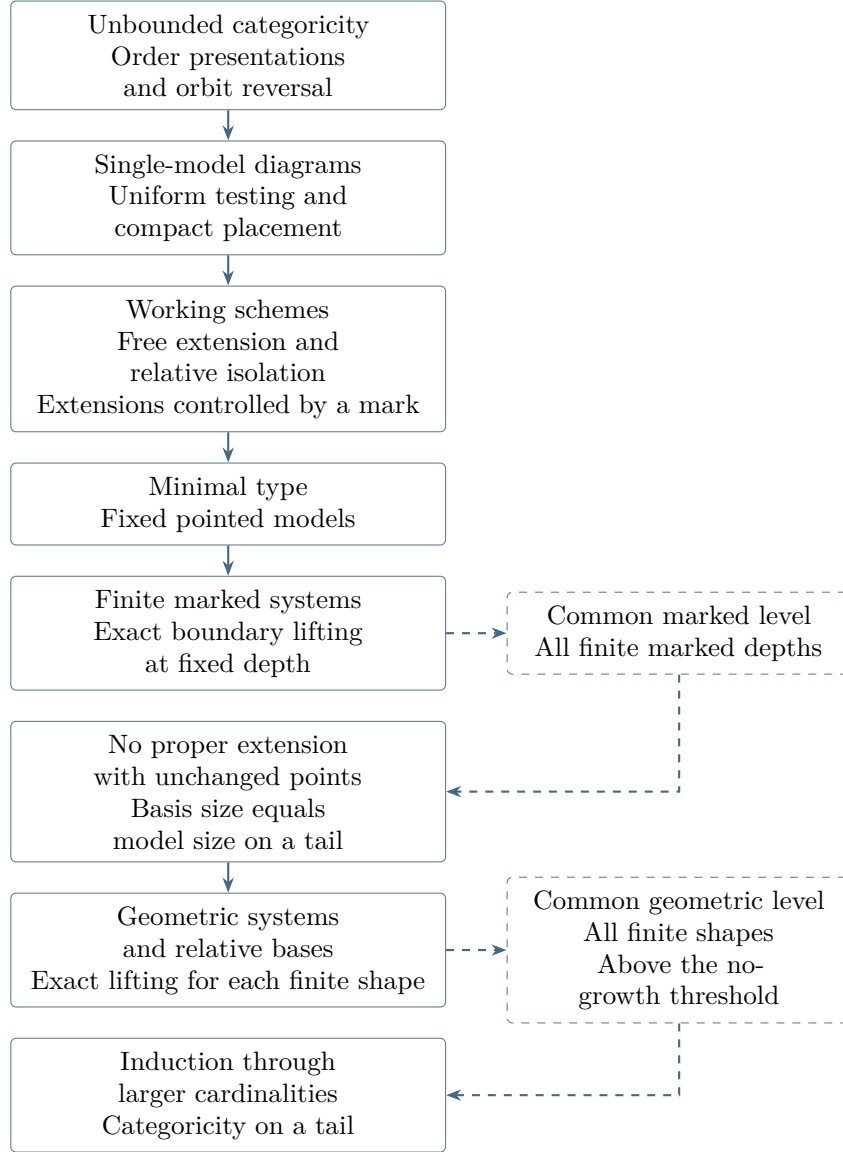


Figure 1: The qualitative proof, ordered by its principal dependencies. The two dashed boxes are separate applications of Section 13. The first makes all finite marked depths available for the no-growth argument over the fixed point. The second is chosen after that argument and supplies the starting cardinal for the final transfer. These classes of common levels need not contain a tail. The small-code reduction then gives the uniform threshold of Theorem 1.2.

family is strong in N , induction first gives $U_\alpha \leq_K N$ for every α , and the chain common-upper-bound clause gives the same conclusion for their union. This completes the induction.

For a diagram (M_i, f_{ij}) indexed by a directed partial order, form the quotient of the disjoint union of its underlying sets by

$$(i, x) \sim (j, y) \iff (\exists k \geq i, j) f_{ik}(x) = f_{jk}(y).$$

Directedness and coherence make this an equivalence relation. The canonical map from each M_i is injective. Interpret any finitary function or relation in a common later model containing representatives of its arguments; coherence and the fact that the maps are structure embeddings make this interpretation well-defined. The images of the M_i are models forming a directed family of strong inclusions. The first assertion puts their union, which is the quotient just constructed, in K and makes all canonical maps strong.

Finally, compatible strong maps $g_i : M_i \rightarrow N$ induce a map on that quotient. It is injective: if two images agree, compare their representatives in a common later M_k and use injectivity of g_k . The same argument gives a structure embedding. Its image is the directed union of the strong submodels $g_i[M_i]$ of N , so the common-upper-bound assertion makes the induced map strong. The quotient construction also gives uniqueness, and hence the stated direct-limit property. $\square$

Lemma 2.2 (Recovery from small models and pairs). *Fix an infinite cardinal λ and a finitary language τ of cardinality at most λ . An AEC in τ with $\text{LS}(K) \leq \lambda$ is determined by its models of cardinality at most λ and its strong inclusions between such models, both taken up to isomorphism. These data have a set code, and the recovered class and strong-submodel relation are uniformly first-order definable in set theory from that code.*

Proof. Represent small structures on ordinals of cardinality at most λ . Represent a small pair $M \subseteq N$ by such a representative for N together with the distinguished subset that is the domain of M . The possible structures and pairs form sets. A code specifies which structure types and which pair types are allowed. Testing whether a given small structure or pair has an allowed type is a set-theoretic condition: quantify over the relevant isomorphisms.

For an actual $M \in K$, let

$$\mathcal{D}(M) = \{A \leq_K M : \|A\| \leq \lambda\}.$$

This is a set. It is nonempty by the LS axiom applied to the empty set, and its union is M by the LS axiom applied to singletons. If $A, B \in \mathcal{D}(M)$, the LS axiom inside M supplies a small strong submodel containing $|A| \cup |B|$. Coherence makes both A and B strong in that submodel. Thus $\mathcal{D}(M)$ is directed.

Conversely, suppose a structure M is the union of a nonempty directed family of small structures whose membership and pair inclusions are allowed by the code of K . These are actual members and strong inclusions of K , so the directed-union axiom gives $M \in K$. Consequently the following condition recovers membership, including for structures not already assumed to belong to K :

M is a union of a nonempty directed family of allowed small models, with all its directed inclusions allowed by the small-pair code.

To recover the strong relation, require a directed presentation $N = \bigcup \mathcal{D}$ of this kind and a nonempty directed subfamily $\mathcal{E} \subseteq \mathcal{D}$ with $M = \bigcup \mathcal{E}$. If $M \leq_K N$, use $\mathcal{D} = \mathcal{D}(N)$ and $\mathcal{E} = \mathcal{D}(M)$; transitivity makes the second a subfamily of the first. In the reverse direction, every member of

$\mathcal{E}$ is strong in N by the union clause for $\mathcal{D}$. The common-upper-bound clause for $\mathcal{E}$ then gives $M \leq_K N$.

Each presentation can be quantified over as a set of substructures of the structure being decoded. All the preceding conditions are thus first-order set-theoretic formulas with the small code as parameter. They recover both the class and the order, as asserted. $\square$

The equivalence below is the usual Hanf-number reduction; compare [Vas17a, Remark 1.3]. We spell out the set coding to make the uniform quantifier explicit.

Proposition 2.3 (The uniform reduction). *The uniform assertion in Theorem 1.2 is equivalent to the following qualitative assertion:*

Every AEC categorical in unboundedly many cardinals is categorical in all cardinals above some bound depending on that AEC.

Proof. At a fixed LS bound λ , the languages can be renamed into a fixed set of symbols with specified finite arities. By Lemma 2.2, the possible AECs in those languages are determined by members of a set of small codes. A code is valid exactly when its decoded class and relation satisfy the AEC axioms with LS bound λ . This is a first-order condition on the code, so the valid codes form a set.

Assume the qualitative assertion. For each valid code c , its categoricity spectrum is either bounded or unbounded. In the bounded case choose the least infinite cardinal b_c strictly above every categoricity cardinal. In the unbounded case choose the least infinite cardinal b_c such that the decoded AEC is categorical in every cardinal at least b_c . These choices are definable, and the second choice exists by the qualitative assertion. Replacement bounds the set of all b_c . Choose a cardinal $\mu(\lambda) \geq \lambda$ strictly above that set of bounds. If an AEC with LS number at most λ is categorical at some $\kappa \geq \mu(\lambda)$, its code cannot be in the bounded-spectrum case. It is therefore categorical at every $\kappa' \geq b_c$, in particular at every $\kappa' \geq \mu(\lambda)$. This is exactly the same threshold in the premise and conclusion.

Conversely, for an AEC categorical unboundedly often, apply the uniform assertion with $\lambda = \text{LS}(K)$. Some categoricity cardinal crosses the resulting threshold, and the asserted tail follows. $\square$

Remark 2.4. Lemma 2.2 also supplies the usual ZFC interpretation of quantification over AECs: one quantifies over their set codes and uses the decoded class predicates. No choice of representatives from a proper class and no class choice axiom is needed. Below we fix one AEC categorical unboundedly often and write $\ell = \text{LS}(K)$. It has arbitrarily large models because categoricity includes existence.

2.1 A finite-arity partition argument

The following end-homogeneous argument belongs to classical partition calculus; compare Erdős and Rado [ER56, Theorem 39 and its corollary]. We prove the finite-arity form and bounds used by the order-template construction.

Lemma 2.5. *For infinite cardinals ξ, ν and a positive integer k , there is an infinite cardinal $R_k(\xi, \nu)$ such that every coloring of the k -element subsets of an ordinal of cardinality at least $R_k(\xi, \nu)$ into ξ colors has a homogeneous subset of cardinality ν . Finitely many specified arities can be made homogeneous simultaneously by enlarging the initial cardinal.*

Proof. For $k = 1$, a set of cardinality greater than $\xi \cdot \nu$ cannot be partitioned into ξ sets all of cardinality less than ν . Suppose the assertion is known for k . Choose an infinite initial ordinal α large enough for that k -ary assertion with target ν , and put $b = \xi^{|\alpha|}$. Consider a coloring c of the $(k + 1)$ -element subsets of an ordinal Ω with $|\Omega| > b$.

Construct a tree of residual subsets of Ω through height α . Its root is Ω . At a nonempty node, name and remove its least point x . The previously named points on its branch, together with x , form an increasing sequence. Partition the remaining residual set according to all values

$$u \mapsto c(s \cup \{u\}), \quad s \text{ a } k\text{-subset of that sequence.}$$

The nonempty parts are the children. At a limit height, a branch has as its residual set the intersection of the earlier residual sets; retain the node precisely when this intersection is nonempty.

At each node there are at most $|\alpha|$ coordinates to record and hence at most b children. The total number of possible nodes through height α is at most $b^{|\alpha|} = b$. This includes the limit nodes, which are specified by their branches. At most b points are removed as node labels. Choose $u \in \Omega$ which is never removed. At every height it belongs to one residual set; at limit heights it belongs to the intersection on that branch. Consequently it determines a branch with α named points and nonempty remainder.

For every k -subset s of these named points, all later named points give the same value of $c(s \cup \{x\})$: that value was fixed by the partition immediately after the last point of s was named. Therefore c on increasing $(k + 1)$ -tuples from the branch depends only on their first k entries. Apply the k -ary induction hypothesis to this induced coloring on the branch. Its homogeneous ν -subset is homogeneous for c as well. For example, the recursion just proved allows $R_{k+1}(\xi, \nu) = (\xi^{R_k(\xi, \nu)})^+$ after increasing the initial R_1 as needed.

For a finite list of arities, work backward to choose a bound large enough for the last desired thinning after all the earlier ones. Successive thinning preserves homogeneity already obtained. This proves the final assertion. $\square$

2.2 A fixed presentation on every order

The construction is an Ehrenfeucht–Mostowski presentation [EM56]; for its AEC form compare [She09, Definition IV.0.8 and Claims IV.0.9, IV.0.12]. We include the finite-support construction in the label notation used below.

Theorem 2.6 (Order presentation). *Suppose K has arbitrarily large models and $\text{LS}(K) = \ell$. There are at most ℓ labels, each of positive finite arity, and a functor E from nonempty linear orders and order embeddings to K and strong maps, with the following properties.*

1. *Every element of $E(I)$ is a label evaluated on an increasing finite list from I . Equality and all structure relations and function values are determined by the labels and the order/equality pattern of the finitely many indices involved.*
2. *$E(I)$ contains distinct distinguished elements $(a_i)_{i \in I}$. An order embedding $f : I \rightarrow J$ sends a_i to $a_{f(i)}$, and its action on any labeled element is position substitution.*

3. For every finite nonempty $s \subseteq I$, the elements labeled on s form a strong submodel $E_s(I)$ of cardinality at most ℓ . For $s \subseteq t$ these submodels are strongly included, and $E(I) = \bigcup_s E_s(I)$.
4. $|I| \leq \|E(I)\| \leq |I| + \ell$. In particular, for every infinite $\kappa \geq \ell$, K has a model of cardinality exactly κ .

Here a subpresentation is identified with its image under the canonical strong map when an inclusion of underlying universes is convenient.

Proof. Take an arbitrarily large $M \in K$ and distinct elements $(a_i)_{i \in A}$ indexed by a large ordinal A . Recursively on the positive finite size of $s \subseteq A$, choose $M_s \leq_K M$ of cardinality at most ℓ containing the displayed elements on s and every M_t for nonempty $t \subsetneq s$. There are only finitely many such t , so the LS axiom applies with bound ℓ . Coherence gives $M_t \leq_K M_s$.

Choose a surjective enumeration $(m_{s,\eta})_{\eta < \ell}$ of each M_s , allowing repetitions. If $s = \{i_0 < \dots < i_{n-1}\}$, require $m_{s,j} = a_{i_j}$ for $j < n$. For an n -element set s , its color is the complete atomic diagram, including negations and equalities, of all the enumerations

$$(m_{t,\eta})_{\eta < \ell}, \quad \emptyset \neq t \subseteq s,$$

with the subsets of s named by their relative positions. This diagram includes formulas involving entries from different enumerations. There are at most ℓ names and atomic formulas for each fixed n , so its possible colors belong to a fixed set of cardinality at most 2^ℓ .

We choose colors d_n , for $n \geq 1$, so that for every finite n there are samples of arbitrarily large cardinality homogeneous in colors $d_1, \dots, d_n$. Suppose $d_1, \dots, d_{n-1}$ have been chosen with this property. For any prescribed target cardinal, choose such a sample large enough for Lemma 2.5 at arity n and thin it to an n -homogeneous set of the target cardinality. The earlier colors are retained. If every possible n -color supported only boundedly large such thinnings, the supremum of the bounds over the set of colors would bound all thinnings, a contradiction. Thus one color d_n supports arbitrarily large thinnings. It restricts on every smaller subset to the previously chosen colors. This proves the recursion and its coherence. It asserts finite-stage realizability in arbitrarily large samples; it does not require an infinite simultaneous homogeneous set inside one sample.

For a nonempty linear order I , introduce symbols (s, η) for each finite nonempty $s \subseteq I$ and $\eta < \ell$, with s listed increasingly. Equate (s, η) and (t, ζ) when the color $d_{|s \cup t|}$ says that the corresponding entries are equal. Coherence makes this independent of passage to a larger finite set of indices. Reflexivity, symmetry, and transitivity hold because any finite instance occurs in a realized color. Form the quotient by this equivalence relation.

Interpret a relation on finitely many quotient elements in a color on the union of their supports. For a function symbol, evaluate it in the enumerated model on that union and use a name for its value. All its arguments belong to that enumerated model, and the model is a substructure, so the value has such a name. Changing representatives or enlarging the finite support gives the same quotient element by the complete atomic diagram and coherence. A constant can be evaluated using any one nonempty finite support. Two such evaluations agree in a color on their union. Thus these rules define a τ -structure $E(I)$.

Let $E_s(I)$ consist of the elements named by (s, η) , $\eta < \ell$. The color $d_{|s|}$ makes this a full enumerated copy of one of the actual models M_s in a sample. Names with support $t \subseteq s$ are included in this copy: their equalities with names in the enumeration on s were recorded. If $s \subseteq u$, the pair $E_s(I) \subseteq E_u(I)$ is isomorphic to an actual pair $M_s \leq_K M_u$ from a realization of $d_{|u|}$. Isomorphism invariance therefore makes the inclusion strong. The family of these small models is directed and covers the constructed structure. Its directed union belongs to K , with all $E_s(I) \leq_K E(I)$.

For an order embedding $f : I \longrightarrow J$, substitution $(s, \eta) \mapsto (f[s], \eta)$ respects every defining diagram and every inequality of names. It therefore induces an embedding $E(f) : E(I) \longrightarrow E(J)$. On each finite-support model it is an isomorphism onto $E_{f[s]}(J) \leq_K E(J)$. The common-upper-bound clause applied to their directed union shows that the whole image is strong in $E(J)$. Substitution also proves $E(\text{id}) = \text{id}$ and $E(g \circ f) = E(g) \circ E(f)$.

Define a_i by the first designated name on the singleton $\{i\}$. The recorded equalities identify this with the designated name for i on any larger finite support. Distinct indices give distinct elements because every relevant sample had that property. The labels are indexed by (n, η) with $n \geq 1$ finite and $\eta < \ell$, so there are at most ℓ of them. Counting finite supports proves the upper bound on size, and the distinct displayed elements prove the lower bound. $\square$

3 Reversing maps on bounded tuples

Throughout this Section, K is categorical unboundedly often, $\ell = \text{LS}(K)$, and E is the presentation of Theorem 2.6. The aim is to reverse a strong self-map on any one fixed bounded tuple in every sufficiently large categoricity model. We first construct index orders with uniformly controlled short-tuple orbits. Homogeneous-order orbit bounds already play a role in Shelah's work; compare [She09, IV.5.1]. The alignment and elimination below establish the form needed at arbitrary categoricity cardinals.

For comparison, [BV15, Fact 1.4 and the discussion of Claim 1.5, pp. 3–4] records syntactic-characterization results under exponentiation or cofinality conditions and a gap in the attempted unrestricted extension. The bounded-tuple reversal needed here is proved directly in Theorem 3.11.

3.1 Finite-support orders and their cuts

Definition 3.1. For an infinite limit ordinal δ , set

$$\Gamma_\delta = \mathbb{Z}^{(\delta)}, \quad H(\delta) = \mathbb{Q}^{(\Gamma_\delta)}.$$

Parentheses denote finite support. Both groups are ordered by their greatest differing coordinate. Write e_α for the integer basis vector at $\alpha < \delta$ and u_g for the rational basis vector at $g \in \Gamma_\delta$. The rank of a nonzero vector is its greatest support coordinate. Unless group operations are explicitly mentioned, $H(\delta)$ is used as a pure linear order; in particular $\text{Aut}(H(\delta))$ means its order automorphism group.

A cut of an order L is a partition (L^-, L^+) with $L^- < L^+$. It is an interior nonprincipal cut when both sides are nonempty, L^- has no greatest element, and L^+ has no least element. Its two characters are $\text{cf}(L^-)$ and $\text{ci}(L^+)$. It is balanced if these cardinals are equal. Cofinality and cointiality here refer to the indicated linear orders.

Lemma 3.2. *The order $H(\delta)$ is dense without endpoints and has cardinality $|\delta|$. Its points have character $\text{cf}(\delta)$ on both sides. All nonempty bounded open intervals have one order type. All left rays have one order type, and all right rays have one order type. An increasing map between ordinal position sets induces an order embedding between the corresponding Γ and H orders by moving positions and padding by zero. Each point of $H(\delta)$ has a code using finitely many ordinal positions and integer and rational coefficients; comparisons are determined by these discrete coefficients and the order/equality pattern of the ordinal positions.*

Proof. Finite-support counting gives $|\Gamma_\delta| = |H(\delta)| = |\delta|$. The group Γ_δ has least positive element e_0 , so every g has predecessor $g - e_0$. Its cofinality and coinitality are both $\text{cf}(\delta)$. Indeed, the vectors e_α for α in a cofinal subset of δ are cofinal, and their negatives are coinital. The supports of fewer than $\text{cf}(\delta)$ vectors are bounded in δ , so such a family is neither cofinal nor coinital.

The order $H(\delta)$ is a divisible nonzero ordered abelian group, and hence is dense and has no endpoints. If (g_i) is coinital in Γ_δ , then (u_{g_i}) is coinital in the positive cone of $H(\delta)$: given a positive vector of rank g , choose $g_i < g$. Conversely, the ranks of fewer than $\text{cf}(\delta)$ positive vectors have a strict lower bound in Γ_δ , and a positive basis vector at that lower rank lies below all of them. Thus the positive cone has coinitality $\text{cf}(\delta)$. Translation and negation give that character at every point on both sides.

Translations move points and therefore identify rays of a fixed direction. We show that any positive vector can be sent to any other by an increasing automorphism. First translate the index group Γ_δ so that their greatest support coordinates agree, say at g . Write the two vectors as

$$au_g + v, \quad bu_g + w, \quad a, b \in \mathbb{Q}_{>0}, \quad \text{rk}(v), \text{rk}(w) < g,$$

allowing zero lower parts. Fix all other basis vectors and replace u_g by

$$\frac{b}{a}u_g + \frac{w - v}{a}.$$

This is an invertible triangular rational-linear map. It preserves the sign of every nonzero vector, since it either fixes the greatest nonzero coordinate or multiplies its coefficient by $b/a > 0$. It sends the first displayed vector to the second. Combining this map with translations sends any ordered pair of distinct points to any other such pair. Restricting gives the interval assertion.

Finally, an increasing map on ordinal positions preserves the greatest differing position of two integer vectors. The induced map on these integer vectors preserves the greatest differing position of rational vectors as well. This proves the embedding claim. Expanding the finitely many integer-vector positions used by a rational vector gives the asserted finite code and comparison rule. $\square$

Lemma 3.3 (Balance of cuts). *Every interior nonprincipal cut of Γ_δ , and every such cut of $H(\delta)$, is balanced.*

Proof. First consider a cut (L, R) of Γ_δ . Among the ranks of $r - l$, with $l \in L$ and $r \in R$, choose the least ordinal α . Fix one crossing pair of that rank and its common head above α . In that head, each fiber with a fixed integer coefficient at α lies wholly on one side: a crossing pair within a fiber would have smaller rank. The resulting cut in the integer coefficients separates two adjacent integers. A point with a head above the fixed head would exceed the fixed upper point, and a point with a head below it would precede the fixed lower point. The two bordering fibers are therefore cofinal and coinital toward the original cut, and their tails are copies of $\mathbb{Z}^{(\alpha)}$. Their end characters agree by negation in that tail group. The case $\alpha = 0$ would give a last lower and a first upper point, and is excluded. This proves balance in Γ_δ .

Now let (L, R) be an interior nonprincipal cut of $H = H(\delta)$, and put

$$T = \{\text{rk}(r - l) : l \in L, r \in R\} \subseteq \Gamma_\delta.$$

This is a final segment: from a crossing pair of rank g , subtract a positive basis vector at any $h > g$ from its lower point. The new pair still crosses and has rank h .

Suppose first that T has a least member g . Fix a crossing head above g . Every fiber with a fixed rational coefficient at g lies on just one side of the cut. Otherwise a crossing difference would have rank below g . The induced cut of $\mathbb{Q}$ is nonempty and proper. If neither coefficient side has a nearest rational, countable rational approximations are cofinal and cointial in the original cut. If the lower coefficients have a greatest member, the entire fiber at that coefficient lies on the lower side and is cofinal there. Its tail, supported below g , has greatest available rank $g - e_0$, and the vectors nu_{g-e_0} and $-nu_{g-e_0}$ show that both its end characters are $\aleph_0$. On the upper side, rational coefficients decreasing to the greatest lower coefficient give cointiality $\aleph_0$. If instead the upper coefficients have a least member, the reversed calculation gives cofinality and cointiality $\aleph_0$. These cases exhaust the rational cuts, so the original cut is balanced.

Suppose that T has no least member, and let $\epsilon = \text{ci}(T)$. Put $S = \Gamma_\delta \setminus T$ and let H_S be the subgroup supported on S . We prove that the lower character is ϵ by separating two cases.

Assume there is $l \in L$ for which

$$\{\text{rk}(r - l) : r \in R\} \text{ is cointial in } T.$$

The coset $l + H_S$ is contained in L , since a crossing within it would have rank in S . It is cofinal in L . In fact, if a point $l' \in L$ lay above the entire coset, $l' - l$ would have positive rank $h \in T$. Choose $r \in R$ with $\text{rk}(r - l) < h$. Then $r - l < l' - l$, contradicting $r > l'$. The set S is nonempty, since otherwise the cofinal coset would be the singleton $\{l\}$. It has no greatest member: if s were greatest, $s + e_0$ would be the least member of T . Balance in Γ_δ , applied to (S, T) , gives $\text{cf}(S) = \epsilon$. The cofinality of H_S at its upper end equals $\text{cf}(S)$. A cofinal set of ranks gives cofinal positive basis vectors, while fewer ranks are bounded and cannot give a cofinal family of vectors. Hence $\text{cf}(L) = \epsilon$ in this case.

In the other case, each $l \in L$ has a bound $t_l \in T$ strictly below all ranks $\text{rk}(r - l)$, $r \in R$. Choose crossing pairs (l_i, r_i) , $i < \epsilon$, whose ranks are cointial in T . For fixed $l \in L$, choose i with $\text{rk}(r_i - l_i) \leq t_l$. If $l_i \leq l$, then $0 < r_i - l \leq r_i - l_i$, whence $\text{rk}(r_i - l) \leq \text{rk}(r_i - l_i) \leq t_l$, a contradiction. Thus $(l_i)_{i < \epsilon}$ is cofinal in L . If fewer than ϵ points of L were cofinal, their bounds t_l would have a common strict lower bound in T , since $\epsilon = \text{ci}(T)$. A crossing pair below that bound would have its lower point above all those points by the same calculation. This is impossible, proving $\text{cf}(L) = \epsilon$.

Apply the proved lower-side calculation after negating the order and exchanging L and R . Negation preserves every difference rank, so it has the same T and ϵ . It yields $\text{ci}(R) = \epsilon$. This proves balance also in the no-least-rank case. $\square$

3.2 The orbit record of a short sequence

For a subset A of a linear order L and a cut $A = A^- \sqcup A^+$, including cuts with an empty side, define its *strict slot* by

$$S_L(A^-, A^+) = \{x \in L \setminus A : A^- < x < A^+\}.$$

Its occupancy is the assertion that this set is nonempty. The order/equality diagram of a sequence identifies its distinct entries and the cuts of their induced order, so the occupancy data can be indexed without choosing an enumeration of the cuts.

Lemma 3.4 (Segments approaching a small cut). *Let L be a dense linear order without endpoints, all of whose points have character κ on both sides. Suppose all bounded open intervals have one order type and every interior nonprincipal cut is balanced. For every infinite regular $\epsilon < \kappa$, the following segment types depend only on ϵ and the indicated direction:*

$$[p, \mathfrak{c}) = \{x \in L_{\mathfrak{c}}^- : p \leq x\}, \quad (\mathfrak{c}, p] = \{x \in L_{\mathfrak{c}}^+ : x \leq p\},$$

where $\mathfrak{c}$ is an interior nonprincipal cut of balanced character ϵ and p belongs to the appropriate side. The isomorphisms can be chosen to match the displayed endpoint.

Proof. We prove both directions simultaneously by induction on the infinite regular cardinal ϵ . Consider two lower-side segments and choose strictly increasing sequences $(p_i)_{i < \epsilon}$ and $(q_i)_{i < \epsilon}$ cofinal in their respective cut sides, with the indicated endpoints at index zero. At a stage $i < \epsilon$, fewer than ϵ preceding points are not cofinal, so one can choose the next point above them and above the next point of a fixed cofinal sequence. This gives the required strictly increasing sequences.

Partition the first segment into blocks as follows:

$$B_0 = \{p_0\}, \quad B_{j+1} = (p_j, p_{j+1}].$$

For a nonzero limit $i < \epsilon$, set

$$D_i = \{x \in L : (\exists j < i) x \leq p_j\}, \quad U_i = L \setminus D_i, \quad \mathfrak{c}_i = (D_i, U_i), \quad B_i = (\mathfrak{c}_i, p_i].$$

The cut $\mathfrak{c}_i$ is interior and nonprincipal. Its lower side has no greatest point and has cofinality $\text{cf}(i)$: the strictly increasing i -sequence is cofinal there, and any smaller proposed cofinal family would give a cofinal set of indices in i . Its upper side contains p_i . If that side had a least point u , the earlier sequence would give $\text{cf}(L_{<u}) = \text{cf}(i) < \kappa$, contradicting point character. Balance therefore gives $\mathfrak{c}_i$ character $\text{cf}(i)$ on both sides. Since $i < \epsilon$, this is a regular cardinal strictly below ϵ .

Make the corresponding blocks from the q_i . Successor blocks are isomorphic by bounded-interval homogeneity, adjoining their greatest points. At a nonzero limit index i , use the *upper-side* induction assertion at $\text{cf}(i)$ to match the blocks. Match the singleton blocks at zero. These are ordered partitions of the two segments. Indeed, for any point in the first segment take the least i such that it is at most p_i ; the three cases for i put it in exactly the block just specified. Gluing the block maps proves the lower-side assertion.

For the upper-side assertion choose strictly decreasing coinitial sequences. The blocks are $\{p_0\}$ and $[p_{j+1}, p_j)$ at successors. At a nonzero limit i , put

$$V_i = \{x \in L : (\exists j < i) p_j \leq x\}, \quad W_i = L \setminus V_i, \quad \mathfrak{d}_i = (W_i, V_i).$$

The set V_i has no least point and has coinitiality $\text{cf}(i)$. The set W_i contains p_i ; if it had a greatest point, that point's upper character would be $\text{cf}(i) < \kappa$. Thus $\mathfrak{d}_i$ is nonprincipal, and balance gives character $\text{cf}(i)$ on both sides. The block at i is $[p_i, \mathfrak{d}_i)$, to which the *lower-side* induction assertion at $\text{cf}(i)$ applies. The least index with $p_i \leq x$ verifies that these blocks, in reverse index order, partition the upper-side segment. For $\epsilon = \aleph_0$ there are no nonzero limit indices below ϵ , so this simultaneous induction starts with just the successor blocks. All displayed endpoint points are matched. $\square$

Theorem 3.5 (Short-sequence order orbits). *Let δ be an infinite limit ordinal and let θ be an infinite cardinal with $\theta < \text{cf}(\delta)$. Two sequences of length at most θ in $H(\delta)$ lie in the same $\text{Aut}(H(\delta))$ -orbit if and only if they have the same order/equality diagram and the same occupancies of all strict slots over their distinct entries.*

Proof. Necessity follows from preservation of order. For sufficiency, the order/equality diagrams give an increasing bijection $f : A \rightarrow B$ between the sets of distinct entries, matching the indexed sequences. We extend it separately on each occupied slot.

Choose a point p in a slot over $A = A^- \sqcup A^+$. The portion of the slot at or below p has one of three forms. If $A^- = \emptyset$, it is $(-\infty, p]$. If A^- has greatest member a , it is $(a, p]$. Otherwise put

$$D = \{x : (\exists a \in A^-) x \leq a\}, \quad U = H(\delta) \setminus D.$$

The cut (D, U) has no greatest lower point, has p on its upper side, and has lower character $\text{cf}(A^-) \leq \theta$. Its upper side has no first point, since such a point would have character at most θ . By Lemma 3.3 this is an interior nonprincipal cut of balanced character $\text{cf}(A^-)$. The slot portion is $((D, U), p]$.

In the first case its type is fixed by ray homogeneity; in the second case by bounded-interval homogeneity; in the third case by Lemma 3.4. The portion at or above p has the corresponding classification using A^+ , a least member when present, and otherwise $\text{ci}(A^+)$. The same descriptions show that the slot has neither endpoint.

The order map f preserves which boundary case occurs and the cofinality or coinitality of the relevant part of A . Thus the corresponding occupied slot over B has matching two halves. Choose an interior point q there and match the two halves with their common point p sent to q . They glue to a slot isomorphism. The occupancy assumption ensures that empty slots also correspond. The entry sets and their occupied slots partition the whole order. Choice for this set of slots and gluing with f produce an increasing bijection of $H(\delta)$, as required. $\square$

3.3 Bounded names for model orbits

We now pass from automorphisms of the index order to automorphisms of its model. At the cofinalities covered by the orbit theorem, a support record determines a model orbit, but different records may determine the same orbit. Only a uniform upper bound on their number is needed for reversal.

Fix an infinite tuple length θ . A presentation of a θ -tuple in $E(H(\delta))$ specifies its labels and their finite index lists. Flatten those lists into an indexed sequence of at most θ points of $H(\delta)$. Its *presentation record* consists of the labels, the order/equality diagram of this flattened sequence, and all its slot occupancies.

Lemma 3.6 (Uniform orbit bound). *There is a fixed set $\mathcal{R}_\theta$ of possible presentation records, depending only on ℓ and θ , with*

$$|\mathcal{R}_\theta| \leq B(\ell, \theta) := 2^{2^{\ell+\theta}}.$$

For every categoricity cardinal $\kappa \geq \max(\ell, \theta^+)$, the set of automorphism orbits of θ -tuples in its model has cardinality at most $B(\ell, \theta)$. This applies also when κ is singular. If $|\delta| = \kappa$ and $\text{cf}(\delta) > \theta$, each realized record in $E(H(\delta))$ determines exactly one such model orbit.

Proof. The finitely many positions belonging to each tuple entry can be indexed in $\theta \times \omega$. Choices of arities and labels number at most $2^{\ell+\theta}$. The flattened order/equality diagrams number at

most 2^θ ; their sets of cuts have size at most 2^θ , and occupancy subsets number at most 2^{2^θ} . The displayed generous bound therefore bounds a universal record set.

For a fixed δ with $\text{cf}(\delta) > \theta$, equal records give an order automorphism matching the flattened lists by Theorem 3.5. Applying E gives an automorphism of the model matching the tuples. Hence a realized record determines one model orbit. Every tuple has a labeled presentation, so the number of orbits is bounded by the number of records.

For any $\kappa \geq \max(\ell, \theta^+)$, the ordinal $\delta = \kappa + \theta^+$, with ordinal addition here, has cardinality κ and cofinality θ^+ . The model $E(H(\delta))$ has cardinality κ and is therefore a copy of the categoricity model. This proves the bound without any condition on $\text{cf}(\kappa)$. $\square$

Choose a model C_κ at each categoricity size needed in a given set-sized construction. On its θ -tuple orbits put $p \rightarrow r$ when some strong self-map sends a representative of p to a representative of r . This relation is reflexive and transitive: compose maps after using an automorphism to align the intermediate representatives. Choose an injection of the orbit set into the fixed cardinal $B = B(\ell, \theta)$, once for each κ . These are its *orbit names*. For any copy of C_κ , a tuple's name is independent of the isomorphism used to C_κ , because two such isomorphisms differ by an automorphism.

For a suitable δ of size κ , let $F_{\kappa, \delta}$ be the partial function from $\mathcal{R}_\theta$ to B sending each realized record to its orbit name. The *profile* of this presentation consists of its named arrow relation and $F_{\kappa, \delta}$. The profiles belong to a fixed set of cardinality at most 2^B ; in particular this bound is independent of κ and δ .

3.4 Preserving every short record in a small subpresentation

Lemma 3.7 (Closure of ordinal supports). *Let $\theta^+ \leq a$ be infinite cardinals and let β be an ordinal of cofinality greater than a . Every subset $S \subseteq \beta$ of size at most a is contained in a set $T_0 \subseteq \beta$ of size at most a with the following property: for every $P \subseteq T_0$ of size at most θ and every finite tuple from β , there is a tuple from T_0 having the same order/equality diagram over P . Moreover T_0 is bounded in β and has cofinality greater than θ in its induced order.*

Proof. Construct increasing sets $S_i \subseteq \beta$ for $i < \theta^+$, starting with S . At a successor stage consider all strict slots over the entire well-ordered set S_i . There are at most $|S_i| + 1 \leq a$ of them. In each finite slot add every point; in each infinite slot add a countably infinite subset. Together with S_i this realizes every finite order/equality pattern over S_i which occurs in β . At limits take unions. The cardinality stays at most a throughout.

Set $T_0 = \bigcup_{i < \theta^+} S_i$. Every set $P \subseteq T_0$ of size at most θ is contained in some S_i , by regularity of θ^+ . For a given finite tuple, keep any coordinates already in S_i and match its other coordinates in their respective slots using the next stage. This proves the asserted property over P . Since $|T_0| \leq a < \text{cf}(\beta)$, T_0 is bounded in β . Each S_i is bounded too, and its uppermost slot is infinite. The next stage therefore contains points above all of S_i . Every at-most- θ subset of T_0 is bounded in this way, proving $\text{cf}(T_0) > \theta$. $\square$

Lemma 3.8 (Appending a long indecomposable segment). *Under the hypotheses of Lemma 3.7, suppose β is additively indecomposable. Let T_0 be as there, and let $\delta < \beta$ be additively indecomposable, with $\text{otp}(T_0) < \delta$ and $\text{cf}(\delta) > \theta$. Place a consecutive segment J of order type δ strictly after T_0 in β , and set $T = T_0 \cup J$. Then $\text{otp}(T) = \delta$, and T has the same finite-pattern matching property over its at-most- θ subsets. Consequently $H(T) \subseteq H(\beta)$ computes correctly the occupancies of every strict slot over any sequence of at most θ of its points.*

Proof. Choose $\alpha < \beta$ above T_0 . Indecomposability of β gives $\alpha + \delta < \beta$, so $J = [\alpha, \alpha + \delta)$ is available. Indecomposability of δ gives $\text{otp}(T_0) + \delta = \delta$.

Fix a parameter subset $P \subseteq T$ of size at most θ and a finite tuple in β . Match the coordinates below α into T_0 over $P \cap T_0$ using Lemma 3.7. Keep coordinates in J fixed. The parameters in J , together with these finitely many retained coordinates, are bounded in J because $\text{cf}(\delta) > \theta$. Match all coordinates at or beyond the end of J by suitably many distinct points of J above that bound, retaining any repetitions. The three portions stay in their correct mutual order, and this gives the required finite tuple over P .

Here $\Gamma_T = \mathbb{Z}^{(T)}$ is identified with its image in Γ_β , and $H(T) = \mathbb{Q}^{(\Gamma_T)}$ with its image in $H(\beta)$. A sequence of at most θ points of $H(T)$ uses at most θ ordinal positions in its finite codes. A witness in $H(\beta)$ to an occupied strict slot uses only finitely many additional ordinal positions. Match those into T over all the parameter positions and retain the integer and rational coefficients. Lemma 3.2 makes every comparison with the sequence unchanged. The resulting point of $H(T)$ witnesses that slot. The reverse implication follows from inclusion. $\square$

Proposition 3.9 (Alignment of model orbits). *Fix the record set and orbit naming at an infinite length θ . Let $a < \mu$ be categoricity cardinals with $a \geq \max(\ell, \theta^+)$. Suppose the following data have one common profile:*

1. *a presentation $E(H(\beta))$ of C_μ , where β is additively indecomposable, $|\beta| = \mu$, and $\text{cf}(\beta) > a$;*
2. *presentations $E(H(\delta))$ of C_a for a cofinal subset of a^+ of additively indecomposable ordinals δ of size a and cofinality greater than θ .*

Then in any copy N of C_μ , every subset of size at most a is contained in $A \leq_K N$ of size a such that the orbit name of every θ -tuple in A , computed as a tuple of C_a , is the same as its name computed in N .

Proof. Identify N with the presentation in the first item. Represent the requested set by labeled elements and collect the ordinal positions in the finite codes of their indices. There are at most a positions. Apply Lemma 3.7 to obtain T_0 . Its order type is below a^+ . Choose a profile occurrence δ from the second item above $\text{otp}(T_0)$. Since $a^+ \leq \text{cf}(\beta) \leq \beta$ and $\delta < a^+$, Lemma 3.8 applies and gives T of order type δ .

The subpresentation $A = E(H(T))$ is strong in N and has cardinality a . It contains the requested set. For any θ -tuple in A , use a labeled presentation there. Its record is identical in $E(H(T))$ and $E(H(\beta))$ by the last assertion of Lemma 3.8. The order isomorphism $\delta \cong T$ identifies its record in the former with the same record for $E(H(\delta))$. The two record-to-name functions belong to the common profile and hence agree. This proves alignment simultaneously for all θ -tuples of A . $\square$

3.5 The elimination argument

Alignment compares every short tuple in a small submodel with its image in a larger model. We now use repeated profiles at two categoricity sizes to rule out an orbit arrow with no reverse. The larger size will absorb the cost of processing all tuples of the small model.

Lemma 3.10. *If κ is an infinite cardinal and $\tau \leq \kappa$ is infinite regular, the additively indecomposable ordinals $\delta < \kappa^+$ with $|\delta| = \kappa$ and $\text{cf}(\delta) = \tau$ are cofinal in κ^+ .*

Proof. Given $\eta < \kappa^+$, choose a limit ordinal $\rho > \eta$ with $\kappa \leq \rho < \kappa^+$ and $\text{cf}(\rho) = \tau$, for example by appending an ordinal segment of type τ beyond $\eta + \kappa$. Set $\delta = \omega^\rho$. It is additively indecomposable. A cofinal sequence of exponents gives a cofinal sequence in ω^ρ by continuity of exponentiation. Fewer than $\text{cf}(\rho)$ ordinals below ω^ρ have all their Cantor-normal-form exponents bounded below ρ , and are therefore bounded in ω^ρ . This proves $\text{cf}(\delta) = \text{cf}(\rho) = \tau$. Every ordinal below ω^ρ has a finite Cantor normal form with exponents below ρ . Thus $|\delta| \leq \max(\aleph_0, |\rho|) = \kappa$. Since $\delta \geq \rho \geq \kappa$ and $\rho > \eta$, we have $|\delta| = \kappa$, $\delta < \kappa^+$, and $\delta > \eta$. $\square$

Theorem 3.11 (Eventual reversal at categoricity cardinals). *For each infinite cardinal θ there is a cardinal $b_K(\theta)$ such that, whenever $\kappa \geq b_K(\theta)$ is a categoricity cardinal, $M \in K$ has size κ , $f : M \rightarrow M$ is strong, and $\bar{a}$ is a tuple of length at most θ in M , there is a strong $g : M \rightarrow M$ with*

$$g(f(a_i)) = a_i \quad \text{for every coordinate } i.$$

The bound concerns all sufficiently large categoricity cardinals, of arbitrary cofinality. No equality of the form $\kappa^\theta = \kappa$ is assumed.

Proof. It suffices to treat length exactly θ ; pad a nonempty shorter tuple by repetitions, and use any strong self-map for the empty tuple. In a categoricity model, the asserted reversal is equivalent to symmetry of the arrow relation on its θ -tuple orbits. Indeed a reverse orbit arrow can be precomposed and postcomposed with automorphisms to match the particular image tuple and original tuple. The converse follows by applying reversal to a map witnessing an arrow.

Suppose that symmetry fails at unboundedly many categoricity cardinals. Let P be an infinite cardinal bounding the profile palette; for example take $P = 2^B$. Choose a strictly increasing list of failing categoricity cardinals $(\kappa_i)_{i < P^+}$. At each step choose it large enough that there is an infinite regular $\tau_i \leq \kappa_i$ exceeding ℓ , θ , and every preceding κ_j , and that

$$\kappa_i > P, \quad \kappa_i > \kappa_j^\theta \quad (j < i).$$

Such choices are possible by unbounded failure: first bound the set of preceding cardinals and their powers, then choose a regular successor cardinal above the first bounds and a failing cardinal beyond all of them. This is a set-sized recursion.

At size κ_i , fix its orbit naming. By Lemma 3.10, the ordinals of size κ_i and cofinality τ_i used there are cofinal in κ_i^+ . One profile occurs cofinally often among them: otherwise the supremum of at most $P < \kappa_i^+$ bounded fibers would be below the regular cardinal κ_i^+ . Since there are P^+ sizes and at most P profiles, choose two sizes $a < \mu$ with the same cofinally recurring profile. They satisfy

$$\theta^+ \leq a, \quad a^\theta \leq \mu, \quad \tau_\mu > a.$$

Their common named arrow relation has $p \rightarrow r$ but no $r \rightarrow p$. Both names occur in both sizes. Proposition 3.9 applies inside every μ -sized model, using any of its cofinal profile occurrences.

We construct θ^+ rounds of an increasing strong chain of μ -sized models. At the start of round $i < \theta^+$, choose an aligned a -sized submodel A_i containing all previous A_j . This is possible because their union has cardinality at most a . They are all strong in the current large model, and coherence will make $A_j \leq_K A_i$ for $j < i$.

Enumerate all θ -tuples of A_i , using a^θ steps. At the step for a tuple $\bar{u}$, if its current large-model orbit cannot reach p , leave it as it is. If it can reach p , compose such an arrow with $p \rightarrow r$ and

realize the composite by a strong self-map of the current model. Regard this map as an inclusion into an isomorphic extension: rename its codomain so that the image of each old element becomes that old element. With the transported identification of the codomain, the tuple now has name r . At every later large stage its orbit remains unable to reach p . Otherwise transitivity from r would give $r \rightarrow p$. An orbit already unable to reach p has the same persistence property.

Take strong unions at every limit step, including between rounds. The entire construction has at most $\theta^+ \cdot a^\theta \leq \mu$ many stages in cardinality. Every large union therefore has size μ and is a copy of C_μ by categoricity. Every transition between such copies gives an arrow of the common relation, so the persistence assertion also holds at limit stages.

The aligned submodels form a strong increasing chain: for $j < i$, their underlying sets are included and both are strong in the large model at the start of round i , so coherence applies. Its union

$$A = \bigcup_{i < \theta^+} A_i$$

belongs to K , has cardinality a , and is a copy of C_a . Any θ -tuple of A is contained in some A_i , since $\text{cf}(\theta^+) > \theta$. After round i it cannot reach p in a large model. At the next aligned small model its small orbit has the same name, and therefore cannot reach p in the common relation. The inclusion of that small model into A gives an arrow from this earlier small orbit to its orbit in A . If the latter could reach p , transitivity would contradict the former exclusion. Thus no tuple in A can have an orbit reaching p .

But $A \cong C_a$, so the orbit named p occurs in A ; it reaches itself by the identity map. This contradiction proves that the failing categoricity cardinals are bounded. A cardinal above their bound is $b_K(\theta)$, and the first paragraph gives the claimed reversals, including all shorter lengths. $\square$

4 Stable diagrams and amalgamation at closed levels

Throughout this section, K is the AEC fixed after Proposition 2.3: it is categorical in unboundedly many cardinals, and $\ell = \text{LS}(K)$. We use the order presentation E of Theorem 2.6. A tuple may have any set as its index set. Comparisons between tuples always specify the identification of their index sets. Reindexing by an ordinal, padding, and repetitions do not change any of the arguments below.

The goal is to turn bounded-tuple reversal into amalgamation at selected levels. We first stabilize tuple comparisons across categoricity sizes, then use the order presentation to test and realize those comparisons over an entire base. Amalgamation will be a conclusion of this process.

4.1 Stabilization of strong-map classes

Fix an infinite cardinal σ . At a categoricity cardinal $\kappa > \ell + \sigma$ where Theorem 3.11 holds for σ -tuples, put

$$(M, a) \equiv_\kappa^\sigma (N, b) \iff \text{some strong map } f : M \rightarrow N \text{ satisfies } f(a) = b.$$

Here $\|M\| = \|N\| = \kappa$, and the lengths of the displayed tuples are at most σ . This is an equivalence relation. Reflexivity is immediate. Categoricity allows one to regard all the maps as maps between copies of one model; reversal then gives symmetry, and composition gives transitivity. Using structures with universe κ , or one fixed representative and its tuples, its classes form a set. Their number, for all the tuple lengths under consideration, is bounded by the fixed cardinal supplied by Lemma 3.6. Denote an infinite such bound by B_σ . It depends on ℓ, σ , not on κ .

Lemma 4.1 (Canonical classes above one categoricity size). *For $\|N\| \geq \kappa$, a tuple a in N has a well-defined $\equiv_\kappa^\sigma$ -class: compute it in any strong κ -submodel containing a . Strong embeddings between models of size at least κ preserve this class.*

Proof. The downward LS axiom supplies the submodel, after adding κ many elements to the requested set if necessary. If $M_0, M_1 \leq_K N$ are two such submodels, apply LS to $|M_0| \cup |M_1|$. It gives $M_2 \leq_K N$ of size κ containing both. Coherence gives $M_i \leq_K M_2$. The two occurrences of a therefore have the same class. For a strong embedding $f : N \rightarrow N'$, the image of a witnessing submodel is a strong κ -submodel of N' , so the same observation proves preservation. All these comparisons take place inside a given ambient model; no amalgamation property is used. $\square$

Proposition 4.2 (Stable diagrams). *There is a definable, nondecreasing cardinal function t , with $t(\sigma) > \ell + \sigma$ a categoricity cardinal for every infinite σ , having the following properties.*

1. *Tuples of length at most σ in models of size at least $t(\sigma)$ have canonical diagrams. Equality of diagrams is preserved by every strong embedding between such models, and there are at most B_σ diagrams at these lengths.*
2. *Diagrams include atomic diagrams, respect reindexing and restriction, and are compatible when the bound σ is increased. We denote equality by $D(a; M) = D(b; N)$, suppressing the bound when it is understood.*
3. *If $\kappa \geq t(\sigma)$ is a categoricity cardinal, equality of diagrams between tuples in κ -models is implemented by a strong map between those models.*
4. *A possible diagram of a pair of tuples (a, c) , of total length at most σ , can be realized over any tuple with the diagram of a in any categoricity model of size at least $t(\sigma)$. “Possible” means that the diagram occurs in a model in the domain of its definition.*

Proof. If $\gamma \geq \kappa$ are categoricity cardinals above the reversal threshold, Lemma 4.1 associates a κ -class to every γ -class. This gives a projection $\pi_{\gamma\kappa}$, and these projections compose. They are surjective. Indeed, a κ -model is isomorphic to $E(I)$ for $|I| = \kappa$; extend I to an order of size γ and use the strong order map. Every tuple class in the smaller model then occurs in the larger one. This argument supplies the needed extension without a no-maximal-model assumption.

These projections are eventually all bijective. Otherwise, construct a sequence of categoricity cardinals with more than B_σ successive noninjective projections, and choose one final categoricity cardinal ρ above the entire sequence. For each noninjective projection, surjectivity supplies a pair of ρ -classes which agree when projected to the earlier stage and differ when projected to the later stage. Pairs selected at distinct stages are distinct: after a pair has separated it remains separated under every later refinement. This gives more than B_σ pairs from a set of size at most B_σ , a contradiction. The construction uses only a set-length sequence of cardinals and choices from sets.

Choose a categoricity cardinal c_σ beyond this stabilization and reversal threshold. The canonical c_σ -class defines the desired diagram on models of size at least c_σ . The bijective projections show that its equality can equally be computed at every later categoricity cardinal. If two tuple bounds are used, compare them at a categoricity cardinal above both thresholds. Restriction of strong maps then gives the same restricted equivalence. Atomic formulas and their negations are preserved because strong maps are structure embeddings.

For definiteness, thresholds can be chosen as least cardinals having the stated eventual properties. Choose $t(\sigma)$ to be the least categoricity cardinal strictly above $\ell + \sigma$ and above all the thresholds

c_τ for infinite $\tau \leq \sigma$. This is a nondecreasing definable function. The preceding arguments prove the first three assertions.

For the fourth, first put a realization of (a, c) into a strong c_σ -submodel. If the originally chosen ambient size is smaller than a desired target categoricity size, this categorical submodel can be extended to that size by the order presentation. If the original ambient size is larger, LS gives a submodel of the desired size containing the realization. Thus the joint diagram has a realization in a model of the target categoricity size. Apply the third assertion to the first tuple and the specified tuple in the target. The resulting strong map transports the whole realization as required. $\square$

Proposition 4.3 (Categoricity at countably cofinal closure points). *Suppose $\zeta > \ell$ is an infinite cardinal of countable cofinality and*

$$t(\sigma) < \zeta \quad \text{for every infinite } \sigma < \zeta.$$

Then K is categorical in ζ . Moreover any diagram match between bounded tuples in two ζ -models extends to an isomorphism between those models. Here and throughout the paper, bounded at a level means of cardinality strictly less than that level.

Proof. Existence follows from $E(I)$ for $|I| = \zeta$. Let M, N have size ζ , and choose countable increasing exhaustions of their underlying sets by sets of size less than ζ . Start with the prescribed bounded diagram match, or with the empty match. At a forward step, enumerate together the current source tuple and the next part of the exhaustion of M . Its length and the length of the current target tuple are bounded by an infinite $\sigma < \zeta$. Put the data on each side inside a strong submodel of size $t(\sigma) < \zeta$. Their current tuples have the same diagram. Proposition 4.2 gives a strong map of these submodels matching those tuples. Its values on the new entries extend the match, and the enlarged tuples again have equal diagrams. At the next step interchange M, N .

Only finitely many previously chosen bounded tuples are used at each step, so their combined length stays below ζ . The union of the countable sequence of partial matches is a bijection $|M| \rightarrow |N|$. Every finite tuple and every value of a finitary function occur at some step, where the atomic diagram is preserved. The bijection is therefore a structure isomorphism. $\square$

The closure cardinals in Proposition 4.3 are unbounded. For example, starting above any specified cardinal, choose a strictly increasing sequence $(\alpha_n : n < \omega)$ such that α_{n+1} exceeds all $t(\sigma)$ for $\sigma \leq \alpha_n$. Its supremum has the required closure and countable cofinality. Including 2^{α_n} in these requirements makes the supremum a strong limit. We will use these stronger closure points below.

4.2 Natural diagrams and interchange of blocks

Definition 4.4 (Natural and placed diagrams). Consider a set-sized list of terms in an order presentation. Each term consists of a label and an increasing finite list of positions. Extend the order to size at least the threshold for the list and compute its diagram there. The result is its *natural diagram*. If $A = E(I)$, a *placement* of an additional tuple d over A is a labeled representation in an order extension of I , retaining the positions and chosen representations of the parameters in A . Its *placed comparison over A* is the natural diagram of (d, a_A) , where a_A is a fixed enumeration of A .

The natural diagram is independent of the extension used. Indeed, linear orders amalgamate over a common suborder: for each cut of the suborder, place the remaining portions of the two

orders in that cut while preserving each portion's order. Extending the amalgam further if necessary gives a common sufficiently large presentation. Strong order maps and Proposition 4.2 identify the two diagrams. The same argument shows that only the labels and the joint order/equality pattern of their supports matter. All diagrams with parameters use matching parameter enumerations. One may instead list all labels on the base order, allowing repetitions: restriction gives an enumeration of the elements, and reinserting the repetitions is forced by their equalities.

A placed comparison over a whole μ -model is evaluated in a sufficiently large presentation. It is not, by this definition, an assertion that the original μ -model itself lies above the threshold for diagrams of length μ .

Lemma 4.5 (Orbits over parameters). *Let σ be infinite and suppose*

$$\chi^\sigma = \chi, \quad \chi \geq \ell + 2^{2^\sigma}.$$

At every sufficiently large categoricity cardinal, there are at most χ automorphism orbits of σ -tuples over any specified set of at most χ parameters.

Proof. Present the categoricity model as $E(H(\delta))$, choosing $|\delta|$ to be its cardinality and $\text{cf}(\delta) > \chi$. This is possible at every sufficiently large cardinal, including singular cardinals. Fix term representations of the parameters, and let $S \subseteq H(\delta)$ be their support set. Its size is at most χ . Every member of S is coded by a finite list of ordinals and integer and rational coefficients. The union of its ordinal supports consequently also has size at most χ .

For one further position, its finite ordinal support has at most χ possible order/equality patterns over these old ordinal supports. The discrete coefficients contribute only countably many choices for each finite code. Such a pattern determines all comparisons of the new position with S . For at most σ new positions, include their mutual comparisons, their identifications with old positions, and the term labels. There are at most $\chi^\sigma = \chi$ choices so far.

To apply Theorem 3.5, also record which strict slots over the old and new positions are realized in $H(\delta)$. The old support S is fixed. A slot over S not containing a new position is unchanged. At most σ old slots contain new positions, and inside each there are at most 2^σ cuts of the new support to test. The new flags therefore have at most $2^{2^\sigma} \leq \chi$ possible values. The full record has at most χ possibilities.

The combined list of old and new positions has size at most $\chi < \text{cf}(\delta)$. Theorem 3.5 extends equality of full records to an order automorphism fixing S pointwise. Its induced automorphism of the presentation fixes the parameters and matches the tuples. Thus the number of structure-automorphism orbits is at most the number of records. $\square$

Lemma 4.6 (Disjoint block transpositions). *Take two set-sized families of labeled terms. In disjoint convex regions of their joint support order, suppose each region consists of two adjacent blocks, one used only by the first family and one only by the second. Interchange the blocks in every such region, retaining their internal orders and all positions outside the regions. The joint natural diagram is unchanged.*

Proof. Suppose the original and modified diagrams were different. Let σ bound the two lists, their supports, and the shared positions. Choose $\chi \geq \ell + 2^{2^\sigma}$ with $\chi^\sigma = \chi$, for instance by taking a sufficiently large power of two.

In each region repeat the pair of blocks along an arbitrary linear order L . If the first family's block was on the left, use the order L for the repetitions; if it was on the right, use the reverse of L . Write d_i for the first-family tuple using the blocks at index i , and e_j for the second-family

tuple using the blocks at index j ; retain all the shared positions outside the regions. For $i < j$, the diagram of (d_i, e_j) is the original one, and for $i > j$ it is the modified one. Comparisons within either family and across distinct regions do not change.

Let ϵ be the least infinite cardinal satisfying $2^\epsilon > \chi$. The binary tree $2^{<\epsilon}$ has at most χ nodes: for $\alpha < \epsilon$ each level has size at most χ , and $\epsilon \leq \chi$. Give it lexicographically ordered divider points, one between the two successor subtrees at each node. Explicitly, encode branch digits by $-1, 1$, and encode the divider at a node s by the sequence s followed by 0 , using $-1 < 0 < 1$ in lexicographic comparison. Adjoin one point for every branch in 2^ϵ . Distinct branch points determine distinct strict cuts over the divider points. Use this as the order L .

The tuples e_j at the divider indices, together with the fixed shared data, use at most χ parameters. If two branch-indexed tuples d_i were in the same automorphism orbit over these parameters, their diagrams with every divider tuple would agree. A divider between the two branches gives the two different diagrams, a contradiction. Hence there are more than χ such orbits. Pad the constructed order to a sufficiently large categoricity size and apply Lemma 4.5. This is impossible. $\square$

4.3 Bounded tests over a well-ordered presentation

The next elementary set-theoretic observation will also be used for the more elaborate layouts of later sections.

Lemma 4.7 (Ordinal-vector witnesses). *Let C be a set, and let U be a definable class of functions $C \rightarrow \text{Ord}$, with fixed set parameters allowed in its definition. There is a set $U_0 \subseteq U$ such that each $u \in U$ dominates some $v \in U_0$ coordinatewise.*

Proof. The assertion is immediate if C is empty or U is empty. Well-order C , and suppose no such set exists. For every ordinal β , the set $U \cap V_\beta$ fails to generate U upwards. There is therefore an element of U which dominates none of its members. Let $h(\beta) > \beta$ be the least ordinal such that one such witness lies in $V_{h(\beta)}$. This is a definable ordinal function.

Take an ordinal length λ satisfying the partition relation for pairs, $|C|$ colors, and an infinite homogeneous set, supplied by Lemma 2.5. Define increasing ordinals β_i , for $i \leq \lambda$, recursively, with $\beta_{i+1} > h(\beta_i)$ and with limit values above all earlier values. Choose for a set-indexed collection of nonempty subsets of V_{β_λ} now selects witnesses u_i which dominate no member of $U \cap V_{\beta_i}$, for $i < \lambda$. For $i < j$, we have $u_i \in V_{\beta_j}$, so some coordinate satisfies $u_i(c) > u_j(c)$. Color the pair $i < j$ by the first such coordinate in the fixed well-order of C . An infinite homogeneous set yields an infinite strictly decreasing sequence of ordinals in that coordinate. This contradiction proves the claim. The rank bounding above ensures that only ordinary set-sized Choice, not a choice principle for proper classes, was used. $\square$

Proposition 4.8 (Uniform testing of placed comparisons). *There is a definable nondecreasing infinite cardinal function g , depending on K and its fixed presentation, such that the following holds. If I is an infinite well-order, $A = E(I)$, and two placements of a tuple of length at most σ have different placed comparisons over A , then their natural diagrams already differ over a subset of A of size at most $g(\sigma)$. The bound is independent of $|I|$.*

Proof. Amalgamate the two support orders over I , and let T be the union of the supports of the two added tuples. Its size is at most σ . Freeze its order, the labels and supports of the two tuples,

and the subset of its positions which belong to I . Up to a relabeling on a fixed set of size at most σ , these frozen data range over a set.

For fixed frozen data, each component of $I \setminus T$ in a strict slot over T is well-ordered. Record its ordinal length. The set of slots is fixed, so these lengths form an ordinal vector. Restrict attention to the vectors that give valid configurations and for which the two placed comparisons are unequal. This is a definable class: order presentations and diagrams are defined from set codes, and diagrams of any specified set length can be evaluated above their thresholds.

Coordinatewise domination gives an order embedding of the smaller configuration into the larger one, fixing T and preserving the base positions. Natural diagrams respect these embeddings. Thus a valid smaller unequal configuration remains a witness in any dominating valid configuration. Apply Lemma 4.7 to this class, obtaining a set of valid witnesses. Equivalently one could apply it to the upward closure and select its generators from valid configurations; no assertion that arbitrary vectors themselves describe well-orders is required.

Bound the sizes of the base orders in this set of witnesses. An embedded witness uses only that bounded number of old positions. All labels on these positions give at most their cardinality plus ℓ elements of A , and the diagrams of the two added tuples with these elements are unequal. Taking a cardinal bound over the set of frozen data gives $g(\sigma)$. Increasing it to an infinite nondecreasing function causes no change in the assertion. When a witness uses only finitely many positions, “all labels” simply means their evaluated terms; no assertion about membership of a finite-order presentation in K is needed. $\square$

Proposition 4.9 (Compact placement). *Let $A = E(I)$, where I is an infinite well-order, and let $\chi > \ell + \sigma$ satisfy $\chi^\sigma = \chi$. Suppose that for every $X \subseteq A$ of size at most χ a possible joint diagram $q_X(d, X)$ is prescribed, where d has length at most σ . Assume these prescriptions restrict coherently and that their parameter diagrams are the natural ones of X . Then one placement of d over I realizes all the prescriptions.*

Proof. Fix one term representation per parameter in A . Write $S_X \subseteq I$ for the supports of X . For one set X , its prescription has a placement. To see this, extend I to a sufficiently large categoricity size and use Proposition 4.2 to realize the possible joint diagram over the actual parameter tuple. The realization has term representations in that presentation.

A support profile over X records the labels for d , its formal support order T_0 of size at most σ , and comparisons and coincidences with S_X . A well-order of size at most χ has at most χ cuts. Hence there are at most $\chi^\sigma = \chi$ profiles. The positions of T_0 in one strict slot over S_X form a cluster. Each profile has a continuation over I which leaves every such cluster intact: insert the cluster at a single cut of I extending its specified cut over S_X .

Call a hit of a new parameter support on T_0 , or the separation of two positions in one cluster by a new parameter support, a new event. For $X' \supseteq X$, all continuations of a profile with no new event over $S_{X'}$ have the same natural diagram over X' . Within each old slot, the whole cluster can only change its position across a block of new parameter positions. The slots are disjoint, so Lemma 4.6 applies simultaneously.

Suppose no placement solves all the prescriptions. Start with any X_0 of size at most χ . Given X_i , choose for every profile over X_i a continuation with no new event over all of I . It fails a prescription on some set of at most χ parameters. Adjoin one such failing set for each profile to obtain X_{i+1} . At limits take unions, and carry out σ^+ stages. Every set constructed, including $X_* = \bigcup_{i < \sigma^+} X_i$, has size at most χ , because $\sigma^+ \leq \chi$.

Take a placement realizing q_{X_*} . At each stage its profile over X_i was considered. If it had no new event between S_{X_i} and $S_{X_{i+1}}$, the preceding no-event comparison would identify its diagram

on the selected failing set with that of the chosen continuation. This contradicts its realization of q_{X^*} . Thus it has a new event at every stage. A support position can be newly hit only once, and a pair can be newly separated only once. There are at most $\sigma + \sigma^2 = \sigma$ such events. They cannot occur at σ^+ different stages. $\square$

Lemma 4.10 (Unique lifts of bounded diagrams). *Let $\mu > \ell$ be a strong-limit cardinal of countable cofinality strictly closed below itself under t and g . Let $A \leq_K B$ have size μ , and identify $A = E(I)$ for a well-order I of size μ . For a tuple d in B of length at most $\sigma < \mu$, the diagrams*

$$D(d, X; B), \quad X \subseteq A, \quad |X| < \mu,$$

lift to one placed comparison over all of A , uniquely as a comparison diagram. Restricting a lifted tuple gives the lift of its restricted diagram.

Proof. The displayed bounded diagrams are defined, since their thresholds lie below μ , and they agree on the parameters with the natural diagrams. For every sufficiently large $\chi < \mu$ with $\chi^\sigma = \chi$, Proposition 4.9 gives a placement realizing the tests of size at most χ . Such χ are cofinal below μ : use $\chi = 2^\alpha$ for $\sigma \leq \alpha < \mu$, enlarging α as necessary.

Once $\chi \geq g(\sigma)$, any two resulting placements have the same full comparison by Proposition 4.8. Fix one of them. Every bounded test is covered by a larger choice of χ , whose placement has that same full comparison. The fixed placement consequently realizes every bounded test. The same argument gives uniqueness. A restriction realizes the restricted tests and hence, by uniqueness, is their lift. $\square$

4.4 Amalgamation and type counts

Theorem 4.11 (Amalgamation at closed levels). *There is an unbounded class of strong-limit cardinals μ of countable cofinality with the following properties.*

1. *K is categorical in μ , and any two μ -sized strong extensions of a μ -model have a strong amalgam of size μ .*
2. *If $A \leq_K B_0, B_1$ all have size μ , and $d_i \in B_i$ are tuples of the same length at most $\sigma < \mu$, there is such an amalgam identifying d_0 with d_1 if and only if*

$$D(d_0, X; B_0) = D(d_1, X; B_1) \quad \text{for every } X \subseteq A, \quad |X| \leq g(\sigma).$$

3. *There are at most μ singleton types over each μ -model, where type means identification in an amalgam over that model.*

One may take every sufficiently large strong-limit cardinal of countable cofinality strictly closed under t and g .

Proof. Such levels are unbounded by the countable closure construction following Proposition 4.3. Categoricity at them has already been proved. Identify a given base with $A = E(I)$, where I is a well-order of size μ . Extend I to an order J whose cardinality γ is a categoricity cardinal at least $t(\mu)$, and let $G = E(J)$. Thus A is strongly included in its specified position in G .

We show that every μ -sized strong extension B of A embeds strongly into G over A . Choose $M_n \leq_K B$, increasing with union B , with $\|M_n\| < \mu$. This follows by LS from a countable exhaustion of B , containing the preceding submodel at each step; coherence makes the chain

strong. Choose compatible enumerations d_n of these models, so that each earlier list is a sublist of the next.

Lemma 4.10 assigns coherent placed comparisons to (d_n, A) . Each is realized over A in G , and the realizations can retain the previous assignments. More explicitly, realize the corresponding term pattern in an order extension of size γ . Its parameter tuple A , and then its tuple (A, d_{n-1}) , have the same diagram as the already assigned ones. Their length is at most μ , so Proposition 4.2 at $\gamma \geq t(\mu)$ supplies the strong map realizing the new list while fixing these previous data.

It remains to verify strongness of the copied small models; atomic agreement by itself would not suffice. Fix n , and bound the length of d_n by an infinite $\sigma_n < \mu$. Choose the categoricity cardinal $\rho = t(\sigma_n) < \mu$. LS puts the original enumeration inside $B'_n \leq_K B$ of size ρ , and the copied enumeration inside $G'_n \leq_K G$ of that size. Coherence gives $M_n \leq_K B'_n$. The copied and original lists have the same bounded diagram; Proposition 4.2 gives a strong map $B'_n \rightarrow G'_n$ with exactly the prescribed values on d_n . Its restriction shows that the copied M_n is a strong submodel of G .

The copied models are nested, and coherence makes their inclusions strong. Their union is a strong submodel of G by the chain and common-upper-bound axioms. It is the image of an embedding of B . Every element of A eventually occurs in some d_n , and its equality with that parameter is part of the lifted comparison. The embedding therefore fixes A . Performing this construction for B_0, B_1 in the same G gives an amalgam. LS supplies a strong μ -submodel of G containing both images; coherence makes it a strong amalgam of the required size.

For the second assertion, any identifying amalgam preserves the displayed bounded diagrams, giving necessity. Conversely, Proposition 4.8 identifies the placed lifts of d_0, d_1 over A . Realize this common lift once in G . On each side start the preceding countable construction with the specified tuple already assigned, and choose every small-model enumeration to contain that tuple. Homogeneity over A and the old assignments works exactly as above. The two resulting embeddings identify the tuples, and the same LS shrinking applies.

For a singleton, a placement uses one label on a finite support. Relative to the well-order I , each support position is either an old position or lies in one of its $|I| + 1$ cuts. Together with the finitely many mutual comparisons and the label, these data have at most μ possibilities. They determine the full natural comparison. The second assertion identifies exactly the types with equal comparison, proving the bound. $\square$

Amalgamation has thus been obtained at the stated levels from the AEC axioms and unbounded categoricity. We have not assumed that all models of intermediate cardinalities amalgamate. The next section records precisely the level-by-level properties that will be required of later categories of finite systems.

5 Working schemes and uniform comparison of diagrams

We isolate the conditions used when the preceding constructions are applied to finite systems of models. The definition includes the hypotheses that must be checked for a new system; it does not infer them merely from categoricity of its vertices.

5.1 Systems and the working package

Convention 5.1 (Closure below a level). An eligible level is an infinite cardinal belonging to the class of cardinals on which the particular construction under consideration has been established. Except when explicitly discussing preliminary levels, we use strong limit levels of countable cofinality. A list is *bounded at μ* if its cardinality is strictly less than μ . If u is a cardinal-valued function, strict closure under u means

$$u(\theta) < \mu \quad \text{for every infinite cardinal } \theta < \mu.$$

Bounds may include fixed parameters, may be increased to infinite cardinals, and may be replaced by monotone majorants.

A system has finitely many vertices, each a K -model, and specified maps between specified vertices. We code it as a many-sorted structure, with one sort for each occurrence of a vertex and function symbols for the specified maps. An element occurring in two vertices therefore has two occurrences in the code, related by the specified map when appropriate. Distinguished lists are coded by names. A system embedding consists of commuting embeddings on its vertices, preserving these names. An *arrow* is such an embedding whose vertex maps are strong, together with any additional condition stated for the scheme. The geometric application will have one such additional condition. No condition on intersections of images is implicit in this definition. An isomorphism means a structural isomorphism onto the entire system.

Definition 5.2 (Working package). Fix an unbounded definable class S of eligible levels, a finite system shape, and a set-sized finite-support term template F . A *working scheme* consists of classes $\mathcal{C}_\mu$, for $\mu \in S$, arrows between their objects, and comparisons D of bounded lists, with the following properties. Every vertex of an object in $\mathcal{C}_\mu$ has cardinality μ .

1. *Structural arrows and unions.* Arrows include all structural isomorphisms, compose, and are invariant under isomorphisms. They are coherent: if X is a subsystem of Y , and both inclusions into Z are arrows, then $X \rightarrow Y$ is an arrow. For an increasing nonempty chain of objects at level μ , whose union has total cardinality μ , the union is an object at that level and the inclusions into it are arrows. If all members of the chain map by the inclusions into one object Z at an eligible level $\nu \geq \mu$, then the inclusion of the union into Z is an arrow.
2. *Presentations and mixed downward closure.* For every well-order I of eligible cardinality μ , $F(I)$ is an object of $\mathcal{C}_\mu$. Its elements are represented by labels applied to finite increasing lists from I ; equality, the symbols of the system structure, and the specified maps are evaluated from the labels and the finite order and equality patterns. Substitution of positions along an order embedding induces an arrow, including when the source and target levels differ. Every object at a level is isomorphic to a well-ordered presentation. If $\nu \leq \mu$ are eligible and A is a set of at most ν vertex occurrences in an object of level μ , there is an arrow inclusion of an object of level ν containing A . Fixed distinguished data are included in this request.
3. *Diagrams, homogeneity, and saturation.* For each index set of bounded cardinality there is an equivalence relation, denoted by equality of D -diagrams, on the corresponding lists in all eligible objects large enough for that index set. It includes the atomic diagram of the entire coded system, respects restrictions and reindexing, and sends every list to a D -equal list under every arrow, including mixed-level arrows. At each level, two bounded lists have the same diagram if and only if a structural isomorphism between their objects takes one list to the other. The number of diagrams at a fixed arity has a set-sized bound

independent of the level. A possible joint diagram of bounded length extending the diagram of an actual bounded list is realized over that list at its level. Here “possible” includes a diagram witnessed at any larger eligible level, provided the total requested length is bounded at the receiving level. On term lists diagrams are natural: they depend only on labels and the order and equality pattern of all positions used, including in comparisons between eligible levels.

4. *Exhaustion criterion for arrows.* Suppose a map is obtained from compatible diagram matches on increasing bounded lists which exhaust its domain object. Include in these lists, for every vertex V , nested enumerations of strong submodels $V_n \leq_K V$ of sizes below the source level, with union V . The image of each such enumeration must enumerate a strong submodel of the corresponding target vertex. Under these conditions the map is an arrow. Component strongness follows from the chain union and common-upper-bound axioms of K ; an additional arrow condition, when present, must be proved for this construction separately.

The empty list is included in the third condition, and has one diagram. Consequently all objects at any one eligible level are isomorphic.

The mixed downward assertion in the second condition usually needs no separate construction. In a chart for the large object, choose finite supports for the requested occurrences. Their union has size at most ν ; enlarge it to a suborder of size ν and use the presentation arrow. If an already given small object is also to be included, put supports for all its occurrences into the request and use coherence.

The assertion about images of small strong submodels in the fourth condition is also a precise obligation, rather than a claim that arbitrary structural embeddings are strong. At one level it follows when the bounded enumeration is carried by a diagram match, since that match is implemented by an isomorphism of the containing objects. For a mixed-level use, first place the smaller containing object in a large presentation by an order arrow. Diagram preservation and an isomorphism at the receiving level then show that the matched small submodel has strong image. Finally, the union of these strong images is strong in the target by the common-upper-bound clause. This proves the componentwise part of the criterion in precisely the situations in which it will be used.

5.2 Refining the eligible levels

The later scheme constructions refine the class of levels on which the working package is available. The following hypothesis ensures that the required refinements still leave an unbounded supply of levels.

Here and below every bound being closed under is given by a fixed set-theoretic definition with set parameters. To see the elementary closure construction explicitly, start above a prescribed cardinal and choose strictly increasing infinite cardinals κ_n such that

$$\kappa_{n+1} > 2^{\kappa_n} \quad \text{and} \quad \kappa_{n+1} > \sup\{u_i(\theta) : \theta \leq \kappa_n, i \in A\},$$

where A is the fixed set of functions to be accommodated. Replacement makes the displayed supremum a set-sized ordinal. The cardinal $\mu = \sup_{n < \omega} \kappa_n$ has countable cofinality, is a strong limit, and is strictly closed under every u_i .

There are two distinct uses of this observation. If a theorem proves its conclusion at every such closure point, it establishes eligibility of μ . If eligibility is already restricted to an unbounded class S , closing under the least member of S above θ ensures only that $S \cap \mu$ is unbounded in

μ ; it does not by itself prove $\mu \in S$. Membership at the limiting cardinal must come from the relevant theorem or from an already established closure property of S . In particular, if S is unbounded and closed under suprema of strictly increasing countable sequences, we may choose the κ_n inside S in the construction above. The resulting μ then belongs to S and has cofinally many smaller members of S . This is the meaning of requiring sufficiently many smaller eligible levels; an arbitrary unbounded class of cardinals is not silently assumed to have this property.

Convention 5.3 (Supply of eligible levels). For the generic scheme refinements below, we use the following *eligible-subclass hypothesis* whenever an unbounded class of refined levels is asserted. The input eligible class S must contain a fixed definable unbounded subclass T closed under suprema of strictly increasing countable sequences. In addition to the input scheme's individual conditions, this is a hypothesis on the supply of eligible levels; arbitrary unboundedness of S does not imply it.

This hypothesis persists through every finite list of the stated refinements. For a fixed definable bound function, choose successive members of T beyond all its values on the ordinals below the previous member, and take their countable supremum. This gives arbitrarily large members of T strictly closed under that bound. The intersection with the bound-closure class is still closed under strictly increasing countable suprema. A threshold only removes an initial segment. For a limit-point refinement of a current class containing such a T , use $\{\mu \in T : \sup(T \cap \mu) = \mu > 0\}$; it is unbounded and has the same countable closure, and is contained in the required limit-point class. Repeat for the finite refinement list, including the finitely many bound functions then required. Thus an unbounded subclass with the same property remains available, including for larger auxiliary-level choices.

Advance notation for common levels. For a definable class A of ordinals, write

$$\text{Lim}(A) = \{\mu : \sup(A \cap \mu) = \mu > 0\}.$$

Here is the hierarchy that will organize the two applications of Section 13. That section constructs one fixed bound majorant Φ for each application. Its initial class G_0 consists of the strong limit cardinals μ of countable cofinality with $\Phi(\alpha) < \mu$ for every ordinal $\alpha < \mu$; subsequent classes are

$$G_{j+1} = G_j \cap \text{Lim}(G_j) \quad (j < \omega), \quad G_\omega = \bigcap_{j < \omega} G_j.$$

This is advance notation: the definition of Φ and the proof that each G_j is unbounded and closed under strictly increasing countable suprema are given in that section. A member of G_{j+1} is thus both a member of G_j and a limit of smaller members of G_j .

In those applications, the input class contains G_j for some finite j . This subclass serves as T , and the finite refinements retain a later G_k . These subclasses supply the required levels; they are not inserted as additional class parameters into the definitions of bounds. Those definitions, given in Section 13, use fixed formulas for the required working assertions and for the finite refinements already described. Without the eligible-subclass hypothesis, the generic proofs assert their conclusions at levels satisfying their displayed bound and limit-point requirements, but do not assert that those levels are unbounded.

5.3 Directed unions and rich orders

Lemma 5.4 (Directed unions and padded presentations). *Assume the structural union and presentation conditions of Definition 5.2 at a level μ . Every nonempty directed family of inclusion subobjects at that level, whose union has total cardinality μ , has an object as its union, with arrow inclusions and the common-upper-bound conclusion, including into an eligible larger level.*

Consequently $F(I)$ is an object whenever I has eligible cardinality μ and contains a well-ordered subset of cardinality μ . We call such an order padded. Order embeddings between padded orders induce arrows, also between different eligible levels. If the diagram conditions of the package hold, diagrams on bounded term lists in padded orders are natural as well.

Proof. We first prove the directed assertion by induction on the cardinality of an indexing set. A finite directed family has a member containing all its members. A countable directed family has a cofinal increasing sequence, obtained by successively choosing an upper bound for the previous choice and the next enumerated member. The chain assertion gives its union and its common-upper-bound property.

For an uncountable indexing cardinal τ , fix a binary operation on the indexing set selecting an upper bound for each pair. Write the indexing set as an increasing union of sets of size less than τ , and close each under this operation. This closure adds only countably many stages and preserves the bound of being smaller than τ . The sets can be chosen nested, and their union is the whole indexing set. By the inductive assertion each corresponding smaller directed family has an object as its union. That union still has cardinality μ : it contains a member of cardinality μ and is contained in the given total union of cardinality μ . The common-upper-bound part of the inductive assertion makes these unions a chain of arrows. The chain assertion now gives the total union. Applying the same common-upper-bound clauses to a specified ambient object proves the last assertion, including the mixed-level version.

Fix a well-ordered subset $J_0 \subseteq I$ of size μ . Let $\mathcal{J}$ consist of all well-ordered subsets of I of size μ . A union of two well-ordered subsets is well-ordered: in any nonempty subset take the smaller of the minima supplied by its two nonempty intersections. Thus $\mathcal{J}$ is directed under inclusion. Every finite subset of I is contained in a member, namely its union with J_0 . The finite-support description of the template identifies

$$F(I) = \bigcup_{J \in \mathcal{J}} F(J).$$

The union has cardinality μ , and the directed assertion applies. For an order embedding $I \rightarrow I'$, first restrict to well-ordered suborders of the source. Their images are well-ordered; if necessary, enlarge each image by a padding subset in the target. The known order arrows and the common-upper-bound assertion give the induced map on the total union as an arrow.

Finally, two occurrences of the same bounded support pattern in orders can be identified by amalgamating the orders over that suborder. Such an amalgamation is obtained cut by cut: retain the common suborder and put the points from each of the two orders in their prescribed cuts, keeping their own orders within each cut. Append a well-order of sufficiently large eligible cardinality to make the amalgam padded. The two order arrows identify the term lists there. Diagram preservation proves their naturalness. $\square$

When the chain assertion for a newly defined scheme is still being proved, the lemma is used only for an underlying scheme whose chain assertion has already been established. Directed unions are not used to presume the assertion under proof.

Lemma 5.5 (Rich padded orders). *Let η be infinite. There is a dense linear order Q_η without endpoints of cardinality 2^η such that every cut given by at most η parameters on either side is filled. Equivalently, every extension of an order pattern with at most η old and at most η added positions can be realized over its old positions. If μ is a strong limit cardinal and $\eta < \mu$, there is a padded order with these properties of cardinality μ . No restriction on $\text{cf}(\mu)$ is needed for this assertion.*

Proof. Put $\xi = 2^\eta$, so that $\xi^\eta = \xi$ and $\eta^+ \leq \xi$. Starting with an order of size ξ , repeatedly fill every cut specified by disjoint sets $A < B$ with $|A| + |B| \leq \eta$, allowing an empty side. At each stage there are at most $\xi^\eta = \xi$ requests; they can be filled in an order extension of size ξ . Perform η^+ stages and take unions at limits. A set of at most η parameters in the final union occurs in one earlier stage, since η^+ is regular. Its requested cut is filled at the next stage. The order is dense and without endpoints because requests with at most two parameters were among those filled. To realize an arbitrary added pattern, well-order its at most η new positions and insert them successively. Each insertion asks for a cut over at most η old or previously inserted positions.

For the second assertion, use the lexicographic sum

$$I_{\mu, \eta} = \sum_{\alpha < \mu \cdot \eta^+} Q_\eta,$$

where the index is ordinal multiplication. Its cardinality is μ , its index has cofinality η^+ , and choosing one point in each summand gives a well-ordered subset of cardinality μ . Here $|Q_\eta| = 2^\eta < \mu$.

Consider a cut $A < B$ with at most η parameters. If B is empty, the indices of the summands meeting A are bounded in $\mu \cdot \eta^+$, and a later summand supplies a point above A . If B is nonempty, let β be the least index of a summand meeting B . All points of A lie in summands of index at most β . In the summand at β , fill the cut between $A \cap Q_\eta^{(\beta)}$ and $B \cap Q_\eta^{(\beta)}$; the first set is allowed to be empty. The resulting point fills the original cut. The same successive-insertion argument realizes added patterns. $\square$

5.4 The initial package and named data

Proposition 5.6 (The package for single models). *At the sufficiently closed levels supplied by Proposition 4.3, the single K -models, with strong maps as arrows and the original term presentation E , form a working scheme. The levels can be refined to those required in Theorem 4.11 without losing this conclusion.*

Proof. The structural conditions are the AEC axioms, including their common-upper-bound clauses. The presentation and its mixed order arrows are supplied by Theorem 2.6. Categoricity at the specified levels identifies every model with each presentation of the same size; supports in such a presentation give mixed downward closure as described after Definition 5.2.

Proposition 4.2 gives the bounded diagrams, their uniform cardinal bounds, restriction compatibility, and preservation under strong maps. Proposition 4.3 gives isomorphism at equal bounded diagrams. For saturation, let a be an actual tuple and let (b, c) witness a possible joint diagram, with $D(a) = D(b)$. Choose an infinite $\sigma < \mu$ bounding both lists. The closure requirements provide a categoricity cardinal $\kappa < \mu$ above σ and the stabilization thresholds for these arities. Put a in a strong κ -submodel of its actual model. Put (b, c) in a strong κ -submodel of its witness, first embedding the witness into a larger eligible presentation if needed. Stabilization identifies the κ -comparison classes of a and b , so a strong map from the second submodel to the first takes b to

a. Its image of c realizes the required joint diagram. Naturality follows from the order arrows and order amalgamation. Images of small strong submodels are strong by the isomorphism argument following Definition 5.2; taking their increasing unions proves the exhaustion criterion. Finally, imposing the additional closure bounds of Theorem 4.11 retains each of these already established assertions. Those additional bounds provide the amalgamation and comparison facts needed later; they are not extra axioms of a working scheme. $\square$

Proposition 5.7 (Naming bounded data). *Let $\mathcal{C}$ be a working scheme, and fix a bounded list c and its diagram, witnessed at some eligible level. After increasing the fixed lower threshold, the objects with a named realization of this diagram, and arrows preserving the names, form a working scheme. This includes naming enumerations of fixed small strong models in the vertices, with their specified maps. The additional lower bound can be chosen in terms of the original scheme and the cardinality of the named list, independently of the level at which that list was first witnessed.*

Proof. Let β be an infinite bound for the length of c , and choose one eligible $\alpha > |F| + \beta$. The diagram of c has a witness at α . If its original witness is at a larger level, use saturation at α with the empty tuple as base. If its witness is at a smaller level, first extend that object to a presentation at a level at least α using an order arrow. This gives a choice of α uniform over all diagrams of the specified arity. Choose a well-ordered chart for this new witness and fix representations of c . Let J be the union of their finite supports, with its inherited well-order. In the presentations $F(J + H)$ name the terms representing c on J . Regard this as a template in the variable order H : any term using positions from both parts becomes a label recording its original label and its finitely many fixed positions from J , applied to its remaining positions from H . Terms supported wholly on J are labels of arity zero. The new collection of labels has cardinality at most $|F| + |J| + \aleph_0$ and is fixed independently of the level. Order maps on H , extended by the identity on J , induce arrows preserving the names.

Naturality gives the prescribed diagram on the named list in every sufficiently large such presentation. Homogeneity for c shows that every object with such a named realization is isomorphic to one of these presentations. Define its diagram on a further list a to be the old diagram on the combined list (c, a) . The old package gives exactly restriction compatibility, the uniform bound, preservation, homogeneity, and saturation for these diagrams. The same support construction gives mixed downward closure, now including the named terms. Chain unions and the exhaustion criterion retain the names, so they are inherited.

For the final assertion suppose a sublist enumerates $M \leq_K V$ in the original witness, which may be at a level larger than α . Embed that original system and the new level- α witness into objects of a common larger eligible level. Mixed preservation and homogeneity there give an isomorphism matching their named lists. The original copy of M has strong image in that large vertex. This image is exactly the substructure enumerated inside the embedded level- α vertex, so coherence makes it strong in that vertex as well. It is a K -model because its complete atomic enumeration is isomorphic to M . Thus strong naming already holds in the uniformly small witness. Above α , order arrows and the same homogeneity argument preserve it. The argument applies to every named vertex model and specified map. Thus the new presentations satisfy the intended strong naming condition, not just the atomic diagram of the enumeration. $\square$

5.5 Two routes to working diagrams

The next statement records the exact diagram input that may be used for a new scheme. In its first route the role of Theorem 3.11 is an explicit hypothesis; it is not assumed for a new category without proof.

Proposition 5.8 (Obtaining the diagram clause). *Suppose a finite-system scheme has the structural and presentation conditions of Definition 5.2, and is categorical at its eligible levels. For an unbounded supply after refinement, assume the eligible-subclass hypothesis of Convention 5.3. Either of the following inputs supplies the diagram clause, after the indicated closure refinements.*

1. *For each infinite σ , at all sufficiently large eligible levels, the relation of being sent to another list by an arrow is reversible on lists of cardinality at most σ , and their orbit sets have a bound depending only on σ and the fixed scheme. In this case take stabilization bounds and require enough smaller levels below the receiving level for the bounded arities in use.*
2. *Same-level arrows preserve isomorphism orbits of bounded lists, and comparisons of such orbits in well-ordered charts are independent of the eligible level. More precisely, for each arity, equivalence of two term patterns by an isomorphism in their respective charts is the same relation on patterns whenever both patterns are bounded at the levels under comparison.*

The componentwise exhaustion condition follows whenever arrows have only componentwise strongness as their additional structural content. For any other arrow condition its exhaustion assertion remains a separate obligation.

Proof. For the first route, fix an arity bound σ . At a sufficiently large eligible cardinal κ , arrow comparison on lists of this arity is reflexive and transitive, and reversal makes it symmetric. A list in a larger object can be put into a level- κ subobject. Two choices of that subobject are included in a third by mixed downward closure and coherence. Their arrow classes therefore agree. This defines the level- κ comparison in all larger objects and shows that it is preserved under mixed arrows.

If $\gamma \geq \kappa$ is another sufficiently large eligible cardinal, there is consequently a projection from the γ comparison classes to the κ comparison classes. This projection is onto: an object at level κ embeds into one at level γ by identifying it with a chart and using an order arrow. The projections compose. Let b bound the numbers of classes. If strict refinement continued arbitrarily high, recursively choose b^+ many successive strict refinements and then an eligible cardinal above all the chosen levels. At each strict refinement some pair of classes at this final level has equal images before the refinement and unequal images after it. A fixed pair can witness this only once, since unequal images cannot merge in a refinement. There are at most $b^2 = b$ pairs, a contradiction. Thus the comparisons stabilize.

Their stabilized equality is canonical, and comparisons at different arities commute with restriction: raise the cardinal beyond the stabilization thresholds for both arities and restrict an arrow implementing the longer comparison. Atomic diagrams are preserved by arrows. A prescribed bounded extension is realized over an actual list in a sufficiently high same-level object by placing a witness and the actual list into objects of one common eligible size and using the reversible comparison to map the witness list to the actual one. This gives the joint extension as well.

At a receiving level μ of countable cofinality, close below μ under these thresholds and under choices of a sufficiently large smaller eligible cardinal for each arity. Enumerate each of two objects by countably many nested bounded lists. To extend a match, choose a smaller eligible cardinal

containing all the old and newly requested entries and above their stabilization thresholds. Put the data into subobjects at that cardinal and implement the equal stabilized diagrams by an arrow in the desired direction. Alternate the direction of extension and include the next enumeration list on each side. The union is a bijection preserving and reflecting every atomic formula, hence an isomorphism of systems. Starting with a given equal-diagram tuple makes it part of that isomorphism. The same smaller-level construction realizes a possible bounded extension diagram over an actual tuple. Order arrows and order amalgamation give naturality. These arguments use both eligibility of μ and the availability of smaller eligible cardinals, as distinguished in Section 5.2.

For the second route, instances of the same well-ordered pattern have the same orbit. Indeed two well-orders amalgamate over a common suborder into a well-order: in each cut of the common well-order concatenate the two remaining well-ordered pieces, then retain the common points between successive cuts. Enlarge to an eligible size if necessary. The induced order arrows and same-level orbit preservation identify the two instances. The assumed independence of orbit comparison from the level therefore defines diagrams on well-ordered patterns. The set of such patterns at a fixed arity gives a uniform bound.

For a mixed arrow from level ν to level μ , include the image of its whole source in a level- ν subpresentation of the target. Coherence factors the arrow through that subpresentation. Same-level orbit preservation computes the comparison there, and the assumed agreement of chart comparisons computes the same diagram in the target. This proves mixed preservation. For saturation, take a well-ordered support pattern witnessing the desired joint diagram in a larger object. Its support has cardinality below the receiving level, so it embeds into a well-order of that level. The old subtuple has the diagram of the prescribed actual tuple; an isomorphism at the receiving level carries the new realization over that tuple. The orbit hypothesis itself supplies homogeneity. Naturality on padded orders follows from Lemma 5.4.

In either route, diagrams on bounded enumerations of small vertex submodels have strong images by homogeneity, with the preliminary order embedding in the mixed-level case. The AEC union axioms give the asserted componentwise exhaustion condition. $\square$

5.6 A set-theoretic code for diagram equality

The next construction supplies one definition of comparison that can be used uniformly with a finite descriptor for the scheme. It does not define truth for a varying family of proper-class predicates.

A code for a finite-support template consists of its set-sized finitary language, sorts and specified-map symbols, its labels and their finite arities, and tables evaluating the function symbols, equality, and relations on finite ordered support patterns. A function table supplies a label and sublist for its output within the union of its inputs' supports; its arity-zero case is included. Adjoin labels of arity zero for closed terms if necessary. This does not increase the infinite bound on the language and label set. Fixed names, including the labels of arity zero in Proposition 5.7, are part of this code. The usual consistency requirements say that the evaluated equality is an equivalence relation and a congruence, that the function tables are total and correctly sorted, and that the evaluations agree under passage to larger finite support patterns. These are tests on sets. A coherent code constructs a structure $F(I)$ on any set-sized order I by taking the quotient of the set of labeled finite lists by the evaluated equality. This construction makes sense whether or not $F(I)$ is an object of a working scheme at that size.

For an arity bound σ , a *pattern* records an indexed list of labels, finite increasing support lists for them, and the order on the union of these supports. Use canonical support universes of

cardinality at most σ and index sets of cardinality at most σ . These codes form a set, denoted by $\mathcal{P}_F(\sigma)$. Two codes representing the same terms need not be identified in advance. The comparison below handles such duplicate representations. When names are represented by reserved support positions instead of labels of arity zero, include their fixed support in every pattern and every support map; the test bounds must then also dominate its cardinality.

Proposition 5.9 (Uniform coding of working diagrams). *There is a fixed first-order set-theoretic definition D_F of an equivalence relation on term patterns, uniform in a coherent finite-support template code F . There is a definable cardinal bound $b_F(\sigma)$ such that, on $\mathcal{P}_F(\sigma)$, a single back-and-forth test with bound at least $b_F(\sigma)$ computes D_F . These definitions do not require the system axioms at the cardinals used for the tests.*

Suppose, in addition, that F presents a working scheme at unboundedly many eligible strong limit levels of countable cofinality. At every eligible level $\mu > |F|$ strictly closed under b_F , the working diagram of a bounded term list is exactly its D_F -comparison class. The same definition therefore computes the diagrams of every such scheme, including the schemes with fixed named data.

Proof. Fix an infinite η at least the cardinality of the template code and of the patterns to be tested. For two set structures A, B in its language, form the set of all pairs of tuples of the same index set of cardinality at most η . Index sets can be taken to be ordinals below η^+ , reindexing concatenations when necessary. Let R_0 consist of the pairs with the same atomic diagram. Having defined R_α , retain in $R_{\alpha+1}$ just those pairs $(a, b) \in R_\alpha$ such that every added tuple of cardinality at most η on either side has a matching added tuple on the other side with the extended pair in R_α . At a limit take the intersection of all preceding relations.

This decreasing recursion stabilizes. If its state set has cardinality τ , there cannot be a strict decrease at every successor stage below τ^+ , since each strict decrease removes a previously unremoved state. Equality at one successor stage gives a fixed point. Write R_∞ for the stabilized relation. In particular, from a pair in R_∞ every allowed extension has a matching extension still in R_∞ ; quantification over all ranks is not being exchanged with an existential choice without this fixed-point argument. Reflexivity and symmetry follow from the construction. Transitivity follows by composing matches: the composition of two relations having the atomic and extension properties again has those properties, and is contained in the largest such relation R_∞ . This also proves invariance under isomorphisms.

Use orders satisfying Lemma 5.5, and realize the two indicated patterns in their term structures. Declare $p \equiv_{F, \eta} q$ when their tuple pair belongs to R_∞ . Every support pattern of size at most η embeds into such an order, by the successive insertion assertion in that lemma.

The definition is independent of the rich orders and of the realizations of an individual pattern. To prove this, relate two term tuples when their representations are carried to each other by an order isomorphism between support sets of size at most η . This relation has atomic matching by the definition of the template. An added tuple has at most η new support positions. Order richness extends the support map to these positions on the other side, in either direction. Thus this relation has the extension property and is contained in R_∞ . Replacing an instance of either pattern by another instance and using transitivity proves independence for the comparison of two possibly different patterns.

If $\eta' \geq \eta$, orders rich for η' are also rich for η , and their back-and-forth test with the larger moves implies the test with the smaller moves. Independence of the chosen orders gives

$$\equiv_{F, \eta'} \subseteq \equiv_{F, \eta}$$

on patterns in the common domain. Define D_F to be agreement at every sufficiently large test bound η .

Here is both uniform definability and the promised single bound. In defining the test at η it is enough to use rich orders of cardinality 2^η on a fixed standard universe, whose existence was proved in Lemma 5.5. All their order relations, the realizations of the patterns, the associated term structures, and the states of the recursion are sets. The recursion up to the successor cardinal of its state set defines R_∞ by an ordinary first-order set-theoretic formula. Independence means that one may quantify over the rich orders and realizations in that set, without making a uniform class choice of an order.

For each pair in $\mathcal{P}_F(\sigma)^2$ which fails some test, take its least failing infinite cardinal above the fixed input bounds. This is a uniquely defined ordinal when it exists. Replacement collects these witnesses into a set. Choose $b_F(\sigma)$ above their supremum, above σ , and above the template and naming bounds. Monotonicity proves

$$p D_F q \iff p \equiv_{F, b_F(\sigma)} q, \quad p, q \in \mathcal{P}_F(\sigma).$$

The same holds at every larger test bound. Taking suprema over smaller arities makes b_F monotone if desired. Notice that the cardinality of the set of patterns justifies taking the set of separation witnesses; it does not assert an a priori numerical upper bound on their ranks. A code failing the coherence tests may, if a total definition on all set codes is desired, be assigned the discrete comparison and a fixed default bound. This convention has no mathematical role for a coherent template.

We now compare with the working diagrams. First suppose two bounded patterns have equal working diagrams. Fix any test bound η . Choose an eligible level large enough for the patterns and η , and use a rich padded order of that level from Lemma 5.5. Lemma 5.4 makes its term structure an object of the scheme. Naturalness transfers the equal working diagrams to realizations of the two patterns there. The package gives an isomorphism taking one realization to the other. It passes the back-and-forth test at η . Independence of rich orders now gives $p \equiv_{F, \eta} q$. Since η was arbitrary, equal working diagrams imply D_F -equivalence.

Conversely fix an eligible μ as in the statement and two objects of that level with bounded lists whose patterns are D_F -equivalent. Choose countable nested bounded enumerations exhausting both objects. We construct compatible matches containing these enumerations, maintaining D_F -equivalence of their term patterns. Suppose the current lists have been matched, and a next bounded list on one side is requested. Let an infinite $\rho < \mu$ bound the cardinality of the combined old and requested lists. Choose

$$b_F(\rho) \leq \eta < \mu.$$

This is possible by strict closure. At the same level μ , form rich padded presentations for this test. Realize the current patterns there. Naturality and homogeneity for the working diagrams give isomorphisms from the current objects to these presentations taking their current lists to the chosen realizations.

Transport the requested list through the first isomorphism. The current pair passes the test at η , so the fixed-point extension property supplies a matching list in the other rich presentation. Choose finite-support term representations for the extended lists. Their patterns pass the η -test; because $\eta \geq b_F(\rho)$, they are D_F -equivalent. Transport the matching list back to the actual target object. The isomorphisms used in this transport preserve working diagrams, which were already shown to imply D_F -equivalence. Transitivity therefore gives the required D_F -equivalence for the extended actual lists, independently of their chosen representations.

Repeat with the sides reversed and with the next lists in the two enumerations. All stages have cardinality below μ ; only their countable union exhausts the objects. Atomic matching ensures that the compatible assignments define a well-defined injective map, and the alternating exhaustion makes it onto every sort. Every atomic instance involves finitely many entries and occurs at one stage, so the resulting bijection is a structural isomorphism. It carries the original list to the other list. The working package therefore identifies their working diagrams, as required. $\square$

Remark 5.10 (What the code does and does not establish). The code F , its finite shape, and its fixed naming data are set parameters. The recursion in Proposition 5.9 is a recursion on a set of states of set structures. Its definition is the same formula for every such parameter, including templates arising after a finite number of later constructions. It therefore introduces no satisfaction predicate for arbitrary class definitions. The bound $b_F(\sigma)$ depends on the full fixed descriptor; names or labels of unbounded size cannot be omitted from that input.

The identification of D_F -equivalence with equality of working diagrams uses an already established working package, including its actual isomorphism assertion. It supplies a uniform way to express that package in subsequent closure tests; it is not a substitute for proving homogeneity, chain closure, or an additional arrow condition for a new scheme. The later simultaneous closure argument must still prove that its descriptors and its bound recipes are uniformly set-theoretically definable and that their requested bounds exist on relevant inputs.

6 The one-dimensional calculus

Throughout this Section, $\mathcal{C}$ is a scheme satisfying the working package of Definition 5.2. Its fixed presentation is denoted by F , and b is an infinite cardinal bounding the set of labels and the finite amount of structural data in the scheme. We use the occurrence sorts of that Definition: a tuple in several vertices is a tuple of tagged occurrences, and a map of systems commutes with every specified structural map. No choice of a hull or a basis is part of a diagram. A level is an eligible strong limit cardinal of countable cofinality. All further closure requirements below are requirements on functions of bounded arities and fixed scheme data. For the supply of unbounded refinements and larger auxiliary levels, we use the eligible-subclass hypothesis of Convention 5.3: the input class contains a fixed definable unbounded subclass closed under suprema of strictly increasing countable sequences. This subclass is used only to supply levels; the bounds below depend on fixed scheme data and bounded arities.

Stationary independence and local character are central to the theory of good frames; see [She09, II.2.1]. The assertions below are proved from the working package for bounded tuples and finite systems. Their role is to supply the independence and descent properties needed for the next isolation construction.

A *query* is an enumerated parameter set Z together with the diagram of the displayed tuple and that enumeration. Repetitions in an enumeration are allowed. Saying that an answer is correct on Z means equality of the whole joint diagram, rather than equality only on finite subtuples. At a level μ , every tuple or query called bounded has cardinality less than μ .

6.1 Testing, placement, and amalgamation

Lemma 6.1 (Parameter orbits for a scheme). *Fix an infinite σ . If $\chi > b + \sigma$ and $\chi^\sigma = \chi$, then, at every sufficiently closed level greater than χ , there are at most χ diagrams of σ -tuples over a fixed parameter tuple of length at most χ . Moreover, natural diagrams are unchanged by simultaneous disjoint block transpositions: within each of a family of disjoint convex regions, interchange an adjoining block belonging exclusively to the first tuple family with one belonging exclusively to the second, retaining each block's internal order.*

Proof. Present the parameter object on a well-order. After choosing one term representation of each parameter, its supporting positions form a well-ordered set of size at most χ . There are at most χ slots and positions relative to this set. Labels, comparisons and coincidences for at most σ additional positions therefore have at most $\chi^\sigma = \chi$ possibilities. Equal profiles give equal natural diagrams. The diagram-isomorphism clause of the working package turns this into the asserted orbit bound over the actual parameters.

Here are the hypotheses needed for the block argument in Lemma 4.6. We have just proved its parameter orbit bound. Repeating the two blocks along an arbitrary order is permitted after padding: the scheme has the union and common-upper-bound clauses, so Lemma 5.4 supplies the padded presentations and their order arrows. Naturalness compares the two patterns at a common higher eligible level. If the transposition changed a diagram, the repeated patterns would distinguish cuts by whether the first block preceded or followed a parameter block. A binary tree of least height ϵ with $2^\epsilon > \chi$ has at most χ nonterminal nodes and more than χ branches. Divider points at those nodes give at most χ parameter blocks and more than χ different diagrams over them. The preceding orbit bound gives a contradiction. The repetition uses only a set of positions, so a higher eligible level can contain the entire test. This proves the assertion also when there are boundedly many simultaneous transpositions. $\square$

Definition 6.2 (Natural parameter layouts). A natural parameter layout consists of finitely many well-ordered segments, a fixed finite set of colors and their prescribed order, and specified families of terms on specified collections of those positions. Its parameters are all the designated terms; vertex sorts may be among the designations. The data must have the following monotonicity property: after freezing bounded extra supports, increasing the ordinal lengths of the remaining slots gives an embedding of layouts that preserves the designated parameter terms. Bounded frozen parameters are permitted and count towards the input arity. The layout, labels and color conventions are fixed set data.

The whole object $F(H)$, with H well-ordered, is a natural layout. Unions of designated vertex sorts generated on specified segments are further examples. An arbitrary subset of the parameters in an unrelated target chart is not asserted to be a natural layout.

Proposition 6.3 (Uniform testing for natural layouts). *For each fixed natural layout there is a monotone cardinal function $g(\sigma)$ with the following property. Two placed comparison types of tuples of length at most σ over that layout are equal if and only if they agree on parameter queries of size at most $g(\sigma)$. The bound is independent of the eligible level. The extra tuple supports may lie in a non-well-ordered extension of the layout. The assertion is uniform when the fixed layout and frozen bounded data range over a fixed set of descriptors of bounded size.*

Proof. Amalgamate the orders containing the two placements over the parameter order. Freeze the labels of the two tuples, their combined ordered support T , the coincidences with old positions,

and the colors. There are set-many such frozen patterns at the prescribed bound. The unfrozen well-ordered parameter positions are described by a vector of ordinal lengths, one coordinate for each strict slot and each prescribed segment. There are set-many coordinates, even when T itself is not well-ordered. By the monotonicity requirement in Definition 6.2, a pointwise increase embeds the old layout and its designated terms into the new one. Unequal comparisons remain unequal because diagrams are preserved by the induced arrows.

For a fixed frozen pattern, apply Lemma 4.7 to the upward closure, among all ordinal vectors, of its valid unequal configurations. It gives a set of generators, which can be chosen from valid unequal configurations. Each generator has a set-sized parameter order. Take a cardinal above the sizes of the terms on all these orders, then above the resulting bounds for all the frozen patterns. An unequal configuration contains one of the generating configurations and is consequently witnessed on at most that many parameters. Increasing the bound monotonically gives g . This is precisely the ordinal-vector proof of Proposition 4.8; its required ingredients here are natural diagrams, their preservation, and the explicit monotonicity of the layout, all verified above. Taking the same supremum over a set of descriptors proves the last assertion. $\square$

Lemma 6.4 (Compact placement for a scheme). *Let P be any chosen parameter set in a well-ordered presentation, with one fixed term representation for each parameter. Let v have length at most σ , and let*

$$\chi > b + \sigma, \quad \chi^\sigma = \chi.$$

Suppose that coherent diagrams for (v, X) are prescribed for every $X \subseteq P$ of size at most χ , and that each individual prescription has a placement. At a sufficiently closed eligible level containing the data, one order enlargement realizes all these prescriptions simultaneously. The same assertion holds with bounded frozen data, after adding its size to σ .

For a padded order that need not be well-ordered, it suffices instead to have locally possible coherent prescriptions on every bounded subset and closure under the profile-count growth used in the proof. The conclusion then holds simultaneously at any one specified bounded test size. All the bounds are uniform in the fixed scheme and layout data.

Proof. For $X \subseteq P$, let S_X be the union of its chosen supports. A profile records the labels of v , its formal ordered support, and all comparisons and coincidences with S_X . If $|X| \leq \chi$, there are at most χ profiles. In each strict slot over S_X , all the new positions there form one cluster. A profile has a continuation over the whole old order with no new event: place each cluster intact into a single strict slot of the old order extending its slot over S_X . Order enlargement supplies positions if necessary. A new event on enlarging X means a new hit of an old position or a separation of two previously unseparated new positions by an old position.

Two continuations with no new event over S_Y , for $Y \supseteq X$, have the same diagram over Y . Indeed, their differences consist of moving whole clusters past blocks of parameter supports. The regions belong to distinct slots, so the block transpositions in Lemma 6.1 apply simultaneously.

If simultaneous realization fails, choose X_0 of size at most χ . For each profile over X_i , choose a no-new-event continuation and one prescribed query where it fails. Adjoin those queries to X_i to obtain X_{i+1} ; at limits take unions. There are at most χ profiles, each chosen query has size at most χ , and $\sigma^+ \leq \chi$. Thus every X_i , including X_{σ^+} , has size at most χ . Choose a placement realizing the prescription on X_{σ^+} . At every earlier step this placement must have a new event, since otherwise the no-event comparison would transfer the chosen failure to it. There are at most σ possible hits and at most σ pairs of new positions. Each event can occur for the first time only once. This contradicts the σ^+ steps.

In a non-well-ordered chart, the number of profiles over X_i is at most $2^{|X_i|+\sigma+b}$. Starting with the desired test size, use this as the next cardinal bound and include a failed query of at most that original test size for every profile. The larger sets X_i have prescriptions because those were supplied on all bounded subsets. More explicitly, start with an infinite β_0 above the desired test size and set $\beta_{i+1} = 2^{\beta_i+\sigma+b}$, taking suprema at limits. Require the single bound $\sup_{i \leq \sigma} \beta_i$ to be strictly below the level. This is closure under the whole indicated recursion, rather than merely closure under each individual exponentiation. It keeps the final set bounded even though the level has countable cofinality. Its prescription is available by hypothesis, and the identical event count is a contradiction. This is a set recursion depending only on the starting arity and fixed data. $\square$

For later reference, fix a convenient explicit threshold for the whole object layout. Increase g so that $g(\sigma) \geq b + \sigma$, and put

$$c(\sigma, \tau) = 2^{g(\sigma)+b+\sigma+\tau}, \quad c(\sigma) = c(\sigma, \aleph_0). \quad (1)$$

These cardinals exceed the displayed arities and satisfy $c(\sigma, \tau)^\sigma = c(\sigma, \tau)$. We close levels under these functions and their successors, as well as under the diagram and rich order bounds at the resulting arities. For a different fixed natural layout, use its testing function in the same formula.

Lemma 6.5 (Placed comparisons of actual tuples). *Suppose $A \rightarrow N$ is an arrow between same-level objects and v is a bounded tuple in N . Its diagrams over bounded subsets of A have a placed comparison over a well-ordered presentation of A . This comparison is unique as a diagram over all of A . Two such actual tuples have equal comparisons if and only if their diagrams agree over queries in A of size at most $g(|v| + \aleph_0)$. The conclusion also holds over any natural parameter layout for which the same coherent local diagrams are given.*

Proof. A single bounded request is possible in a presentation: the parameter diagram is preserved by the arrow, and saturation realizes its desired extension over the actual parameters in a level object. Apply Lemma 6.4 at $\chi = c(|v| + \aleph_0, \tau)$, for any desired bounded query size τ . Two outputs agree on all tests of size at most $g(|v| + \aleph_0)$, and Proposition 6.3 identifies their full placed comparisons. For any bounded query, choose τ large enough to include it. The unique comparison therefore computes every actual bounded diagram. This argument does not require an amalgamation relation on types. $\square$

Theorem 6.6 (Amalgamation and types for a scheme). *At sufficiently closed levels, $\mathcal{C}$ has amalgamation. If $A \rightarrow B_0, B_1$ are arrows and v_i are equally indexed bounded tuples in B_i , then an amalgam over A identifying v_0 and v_1 exists if and only if their diagrams agree on all parameter queries in A of size at most $g(|v_i| + \aleph_0)$.*

Consequently $\text{tp}(v/A; B)$, defined by identification in an amalgam over A , is an equivalence class. It is determined by these bounded diagrams and has an order placement. Restriction of a type means restriction of its actual tuple and parameter diagrams.

Proof. Write arrows as inclusions by taking renamed copies. We prove the stronger construction with a prescribed identification of bounded tuples; omit the tuples for ordinary amalgamation. Let the target initially be $N_0 = B_1$. Choose increasing bounded lists d_n exhausting B_0 , starting with $d_0 = v_0$ (or the empty list when no identification is prescribed). In each vertex also choose an increasing countable exhaustion by strong submodels of cardinality less than the level, and include their enumerations in the d_n 's. Such exhaustions exist by the component LS property, coherence and $\text{cf}(\mu) = \omega$.

We construct targets N_n and compatible images of d_n . The invariant is equality of all bounded joint diagrams of (d_n, A) with their original diagrams in B_0 . For the prescribed initial tuple this is the hypothesis and Lemma 6.5; the empty initial tuple is automatic.

Suppose an image of d_n has been chosen. In a well-ordered chart of N_n , regard A and this whole image as the parameter set P . For a bounded subset of P , its required joint diagram with d_{n+1} is read in B_0 , interpreting the old image as d_n . This prescription is well-defined on repetitions and coincidences by the invariant. It is locally possible: equality on the old parameters and saturation give a realization in N_n . Choose a test cardinal of the form (1), large enough also for the length of d_n , and apply Lemma 6.4. The resulting tuple lies in an order enlargement N_{n+1} and retains the old assignments.

To verify the invariant, use a natural well-ordered chart of the *source* A . Both the desired tuple and the newly placed tuple have comparisons there by Lemma 6.5. Their agreement on the chosen tests includes the testing bound for the joint arity, so Proposition 6.3 makes their comparisons equal. They therefore agree on every bounded query in A . We have not applied uniform layout testing to the possibly unnatural image of A in the target chart; compact placement alone was used on that image.

Take the union of the countable target chain. Its total size is μ , so the union and upper-bound clauses of the package give an object and an arrow from B_1 . The assignments define a map from B_0 into this union. They match diagrams on increasing lists that exhaust B_0 and include the component strong-submodel exhaustions. The image of each such small submodel is strong: the matching diagram of its enumeration is implemented by an isomorphism between its original containing object and the target object. The arrow criterion of the package consequently makes this map an arrow, including any extra basis condition already checked for the current scheme. The initial tuples have the same image. Necessity of diagram equality follows from preservation in any amalgam.

Amalgamation makes the identification relation transitive: amalgamate two witnesses over their common intervening object. Reflexivity and symmetry follow from identity maps and interchange of the two legs. The equivalence and testing descriptions now follow from the proved criterion and Lemma 6.5. $\square$

Corollary 6.7 (Realization into a larger level). *Let $\nu < \mu$ be sufficiently closed eligible levels. If $A \rightarrow B$ are ν -objects and $A \rightarrow N$ is an arrow into a μ -object, then there is an arrow $B \rightarrow N$ extending the given arrow on A . One may prescribe the image of a tuple of length less than ν whose diagrams over queries in A below ν already match. Types of tuples of length less than ν over A obtained in a larger object are actual ν -types, and the mixed diagram comparisons at those arities agree.*

Proof. Use the countable exhaustion in the preceding proof, with all source lists and testing thresholds below ν . Perform the compact placement in a chart of N . The placed tuple, all of A , and the previously assigned list together have cardinality at most ν , which is bounded at μ . Saturation at μ therefore realizes their whole joint diagram back in the actual N , fixing the existing parameters. Continue there, without enlarging the final target. For component strongness, first place the source object at level μ by a presentation arrow. Mixed preservation and an isomorphism of μ -objects implementing the matching enumeration show that every small source strong submodel has strong image in N . The exhaustion criterion now gives the arrow.

To interpret a tuple of length less than ν over A inside a larger object, take a ν -subobject containing A and the tuple, using mixed LS. Coherence makes it an extension of A . Mixed

preservation identifies its diagrams with those in the larger object. The testing criterion shows independence of the chosen subobject. This also justifies the source-level type comparisons used during the construction. $\square$

6.2 Approximations and stationary free extensions

Amalgamation has made types over a whole object available. We now choose their free extensions by means of sequences in the base: on each bounded parameter query, almost all entries of the sequence give the same answer. The next two results construct these sequences and show that their answers determine an actual type, independently of the choices.

Definition 6.8 (Approximation sequence). An approximation sequence for $p = \text{tp}(v/A)$ at query size τ is an indexed family $I = (v_i : i < \kappa)$ in A , each v_i having the same index set as v , such that the concatenation of the family is bounded and, for every $Z \subseteq A$ of size at most τ ,

$$|\{i < \kappa : D(v_i, Z) \neq p \upharpoonright Z\}| \leq |Z| + \aleph_0. \quad (2)$$

Here $p \upharpoonright Z$ denotes its joint diagram, and a sequence is always indexed: different indices may carry the same tuple. We use the same terminology for all projections of the displayed tuple. Whenever $\kappa > |Z| + \aleph_0$, the unique value assumed outside at most $|Z| + \aleph_0$ indices is called its *majority value* on Z .

Lemma 6.9 (Uniform majority approximations). *Let $|v| \leq \sigma$ and let $\tau \geq \aleph_0$ be bounded. At sufficiently closed levels there are approximation sequences for $p = \text{tp}(v/A)$ at size τ , of every prescribed bounded infinite length κ . They satisfy (2) simultaneously for all subtuples of v . The construction has bounds depending only on b, σ, τ, κ , rather than on the level or the type.*

In particular, take $\tau = c(\sigma)$ and $\kappa = c(\sigma)^+$. The concatenated sequence then has a uniform bounded length depending only on σ and the fixed scheme. Its diagram is a code for p in the precise transport sense proved in Theorem 6.10 below.

Proof. By the rich-order and padding constructions of Lemmas 5.5 and 5.4, present A on a padded dense order sufficiently saturated for every support size used below. The presentation is an object of the current scheme; categoricity in the package gives an isomorphism with A . Lemma 6.5, or its compact-placement proof in the padded chart, places p correctly at size τ .

Keep every extra supporting position that belongs to the old chart. Partition the other supporting positions into clusters according to their strict cut over the entire old order. For every pair of distinct cuts choose an old position separating them. Include the shared supporting positions as well. This gives a bounded anchor set of size at most $\sigma + \aleph_0$. Each unshared cluster now belongs to a different strict slot over the anchors.

In each such anchor slot, choose κ pairwise disjoint convex regions in the old order and, in the i th region, put a copy of the corresponding cluster with its internal order. The prescribed richness supplies these patterns; its required arity is bounded in terms of $\sigma + \kappa$. Use the original labels and the unchanged shared positions to obtain $v_i \in A$.

Choose a query Z and one finite support for each of its elements. If these supports avoid every region used for the i th copies, then each copied cluster and the original cluster can differ relative to the query only by passing whole blocks of parameter positions. There are no hits or separations inside a cluster. Their anchor slots are disjoint, so Lemma 6.1 gives $D(v_i, Z) = p \upharpoonright Z$. Every supporting position of Z belongs to at most one of the regions, and therefore excludes at

most one index i . Finitarity of supports gives at most $|Z| + \aleph_0$ excluded indices. The same set of good indices works for every subtuple of v . Only bounded support patterns, the fixed labels, rich orders at bounded arities, and the compact-placement bounds were used. This proves the uniformity assertions. $\square$

Theorem 6.10 (Stationary free extension). *Let $A \rightarrow B$ be an arrow at a sufficiently closed level and let $p = \text{tp}(v/A)$, where $|v| \leq \sigma$. There is a specified actual type p^B over B , called its free lift, extending p . At every desired query size whose approximation and testing costs are bounded at the originating level, it is computed by transporting sufficiently long approximations in A . At one level this includes every bounded query size; in a mixed-level use the costs must remain below the level of A . It has the following properties.*

1. *The answer is independent of the approximation sequence, its length, and the auxiliary test size used in its construction.*
2. *Free lifts commute with restriction of the tuple, are invariant under isomorphisms, and satisfy $(p^B)^C = p^C$ whenever $A \rightarrow B \rightarrow C$.*
3. *If A has an eligible size ν and B has a larger eligible size, the construction still works for $|v| < \nu$, with its construction thresholds and approximation lists below ν . The restriction to A is the actual ν -type of Corollary 6.7.*
4. *A sequence from Lemma 6.9 at the fixed threshold $c(\sigma)$ is bounded code data. If an equally indexed sequence in any sufficiently closed eligible object has the same diagram, it determines a unique corresponding type there by the same majority prescription. Arrows carrying one such sequence to another carry the corresponding type to its free lift. In particular an isomorphism carrying the code sequence transports the type.*

We say that an actual type over B is free from A when it is the free lift of its restriction to A .

Proof. First fix a query bound τ and take an approximation sequence $I = (v_i : i < \kappa)$ in A with $\kappa > \tau + \aleph_0$. Given a query Z in B of size at most τ , saturation moves it into A over the entire sequence: there is $Z_0 \subseteq A$, with the same enumeration, such that

$$D(I, Z) = D(I, Z_0). \quad (3)$$

The total length $|I| + |Z|$ is bounded. For mixed levels it is bounded at the originating level, and the mixed saturation clause applies. Define the answer on Z to be the majority of $D(v_i, Z)$, equivalently the answer $p \upharpoonright Z_0$. Equation (3) preserves every entry of the sequence, so this definition is independent of the chosen Z_0 and retains the same exception estimate.

If I' is another sufficiently long approximation sequence, move Z into A over $I \cup I'$. On that moved query both majorities compute the original p . Thus the answers agree. This proof requires saturation over the concatenation of the sequences; it does not require either approximation estimate to hold for a query containing that concatenation. The averaging queries still have size at most τ . The identical argument compares different query bounds on their common domain. It also proves coherence under restrictions and reindexing of queries.

We next obtain an actual type. Use $\tau = c(\sigma)$ and $\kappa = \tau^+$. In a well-ordered chart of B prescribe the majority answers on every query of size at most τ . For each single query these are realized by some $v_i \in A \subseteq B$, since fewer than κ indices are exceptional. Thus they have local placements. Lemma 6.4 gives a placement realizing all of them in an order enlargement of B . This

is an actual tuple in an extension object, and Theorem 6.6 makes its type unique, since $\tau \geq g(\sigma)$. This type is p^B . Its restriction to A equals p by the same testing criterion.

For a larger desired bounded query size τ' , repeat with a power closed bound at least $c(\sigma, \tau')$. The comparison of sequences just proved identifies the resulting type with p^B on the initial testing threshold and hence everywhere. In a mixed-level use, this additional direct majority evaluation is asserted only when its costs remain below the originating level. The type itself is already actual over B , and its diagrams at larger arities are those of this actual type; no averaging inside a smaller model at unbounded arities is being asserted.

Restriction to a subtuple commutes with every majority evaluation. Choose the threshold large enough for both arities and use testing to obtain equality of types. Isomorphism invariance follows by applying the isomorphism to the sequences and their query diagrams. For transitivity, let $J \subseteq B$ be a canonical approximation sequence for p^B at a sufficient fixed threshold, and retain an original sequence $I \subseteq A$ at that threshold. The latter computes p^B on queries in B by the construction. Move a query in C into B over $I \cup J$. The two sequences give the same answer there, because each computes p^B . The first also computes p^C , and the second computes $(p^B)^C$. Testing proves transitivity. All these sequence lengths can be chosen as the uniform bounds in Lemma 6.9, below the smallest originating level. This proves the mixed-level assertions as well.

Finally, suppose $J \subseteq B$ has the same whole diagram as a fixed code $I \subseteq A$ at threshold $c(\sigma)$. For each query Z in B of that size, saturation realizes $D(J, Z)$ over I in A . The resulting majority is therefore well-defined, with the same exception estimate. For each query one of the entries of J realizes the answer in B . Coherence, compact placement and testing give an actual unique type, as in the preceding construction. Moving queries over code sequences shows that an arrow carrying J into a larger object gives exactly its free lift there. The code length and every threshold used here depend only on σ and the fixed scheme, which proves the final assertion. $\square$

Lemma 6.11 (An average criterion for descent). *Let $A \rightarrow B$ have eligible sizes $\nu \leq \mu$, and let $q = \text{tp}(d/B)$ be actual, with $|d| = \sigma < \nu$. Choose $\chi \geq c(\sigma)$ below ν . Suppose that a bounded sequence $I = (d_i : i < \kappa)$ in A , with $\chi + \aleph_0 < \kappa < \nu$, has majority value $q \restriction Z$ on every query $Z \subseteq B$ of size at most χ , with at most $|Z| + \aleph_0$ exceptions. Then q is the free lift of its actual ν -restriction to A .*

Proof. Let $p = q \restriction A$ and choose a canonical sequence $J \subseteq A$ approximating p at size χ . For a query $Z \subseteq B$ of that size, mixed saturation moves it to Z_0 in A over the whole $I \cup J$. The total length is less than ν . The majority from I on Z computes q , and on Z_0 computes p . These answers agree by diagram preservation. The majority from J on Z_0 also computes p , and on Z computes p^B . Hence q and p^B agree on all χ -queries. Apply the testing criterion. The same proof works for an arbitrary longer indexed family whenever a bounded subsequence of the displayed length retains the exception estimate. $\square$

6.3 Products and local character

Proposition 6.12 (Products of bounded types). *For bounded types $p(x), q(y)$ over A , realize p in an extension $B \geq A$ and then realize q^B in a further extension. The joint type of (x, y) over A , denoted $p \otimes q$, is independent of all choices. Binary products are symmetric, with the indicated permutation of coordinates. Finite products associate, are permutation invariant, and commute with coordinate projections. Furthermore*

$$(p \otimes q)^B = p^B \otimes q^B. \quad (4)$$

These assertions hold across eligible levels when all originating arities and uniform testing costs are bounded at the smallest level.

If x/B is free from A and x, d have product type over B , then x and (d, e) have product type over A for every bounded tuple e from B .

Proof. Two realizations of p can be identified over A by amalgamation. In a common model containing their witnessing extensions, the free lifts of q agree by stationarity and restriction. A second type-identifying amalgamation identifies the realizations of q . This proves independence of the choices in the definition.

We give the cardinal calculation for symmetry. Fix a bounded query Z in A , and put $e = |Z| + |x| + |y| + \aleph_0$. Choose approximation sequences $(a_i : i < \kappa)$ for p and $(b_j : j < \lambda)$ for q , sufficient for queries of size e , where

$$\kappa = e^+, \quad \lambda = \kappa^+.$$

All these cardinals lie below a sufficiently closed level. For the array $M_{ij} = D(a_i, b_j, Z)$, first realize x and freely realize y . The resulting answer R is obtained by first taking the majority in i and then in j . Except at at most e columns, the column majority is R , and in each such column at most e entries differ from R . Reversing the realization order gives an answer S : except at at most e rows, its row majority is S , with at most e exceptions in each such row. The bounds concern the size of (x, Z) or (y, Z) , not the lengths of the sequences.

There are κ good rows. Their exceptional columns together number at most $\kappa e = \kappa$. Since $\lambda > \kappa$, choose a column outside this set and outside the at most e bad columns. Every good-row entry in that column equals S , whereas all but at most e entries equal R . As $\kappa > e$, some entry has both values. Thus $R = S$. This explains why two equally sized sequences would not suffice for the argument. Testing gives symmetry of the joint types.

For (4), take the two sequences just used in A and a sequence $K \subseteq A$ approximating the actual old product. Choose their accuracy at a common sufficient test threshold. Given a query Z in the enlarged base B , move it into A over these three whole sequences. Both successive free evaluations over B are the iterated majority of the same array $D(a_i, b_j, Z)$. Transitivity permits the original sequence for the second factor to be used after the first factor has been realized. The query movement identifies this array with the old array in A , where it computes $p \otimes q$. The sequence K evaluates that old product and computes its free lift on the original query. The answers therefore agree. Testing proves the formula. All the lists fixed during the query movement are bounded, including in the mixed case.

Now realize x, y, z successively along a chain of free lifts from A . The first two form $p \otimes q$, and z is free over a model containing them, so the joint type is $(p \otimes q) \otimes r$. Over a model containing x , the last two have type $q^B \otimes r^B = (q \otimes r)^B$, by the formula just proved. Thus the same joint type is $p \otimes (q \otimes r)$. This proves associativity. Projection follows from the projection property of free lifts

in each successive realization. Associativity and the binary symmetry permit every permutation of a finite list of factors.

For the final assertion, use symmetry and a type-identifying amalgam to realize the given joint type by first putting d in a model containing B and then realizing x freely over it from B . Since x/B is itself free from A , transitivity makes this last realization free from A . The tuple (d, e) is in the intervening model, so the definition of product gives the assertion. The amalgam identifies the full original tuple over B , and hence retains its actual joint type over A . $\square$

Proposition 6.13 (Local character and continuity). *Let $(A_i : i < \delta)$ be an increasing chain of objects at one eligible level μ , whose union A has size μ , and let δ be a nonzero limit ordinal. If $p = \text{tp}(v/A)$ and either v is finite or $|v| < \text{cf}(\delta)$, then p is free from some A_i . If, in addition, every restriction $\text{tp}(v/A_i)$ is free from a fixed base $X \leq A_0$, then p is free from X .*

Proof. Pass to a cofinal chain indexed by the regular cardinal $\gamma = \text{cf}(\delta)$. Choose a sufficient testing size $\chi \geq c(|v| + \aleph_0)$ below μ . If γ is larger than the size of an approximation sequence for p at that threshold, the whole sequence lies in one stage. Its exception estimate is still correct on all queries in A , so Lemma 6.11, at the same level, applies.

In the other case $\gamma < \mu$. Choose a regular $\kappa < \mu$ above $\gamma, \chi, |v|$ and the needed bounds, and take κ approximations inside A . Every whole approximation tuple belongs to some stage: this uses finiteness, or the hypothesis $|v| < \gamma$. If every stage contained fewer than κ of the tuples, regularity of κ and $\gamma < \kappa$ would make their union contain fewer than κ tuples. Thus one stage contains κ of them. Pass to those indices. The exception estimate survives unchanged, and Lemma 6.11 again identifies p as the free lift from that stage.

For the final assertion, local character makes p free from some A_i . Its restriction there is free from X by hypothesis. Transitivity in Theorem 6.10 makes p free from X . $\square$

6.4 Universal strips and approximate retractions

The remaining construction provides maps from an extension back into its base which fix each individual base element at all but finitely many indices. It will allow isolation to descend from a larger level without requiring a bounded tuple to lie in one stage of a countable exhaustion.

Lemma 6.14 (Closure of a bounded support set). *Let $J \subseteq J^*$ be linear orders, let V be a convex strip in J^* , and put $J_0 = J \setminus V$. Allow any fixed finite coloring of positions in J_0 , including their side of V . Let $T^* \subseteq J^*$ and $S_0 \subseteq J_0$ be bounded sets. Given an infinite χ , choose an infinite λ with*

$$\lambda \geq \chi + |T^*| + |S_0|, \quad \kappa = 2^\lambda.$$

There is $S \subseteq J_0$, of size at most κ , containing $S_0 \cup (T^ \cap J_0)$, with this property: for every $R \subseteq S$ of size at most χ and every indexed tuple Q of at most χ positions from J_0 , there is an equally indexed tuple Q' in S having the same order, equality and color pattern over $R \cup T^*$. In particular all positions named by R are fixed whenever they occur in Q .*

At a strong limit level with the displayed input bounded, S is bounded. The assertion needs no well-order hypothesis on J .

Proof. Build S_α for $\alpha < \chi^+$, starting with the required set. At a successor stage, for every $R \subseteq S_\alpha$ of size at most χ , and every order, equality and color pattern of at most χ further positions over $R \cup T^*$ that is actually realized in J_0 , choose one realizing tuple and adjoin its positions. Use

unions at limits. If $|S_\alpha| \leq \kappa$, the number of choices of R is at most $\kappa^\chi = \kappa$, and the number of patterns for each R is at most $2^\lambda = \kappa$. Each chosen tuple adds at most χ positions. Consequently every stage and their union have size at most κ .

For R contained in the final union, regularity of χ^+ puts all of R into a single stage. The actual tuple Q witnesses that its pattern was among the requirements processed at the next stage. Its chosen witness is the required Q' . Equality with elements of R and T^* is part of the pattern, so it is preserved literally. This also covers arbitrary enumerations and repetitions in Q . $\square$

Proposition 6.15 (Universality of a fresh strip). *Let $\mathcal{C}$ already satisfy the working package. In addition to its fixed template F , one may use a presentation G with the following properties: it has a fixed set of finite-support labels; every presentation $G(J)$ used below, for an order of size μ , is an object of $\mathcal{C}$ at level μ ; order embeddings give $\mathcal{C}$ -arrows; and the diagrams of its term lists are natural, including in the order enlargements and padded orders used. Assume also that these labels represent every element of $G(J)$. Require μ to exceed the cardinality of the label set of G and to be closed under the bounds below with that cardinal included in their fixed input data.*

Put $A = G(J)$. Insert into J a fresh convex strip of size μ with countably many disjoint convex substrips, each of size μ , and denote the resulting object by A^+ . Then every same-level extension of A embeds into A^+ over A . In particular this applies to $G = F$ and to well-ordered presentations. No separate union or categoricity assertion for a category of G -presentations is required.

Proof. We first realize a bounded type $p(v/A)$ using just one fresh substrip V . Choose a testing threshold $\chi \geq c(|v| + \aleph_0)$ large enough also for $|v|$ and the number of labels of G . Take a sequence I in A approximating p at query size χ . Represent its entries by G -terms and collect their old supports in $S_0 \subseteq J$. Apply Lemma 6.14, with the side of V recorded, to obtain a bounded $S \subseteq J$. Its closure permits every at-most- χ old query pattern to move into S over any at-most- χ subset of S . The approximation supports, which may be larger than χ , can instead be included as the fixed set T^* in that Lemma. Thus all of them are fixed during every such movement.

Let P_S be the set of all G -terms on S . It is bounded, since G has a bounded set of finite-support labels. The natural parameter diagram of P_S is the same in A and in $G(S \cup V)$. The latter is a level object because $|V| = \mu$. Saturation there realizes the desired whole joint diagram of (v, P_S) , giving a tuple w whose supporting positions belong to $S \cup V$. Only at most $|v| + \aleph_0 \leq \chi$ of its supporting positions are old; call them $R \subseteq S$.

For any query Z in A of size at most χ , move its chosen support tuple into S over R and the fixed approximation supports, preserving the side of V . Let Z' be the resulting term tuple. All comparisons with the new supporting positions of w in V are forced by that side. Naturalness therefore gives

$$D(w, Z) = D(w, Z'), \quad D(I, Z) = D(I, Z').$$

The majority from I computes p on both queries. The chosen realization of (v, P_S) gives $D(w, Z') = p \upharpoonright Z'$. It follows that $D(w, Z) = p \upharpoonright Z$. Testing of actual types, using a natural F -chart of A if G differs from F , proves $\text{tp}(w/A) = p$. Naturality of G was used only for the displayed support movements; the testing and saturation are those already proved in $\mathcal{C}$.

Now let $B \geq A$ be arbitrary and choose increasing bounded lists d_n exhausting it, including countable strong-submodel exhaustions in every vertex. Use the first substrip to realize the type of d_0 over A . At subsequent step n , the target base consists of A and the whole previously used substrips. Suppose d_n has already been assigned there with its correct type over A . Theorem 6.6 amalgamates B with this target base over A , identifying that entire assigned tuple. In this amalgam, d_{n+1} has an actual bounded type over the target base. Apply the one-substrip construction to the

next fresh substrip. Since its type includes equality with the already assigned coordinates, the new assignment extends the previous one. The final union of the assignments lands in A^+ . The arrow criterion applies to the included strong-submodel exhaustions, so it is an arrow $B \rightarrow A^+$ over A . $\square$

Corollary 6.16 (Approximate retractions). *For same-level $A \leq B$ there are arrows $(r_i : B \rightarrow A : i < \mu)$ such that, for every $a \in A$,*

$$|\{i < \mu : r_i(a) \neq a\}| < \aleph_0.$$

The maps are not asserted to fix all of A at once.

Proof. Present A by F on a well-order consisting of μ disjoint convex blocks J_i , each of cardinality μ and itself containing countably many convex substrips of that size. The whole order still has cardinality μ , and the working package identifies its presentation with A . Let A_i be the presentation on the complement of J_i . It is a level subobject of A . Proposition 6.15, with J_i as the inserted strip, embeds B into A over A_i ; call the arrow r_i . Choose one finite supporting set for a . It meets only finitely many J_i . For every other index, $a \in A_i$ and $r_i(a) = a$. $\square$

Lemma 6.17 (Descent through approximate retractions). *Let $A \leq U$ be objects at an eligible level μ , and let $A' \leq U'$ be objects at an eligible level $\nu \leq \mu$, with compatible arrows $A' \leq A$ and $U' \leq U$. Suppose there are arrows $(r_i : U \rightarrow A : i < \nu)$ such that every $a \in A$ is fixed for all but finitely many i , and $r_i[U'] \subseteq A'$ for all $i < \nu$. Then for every tuple d in U' of length less than ν , its actual type over A is the free lift of $\text{tp}(d/A'; U')$. All comparisons are comparisons of the working diagrams, including mixed-level diagrams at arities bounded at ν .*

Proof. Coherence makes $r_i|_{U'} : U' \rightarrow A'$ an arrow: its image and A' are subobjects of A , and its image is contained in A' . Put $\sigma = |d| + \aleph_0$, choose $\chi \geq c(\sigma)$ below ν , and choose $\chi^+ \leq \kappa < \nu$. Retain any κ indices and form $I = (r_i(d))$ on them. This is a bounded concatenated list in A' .

For a query $Z \subseteq A$ of size at most χ , at most $|Z| + \aleph_0$ indices fail to fix every element of Z . At every other index, arrow preservation gives

$$D(r_i(d), Z) = D(r_i(d), r_i(Z)) = D(d, Z).$$

Thus I has the required majority estimate for the actual type over A . On queries in A' the same identity is a comparison at level ν , using the restricted arrows. Mixed preservation identifies it with the larger-level identity.

Apply Lemma 6.11 to $A' \leq A$. Its restriction type is precisely $\text{tp}(d/A'; U')$, by Corollary 6.7. The conclusion is therefore the stated free lifting of the actual type, including tuples that span a chosen resolution of U' . $\square$

The preceding calculus will be applied only after the working package has been established for the relevant scheme. Its approximation, testing and support-closure bounds are functions of fixed scheme data and bounded input arities. This is the uniformity needed when the schemes are iterated at finite depths.

7 Isolation and dominated extensions

Throughout this section, $\mathcal{C}$ is a working scheme in the sense of Definition 5.2. We use the refinements of the eligible levels required in Section 6. All objects and extensions in a displayed type are at one eligible level, unless a change of level is stated. We retain the eligible-subclass hypothesis of Convention 5.3; every larger auxiliary sufficiently closed level below is chosen in its unbounded refined subclass. A type over an object at a smaller eligible level is computed in a section of that level; Corollary 6.7 makes this independent of the section. A tuple is bounded when its cardinality is less than the level of its base. We write p_B^{fr} for the free lift p^B of Theorem 6.10 when its role needs emphasis.

We will repeatedly use the following convention concerning bounds. A request involving a tuple of cardinality σ is first enlarged to an infinite arity at least σ . The testing, compact-placement, and approximation bounds of Section 6 are then applied at that arity. There are only finitely many such enlargements in each argument below. A source level is required to be strictly closed under those bound functions. Thus, when a tuple is bounded at a smaller level, every test used to determine its type may be taken below that smaller level, even if the receiving object is at a much larger eligible level. This does not assert saturation over an unbounded parameter set.

Our goal is an extension controlled by a given mark, with an embedding into every ambient object realizing that mark's type. Relative isolation will give this control on bounded tuples. The main existence argument first treats one bound on tuple length and then obtains all bounded lengths in a single extension.

7.1 Relative isolation

Definition 7.1. Let $q = \text{tp}(u, c/A)$ and $p = \text{tp}(u/A)$, where the concatenation (u, c) is bounded. We say that q is *isolated relative to u* if there is a bounded set $Z \subseteq A$ such that, for every $q' = \text{tp}(u', c'/A)$,

$$\text{tp}(u'/A) = p \quad \text{and} \quad D(u', c', Z) = D(u, c, Z) \implies q' = q.$$

Here and below a parameter set in a diagram is supplied with a fixed enumeration, and the same enumeration is used on both sides. We call Z an *isolating fragment*. A projection retains the entire tuple u unless explicitly stated otherwise.

Lemma 7.2. *The following hold at sufficiently closed eligible levels.*

1. *A projection of a relatively isolated type is relatively isolated.*
2. *Suppose $A \leq B$, $q = \text{tp}(u, c/A)$ is isolated relative to u , and an actual coupling over B satisfies*

$$\text{tp}(u/B) = \text{tp}(u/A)_B^{\text{fr}}.$$

Then

$$\text{tp}(u, c/B) = q_B^{\text{fr}}.$$

Moreover this type is isolated relative to u , using the same isolating fragment as q .

The second assertion also holds when A is at a smaller eligible level than B , provided (u, c) and the required uniform bounds are below the level of A .

Proof. For the first assertion, write $c = (c_0, c_1)$ and let Z isolate q . Suppose (u', c'_0) has the prescribed marginal on u and the same diagram as (u, c_0) over Z . In a model containing A, u', c'_0 , bounded saturation extends this diagram by a tuple c'_1 so that

$$D(u', c'_0, c'_1, Z) = D(u, c_0, c_1, Z).$$

Isolation identifies the full type over A , and its projection identifies $\text{tp}(u', c'_0/A)$.

For the second assertion, fix an isolating fragment Z . It is enough to compare the actual joint type over B with q_B^{fr} on the uniform testing arity for (u, c) . Enlarge that arity to include Z and choose approximation lists in A for both q and $p = \text{tp}(u/A)$, long enough to evaluate all the comparisons just specified. Let t be a test tuple from B at this enlarged arity. Bounded saturation moves t to a tuple t' in A , preserving its diagram over Z and both approximation lists.

Choose a realization u_0 of p over A . The free evaluation of p gives

$$D(u, t, Z) = D(u_0, t', Z).$$

Saturation over the bounded list (u_0, t', Z) supplies c_0 such that

$$D(u_0, c_0, t', Z) = D(u, c, t, Z).$$

In particular (u_0, c_0) agrees with q on Z and has marginal p . Isolation gives $\text{tp}(u_0, c_0/A) = q$. Consequently the actual answer to the query t is the answer that q gives to t' . Because the query was moved over the approximation list for q , that is also the free evaluation of q on t . The testing theorem identifies the types over B .

Now let a competing type over B have the same marginal on u and agree with the displayed type on Z . Its restriction to A is q . The argument just given identifies it with q_B^{fr} , proving isolation there. In the mixed-level case, perform these comparisons at the uniform test bound below the source level. Mixed realization supplies the required sections, and the same argument applies to those diagrams. $\square$

Proposition 7.3. *There is a bound function h , depending only on the fixed working scheme, with the following property. Put $\sigma = |u, c| + \aleph_0$. At sufficiently closed eligible levels, precisely one of the following alternatives holds for $q = \text{tp}(u, c/A)$:*

1. q is isolated relative to u by a fragment of size at most $h(\sigma)$;
2. there are an extension $A \leq B$ at the same level and an extension q_B of q such that

$$q_B \neq q_B^{\text{fr}}, \quad q_B \upharpoonright u = \text{tp}(u/A)_B^{\text{fr}}.$$

In particular, every relatively isolated type has an isolating fragment bounded by $h(\sigma)$, uniformly in the level and in the type.

Proof. Choose an infinite χ large enough for the testing theorem at arity σ , for the approximation lists for q and its marginal p , and for the finitely many joint comparisons made below. These choices are uniform in σ . Increasing χ if necessary, all the supports in the chosen approximation lists have cardinality at most χ .

Present $A = F(I)$ on a well-order of the level cardinality μ . We may choose I to have more than σ disjoint consecutive blocks, each of order type and cardinality μ : this is still a well-order of cardinality μ , and all such presentations are objects at this level. Let J_0 consist of the supports

of the approximation lists, together with an initial segment of order type χ^+ in every candidate block. Its size has a bound depending only on σ . Let Z_0 be the set of all term values on J_0 . Since the label set is fixed, the same is true of $|Z_0|$.

If Z_0 isolates q , we have the first alternative. Suppose it does not. Choose $q^1 \neq q^0 = q$ with the same marginal p and the same diagram on Z_0 . By Lemma 6.4, give both types placements over the displayed presentation of A , and amalgamate their support orders over I . The combined new support has cardinality at most σ . Each new support position either hits the interior of one candidate block or determines an interior cut of at most one such block. We can therefore choose a candidate block J whose interior contains neither a hit nor a cut of either placement.

Set $A_* = F(J_0 \cup J) \leq A$. This object has cardinality μ . The restrictions of q^0 and q^1 to A_* are equal. To check this, take a test at the fixed sufficient arity. Move every position of the test lying in J into the initial χ^+ part of J , preserving the order and equality pattern of the test. This can be done above any positions of that initial part which need to remain in the pattern: fewer than χ^+ positions have been specified. Leave the positions outside J unchanged. No position of either candidate separates the interior of J , so the move preserves the natural diagram with either candidate. The moved query uses only J_0 and therefore has the same answer for q^0 and q^1 . The testing theorem proves equality over A_* .

The approximation lists for q^0 and p lie in A_* . Their majority evaluations show that q^0 and p over A are free from A_* . Also, an isomorphism $f : A_* \simeq A$ may be chosen to fix those bounded lists, by bounded diagram homogeneity. The transport assertion for free types then gives

$$f(q^0 \upharpoonright A_*) = q.$$

Transport the inclusion $A_* \leq A$ along f , viewing its larger object as an extension B of the original base A . Transport q^1 as well. Its restriction is q because the two restrictions to A_* agree. Its marginal is the free lift of p . Its joint type is not free, since the unique free lift before transport was q^0 , distinct from q^1 . This is the second alternative.

Choose $h(\sigma)$ to bound the size of Z_0 for these uniform choices. Lemma 7.2 excludes the second alternative when q is isolated, even if an initially given isolating fragment was larger. This proves both exclusivity and the asserted uniform bound. $\square$

7.2 Primeness and descent of isolation

Definition 7.4. Let $X \leq Y$ be same-level objects and let b be a bounded tuple in Y . We say that Y is *all-dominated over $X + b$* if, for every bounded tuple c from Y , the type $\text{tp}(b, c/X)$ is isolated relative to b .

The quantifier over all bounded tuples is part of this definition. It is not enough to verify isolation for individual entries, or for the tuples in one chosen countable exhaustion of Y .

Proposition 7.5. *Let Y be all-dominated over $X + b$ at a sufficiently closed eligible level μ .*

1. *If an extension N of X contains a tuple b' of the same type over X , there is an arrow $Y \longrightarrow N$ over X sending b to b' . The target may be at a larger eligible level. A prescribed bounded tuple match can be retained whenever its joint type with b over X already agrees.*
2. *Two all-dominated objects over the same data $X + b$ are isomorphic over those data. The isomorphism may retain a prescribed bounded tuple match of the kind just described.*

3. In an actual coupling, if a bounded tuple x has product with b over X , then $\text{tp}(x/Y)$ is free from X .

Proof. Choose increasing bounded lists c_n , $n < \omega$, exhausting Y ; include b , the prescribed initial data, and exhaustions of every vertex by smaller strong submodels. Suppose the image of c_n has already been chosen with the correct full type over X . Let $Z_{n+1} \subseteq X$ isolate $\text{tp}(b, c_{n+1}/X)$. The old image and the source list have the same diagram over Z_{n+1} . Saturation in N extends this bounded match to (b, c_{n+1}, Z_{n+1}) . Relative isolation makes its full type over X correct. Taking the union gives a map on all of Y fixing X : whenever an entry of X occurs in a list, its equality with the corresponding parameter is part of its full type. The strong-submodel exhaustions and the map criterion in the working package show that this map is an arrow. This proves the first assertion, also into a larger target because every saturation request is bounded at the source level.

For the second assertion, perform the same construction alternately on the two sides, at each step including the next list in the chosen exhaustion on the side to be covered. Both objects have isolation for the enlarged bounded list. The construction therefore continues in both directions and its union is an onto structural isomorphism.

For the third assertion, use symmetry and the definition of product to choose a model $W \geq X$ containing x over which b is free from X . Type-identifying amalgamation brings W and the actual coupling into a common extension, identifying x and b over X . For every bounded c in Y , Lemma 7.2 says that $(b, c)/W$ is free from X . Hence x and (b, c) have product over X . Symmetry and projection give the free evaluation of x on every bounded test c from Y . The testing theorem identifies $\text{tp}(x/Y)$ with the free lift of $\text{tp}(x/X)$. $\square$

Lemma 7.6. Suppose $A \leq B$ are objects at eligible levels $\nu \leq \mu$, $|u, c| < \nu$, and the actual type

$$q_B = \text{tp}(u, c/B)$$

is the free lift of $q_A = \text{tp}(u, c/A)$. If q_B is isolated relative to u , then q_A is isolated relative to u . Here ν is sufficiently closed for the uniform bounds at $|u, c| + \aleph_0$.

Proof. By Proposition 7.3, choose a large-base isolating fragment $W \subseteq B$ with

$$|W| \leq h(|u, c| + \aleph_0) < \nu.$$

Let w enumerate W . The type $\text{tp}(w/A)$ is a small-level type, computed by taking a level- ν section containing A and w . Choose a sufficiently long bounded approximation list $(w_i)_{i < \tau}$ for this type inside A , where $\tau < \nu$. Enlarge its testing bound to include the arities of w and (u, c) and choose τ larger than the two exception bounds that will occur below. Put

$$Z = \bigcup_{i < \tau} \text{ran}(w_i) \subseteq A.$$

This is a bounded fragment.

Let q'_A have the same marginal on u as q_A and agree with it over Z . Lift q'_A freely to B . The marginal on u then agrees with the marginal of q_B . In either free coupling, symmetry identifies the joint diagram with w by the majority evaluation of $\text{tp}(w/A)$ against the candidate tuple. Thus all but the prescribed bounded number of $i < \tau$ give the diagram of that candidate with w . Choose one index outside the two exception sets. The candidates agree on w_i , because w_i lies in Z , so their free lifts agree on w . Isolation of q_B identifies the free lifts, and restriction identifies q'_A with q_A . Every approximation and comparison here has size below ν ; no approximation to the whole of B is used. $\square$

7.3 Existence with a prescribed bound on tuple length

We first isolate the continuity requirement on the marked marginal. This will let us identify exactly where the proof for a finite mark must be supplemented for a long independent list.

Definition 7.7. A bounded marginal type p satisfies the *continuity condition used here* if the following holds at every sufficiently closed eligible level to which p is freely lifted. Suppose

$$X_0 \leq X_1 \leq \cdots \leq X_i \leq \cdots \quad (i < \delta)$$

is a continuous chain at that level, its union is an object at that level, and a fixed realization b in a coupled extension satisfies

$$\text{tp}(b/X_i) = \text{tp}(b/X_0)_{X_i}^{\text{fr}} \quad (i < \delta).$$

Then the same equality holds over $\bigcup_{i < \delta} X_i$. The type at X_0 is required to be the appropriate free lift of p .

Every finite-tuple marginal has this property. At a limit, local character for the finite tuple gives freeness from some earlier stage, and transitivity gives freeness from X_0 . This is a direct application of Proposition 6.13, including when the chain has small cofinality.

Proposition 7.8. *Let λ be a sufficiently closed eligible level, let X be an object there, and let $p = \text{tp}(b/X)$ be a finite-tuple type. For every infinite $\theta < \lambda$ there is a same-level extension $X \leq U$ containing a realization of p such that*

$$\text{tp}(b, c/X) \text{ is isolated relative to } b \text{ for every } c \in U \text{ with } |c| \leq \theta.$$

More generally the assertion holds for a bounded mark b satisfying Definition 7.7, with $|b| \leq \theta$. The closure requirements at the output level are uniform functions of bounded arities; they do not require θ to have been fixed before the level was chosen.

Proof. We prove the more general statement. Fix a bounded approximation list I_p in X which determines p at a sufficient testing arity. Choose a much larger eligible level κ and a cardinal ρ such that

$$\lambda < \rho = \rho^\lambda, \quad \rho^+ < \kappa.$$

Require ρ to dominate the approximation and small-hull costs at arity λ , and choose κ after those set-sized bounds. Such choices exist: first choose a sufficiently large power 2^ξ with $\xi \geq \lambda$ for ρ , and then choose an eligible κ above ρ^+ and all remaining bounds. Extend X by a presentation arrow to a level- κ object X_0 , and choose $Y_0 \geq X_0$ containing a realization b of the free lift of p .

We construct continuous pairs

$$X_i \leq Y_i \quad (i \leq \rho^+)$$

at level κ , with a single fixed b in all the upper objects. The invariant on the marked marginal is

$$\text{tp}(b/X_i) = \text{tp}(b/X_0)_{X_i}^{\text{fr}}. \quad (5)$$

For $i < \rho^+$, we also keep increasing continuous sets

$$R_i \subseteq X_i, \quad S_i \subseteq Y_i, \quad R_i \subseteq S_i, \quad |R_i|, |S_i| \leq \rho,$$

with $I_p \subseteq R_i$ and $b \subseteq S_i$. They are bookkeeping sets, not additional structure on the objects.

Here are the two closure requirements at a stage i with $\text{cf}(i) > \lambda$.

1. Any specified at-most- λ sets of base and top parameters from R_i, S_i are contained in a pair $A \leq V$ of level- λ subobjects of $X_i \leq Y_i$, with $A \subseteq R_i$ and $V \subseteq S_i$.
2. Every tuple d of length at most λ from S_i has a sequence of at least λ copies inside R_i computing its actual type over X_i on all tests of size less than λ . For every subtuple and each such parameter test Z , all but at most $|Z| + \aleph_0$ of the copies give its joint D -diagram.

We explain how to arrange the requirements; in particular, they do not amount to assuming a small saturated submodel. At successor stages, process all at-most- λ lists in the current parameter sets. There are at most $\rho^\lambda = \rho$ of them. Mixed Löwenheim–Skolem supplies the requested level- λ pairs. Lemma 6.9 supplies copies of each top list in the current base, at the test sizes below λ and with the displayed exception estimate. Their total number of coordinates is at most ρ . Close the sets under these operations for λ^+ steps; the size is still ρ . If new elements are added later, repeat these successor closures. At a limit i of cofinality greater than λ , every requested list was contained at an earlier stage. Local character applied to that list, whose length is at most $\lambda < \text{cf}(i)$, says that its actual type over X_i is free from some earlier stage. Approximations placed in a later intervening successor base still compute it over X_i by the free-extension theorem. Their copies lie in R_i . The same cofinality argument places every small-hull request in an earlier closure. This proves both requirements at the indicated limits.

At any such stage, if some list d of length at most λ from S_i makes $\text{tp}(b, d/X_i)$ nonisolated relative to b , apply Proposition 7.3. It gives a same-level extension of X_i in which this joint type has a nonfree extension while the b -marginal remains free. By type-identifying amalgamation, extend Y_i into a common Y_{i+1} , identifying the old (b, d) with that realization. The new base is the extension just chosen. Transitivity preserves (5). At other stages use compatible extensions and the bookkeeping closures. At limits use the working scheme's unions; Definition 7.7 preserves the marginal invariant. All these unions have size κ , since $\rho^+ < \kappa$.

This process must reach an indicated stage at which every list in the second closure requirement is isolated jointly with b . Suppose not, and choose a failing list d_i at every stage with $\text{cf}(i) > \lambda$. The set of these stages is stationary in the regular cardinal ρ^+ . Indeed, in any closed unbounded subset of ρ^+ take a continuous increasing sequence of length λ^+ and its supremum; the supremum is still below ρ^+ , is in the closed set, and has cofinality λ^+ .

Because S_i is continuous and $|d_i| \leq \lambda < \text{cf}(i)$, there is $j(i) < i$ with $d_i \subseteq S_{j(i)}$. Fodor's pressing-down argument [Fod56] makes $j(i)$ constant on a stationary subset. For completeness, if every fiber of a regressive map on a stationary subset of a regular cardinal were nonstationary, choose a closed unbounded set avoiding each fiber. Their diagonal intersection is closed unbounded and gives a contradiction at a point of the stationary set. The diagonal intersection is unbounded by taking an increasing countable sequence whose next point is in all the clubs indexed below the previous point; the needed intersections of fewer than the regular cardinal many clubs are closed unbounded, by interleaving their choices and taking a limit. Thus this application uses only set-indexed choices in ZFC.

There are at most $\rho^\lambda = \rho$ possible lists in the resulting fixed S_j . Partitioning the stationary subset into these ρ possibilities retains a stationary piece: otherwise the intersection of $\rho < \rho^+$ clubs avoiding the pieces is a contradiction. Consequently one fixed list d witnesses nonfree extensions at stationarily many stages. In the final union, the tuple (b, d) has length at most $\lambda < \rho^+$. Local character makes its type free from some X_j . Its restrictions at every later successor are then free from their preceding base, contradicting the chosen stationarily many nonfree extensions.

Fix a stage i at which all the required lists are isolated. We next shrink, and this step must

preserve freeness for the tuples on which isolation will be transferred. Construct increasing pairs

$$A_\alpha \leq U_\alpha \quad (\alpha < \theta^+)$$

of level- λ subobjects inside R_i, S_i . Keep I_p in every base and b in every top. At successor stage $\alpha + 1$, take a tuple enumerating U_α . By the second closure requirement it has λ approximation copies in R_i . Include every coordinate of these copies in $A_{\alpha+1}$, and include A_α, U_α in the next pair, using the first closure requirement. The requests have cardinality λ . At proper limits take unions; the upper bound clause into X_i, Y_i keeps them strong there. Finally put

$$A = \bigcup_{\alpha < \theta^+} A_\alpha, \quad U = \bigcup_{\alpha < \theta^+} U_\alpha.$$

Because λ is strong limit and $\theta < \lambda$, we have $\theta^+ < \lambda$. Both unions are level- λ objects, and the common upper bound clause gives $A \leq U$.

Let c have length at most θ in U . Regularity of θ^+ puts (b, c) inside some U_α . The next base contains λ copies of its type over X_i , with the exception estimates above. For any fixed sufficient testing arity below λ , select a bounded subsequence longer than that arity's exception bound. These copies are inside A and compute the actual large-base type. The majority characterization of free extension now says that

$$\text{tp}(b, c/X_i) = \text{tp}(b, c/A)_{X_i}^{\text{fr}}.$$

Explicitly, move a large-base query into A over the selected copies and over canonical approximations for the small type. Choose an index outside the exception sets for the query and its moved copy. The actual answer therefore equals the small free evaluation. The testing theorem identifies the whole type. The bounds needed for this argument are below λ , since $|b, c| \leq \theta$.

The large type is isolated by the choice of i . Its uniform fragment has size at most $h(|b, c| + \aleph_0) < \lambda$, so Lemma 7.6 gives isolation over A . Finally, I_p still determines the marginal on b : it did so at X_0 , the matrix preserved its free lift, and the same approximation answers hold on restriction to A . Bounded diagram homogeneity gives an isomorphism $A \simeq X$ carrying these anchors to themselves. Transport of approximation evaluations carries $\text{tp}(b/A)$ to p . Renaming U along this isomorphism gives the asserted extension of the original X . $\square$

7.4 All bounded lengths at one level

The preceding construction isolates tuples through one prescribed length. All-domination requires this for every bounded tuple in a single object. We obtain it by descending from a larger level with one common family of approximate retractions; the argument therefore also covers tuples that span the countable descent construction.

Theorem 7.9. *At sufficiently closed eligible levels, every finite-tuple type $p = \text{tp}(b/X)$ has an all-dominated realization: there is $Y \geq X$ all-dominated over $X + b$ realizing p . More generally this holds for a bounded marginal satisfying Definition 7.7. An all-dominated realization embeds over X, b into any actual ambient containing a realization of that type, with the mixed-level conventions of Proposition 7.5.*

Proof. Write ν for the desired level and fix sufficient bounded anchors $I_p \subseteq X$ for the input marginal. Choose a much larger eligible $\lambda > \nu$ and lift the type to a level- λ base by a presentation

arrow. Apply Proposition 7.8 at level λ with $\theta = \nu$. We obtain a pair $A \leq U$ of level λ such that every tuple of length at most ν in U , jointly with b , has isolation relative to b over A . The original anchors may be retained in A throughout that construction.

Choose ν distinct indices from the family in Corollary 6.16, and write the corresponding approximate retractions as

$$r_j : U \longrightarrow A \quad (j < \nu)$$

such that each $a \in A$ is fixed by all but finitely many of them. Use mixed Löwenheim–Skolem to choose level- ν subobjects $A' \leq U'$ of $A \leq U$ which contain I_p, b in their indicated places and satisfy

$$r_j[U'] \subseteq A' \quad (j < \nu). \quad (6)$$

Here is the closure construction. Start with level- ν sections containing the desired entries. From the current top of size ν , adjoin to the next base all its images under the ν selected maps; there are at most $\nu \cdot \nu = \nu$ coordinates. Enlarge the next top to contain that base and the old top. Iterate countably many times and take unions. Same-level chain closure and the common upper bound clause into A, U give the pair and (6).

Fix any bounded tuple d in U' . The tuple (b, d) is still bounded at ν . The maps in (6) place all its images in A' . For a test $Z \subseteq A$, all but at most $|Z| + \aleph_0$ of the maps fix Z pointwise, and hence for all those indices

$$D(r_j(b, d), Z) = D(b, d, Z).$$

At a fixed testing arity below ν , choose a bounded subsequence of indices longer than the exception bound. Lemma 6.17 gives

$$\text{tp}(b, d/A) = \text{tp}(b, d/A')^{\text{fr}}_A.$$

The type on the left is isolated, by the length- ν assertion at level λ . Its uniform fragment bound for the actual arity $|b, d| < \nu$ is below ν . Downward transfer gives isolation of $\text{tp}(b, d/A')$ relative to b .

This argument applies separately to every bounded $d \subseteq U'$; it makes no assumption that d lies in one stage of the countable construction. Thus U' is all-dominated over $A' + b$. The retained anchors and bounded diagram homogeneity identify its marginal with the original p by an isomorphism $A' \simeq X$. Transport gives the required object over X . Its embedding into an actual ambient is Proposition 7.5. $\square$

7.5 Independent lists and coherent witnesses

Finite marks already have all-dominated extensions. To obtain the same conclusion for a bounded independent list of finite marks, we must first show that its finite product data determine one whole type and that this type persists along the actual coupled chains used above.

Definition 7.10. A list $(b_i)_{i < \delta}$ of finite tuples is *product-independent over A* if each finite sublist has the product of its marginal types over A . A *witnessing chain* is a continuous chain $(M_i)_{i \leq \delta}$ of objects, with $M_0 = A$, such that b_i lies in M_{i+1} and

$$\text{tp}(b_i/M_i) = \text{tp}(b_i/A)^{\text{fr}}_{M_i}.$$

It is an *all-dominated-increment chain* if, in addition, M_{i+1} is all-dominated over $M_i + b_i$. It uses at level μ , the chain has total size μ , in particular $|\delta| \leq \mu$.

Lemma 7.11. *Suppose $(M_i)_{i \leq \delta}$ is an all-dominated-increment chain over A for finite tuples $(b_i)_{i < \delta}$. If a finite tuple x in an actual coupling has product over A with every finite selection of the accumulated b_i , then $\text{tp}(x/M_\delta)$ is free from A .*

Proof. Induct on the stage. At a successor $i + 1$, the finite tuple (x, b_i) has product with every finite selection of earlier marks, by the finite product rules. The induction hypothesis applied to this tuple says that its joint type over M_i is free from A . Products lift freely by Proposition 6.12; hence x and b_i still have product over M_i . Proposition 7.5, applied to the all-dominated increment M_{i+1} , makes x/M_{i+1} free from M_i . Transitivity makes it free from A .

At a limit j , local character for the finite tuple x gives a proper stage $i < j$ from which x/M_j is free. The induction hypothesis and transitivity give freeness from A . All types are computed in the actual coupled ambient, so this argument asserts continuity of the specified realization, not merely existence of some independent realization at the limit. $\square$

Lemma 7.12. *The following hold at sufficiently closed eligible levels.*

1. *Every actual product-independent list of finite tuples admits an all-dominated-increment witnessing chain, retaining its whole joint configuration in an extension of the given ambient.*
2. *Two lists supplied with witnessing chains have a unique whole amalgamation type over A , determined by the successive marginal types over A . This assertion also permits bounded, rather than finite, individual tuples in the witnessing chains. For a list of total length at most the level, the assertion means that one amalgam identifies the entire lists; for a bounded total list it is an equality of the bounded-list types already defined.*

Proof. For the first assertion, maintain a common ambient containing the original list and the already constructed witnessing chain. At stage i , finite domination makes the actual b_i free over M_i from A : its products with finite selections of previous marks are still the original ones, since arrows preserve types. Theorem 7.9 supplies an all-dominated realization over $M_i + b_i$. Primeness embeds it into a compatible actual ambient over this tuple; alternatively, type-identifying amalgamation supplies such an ambient while retaining the original list. At limits take the coherent unions. Finite domination continues to apply, and the chain has the stipulated total size.

For uniqueness, let M_i^0, M_i^1 be the two witnessing chains. Build common objects N_i and coherent arrows from M_i^0, M_i^1 into N_i , agreeing on A and identifying all marks already processed. At a successor, take the free lift over N_i of the next common marginal type over A , with realization c . Its restriction to either image of M_i^r is the prescribed type of b_i^r , by transitivity. Amalgamate the next stage M_{i+1}^0 with this realization over its old stage, identifying b_i^0 with c . Then amalgamate M_{i+1}^1 in the same way, identifying its mark with the same c . The first amalgamation preserves the type of c over the other old-stage image, so the second identification is legitimate. Retain the old common object throughout. At limits, coherent unions give the required arrows. The final common ambient identifies every mark. Only bounded individual marginal types were used in a single step, which proves the stated version for bounded individual tuples as well. $\square$

Theorem 7.13. *Let $b = (b_i)_{i < \delta}$ be a bounded product-independent list of finite tuples over X at a sufficiently closed eligible level.*

1. *Its whole type is determined by the product data on finite sublists. Its free lift is product-independent and is the unique whole type with the lifted finite data.*
2. *The marked marginal $\text{tp}(b/X)$ satisfies Definition 7.7.*

3. *There is an all-dominated object over $X + b$, and it embeds over these data into an actual ambient containing the given list.*

Proof. Lemma 7.12 supplies witnesses for any actual list with the specified finite products and identifies their whole types. The finite projections of a free lift are the free lifts of the finite projections, by Proposition 6.12. They are therefore the required products of the lifted marginal types. Applying witnessed uniqueness at the receiving base proves the first assertion, including its mixed-level interpretation.

For continuity, take the actual coupled chain in Definition 7.7. Each finite sublist of the fixed mark is a finite tuple. Its actual free evaluations at the proper stages persist at the union by local character and transitivity. These finite data are precisely the finite data of the free lift of the original whole marginal. By the first assertion they determine its whole type at the union. This proves the second assertion without applying finite-tuple local character directly to the possibly long mark.

We may now apply the general forms of Proposition 7.8 and Theorem 7.9. To check the additional arity conditions, choose the auxiliary θ at least $|b|$. The parameter-set scheduling then handles (b, d) at length at most λ , and the final stationary contradiction uses length $\lambda < \rho^+$. At limits of the matrix construction, the continuity just proved preserves the fixed marked marginal. All approximation and isolation bounds are evaluated at the actual bounded joint arities. The all-length descent uses the same retractions on the concatenations (b, d) , which remain bounded at the desired level. Thus all hypotheses of the two existence results hold. Primeness gives the assertion about the actual ambient. $\square$

8 Adjoining a marked increment

Throughout this Section, $\mathcal{C}$ is a working scheme whose arrows are precisely the componentwise strong embeddings preserving its specified maps and distinguished data. In particular, there is no additional condition on intersections or on bases. We use the working package of Definition 5.2, and the calculus and isolation results of the preceding two Sections, at sufficiently closed eligible levels. All refinements in this Section have fixed set parameters: the input scheme and template, the prescription below, and, after it is chosen, the new template. An unbounded output class is asserted under the eligible-subclass hypothesis of Convention 5.3. This supplies both arbitrarily large closed samples and levels that remain eligible while having cofinally many smaller template-valid levels. Theorem 13.5 supplies the simultaneous closure bookkeeping.

Definition 8.1 (A stationary prescription and its marked pairs). A stationary finite prescription for $\mathcal{C}$ assigns to every object X a type p_X of a finite tuple, with specified component sorts. The assignment commutes with isomorphisms, and, for every arrow $X \rightarrow X'$, including an arrow between eligible levels, $p_{X'}$ is the free extension of p_X , after identifying the base by that arrow. The prescription is given by fixed set-coded data of the kind used in the applications.

An object of $\mathcal{C}^b$ at level μ is a triple

$$(X, Y, b), \quad X \leq Y,$$

where X, Y are objects of $\mathcal{C}$ at level μ , $b \in Y$ realizes p_X , and Y is all-dominated over $X + b$. The inclusions and the mark are part of the structure. An arrow is a commuting pair of componentwise strong embeddings taking the mark to the mark. The notation $X \leq Y$ records an actual inclusion; arbitrary arrows are put into this form by replacing the codomain with an isomorphic copy.

Recall that all-domination quantifies over *every* tuple of length strictly below the level. It is not just domination for the members of one fixed exhaustion of Y .

Lemma 8.2 (The pair isomorphism test). *After a further closure refinement, marked pairs exist over every object X . An isomorphism $e : X \cong X'$ extends to an isomorphism*

$$(X, Y, b) \cong (X', Y', b').$$

More precisely, for bounded tuples t and t' in the respective pair structures, such an extension can take t to t' if and only if e takes the indicated base entries to their counterparts and transports

$$\text{tp}(t, b/X) \quad \text{to} \quad \text{tp}(t', b'/X').$$

Entries are occurrence-tagged when more than one component is used.

Same-level arrows preserve bounded pair orbits. Moreover, comparisons of bounded pair tuples are preserved and reflected when two pairs at one common small level are mapped to two pairs at a common larger eligible level. All tuple lengths and the bounds needed for the comparison are strictly below the smaller level.

Proof. Existence is Theorem 7.9 applied to the finite prescription. The pointed existence and uniqueness assertions follow from Proposition 7.5: its back-and-forth form retains prescribed bounded tuples with the same type together with the mark. Necessity of the displayed type condition follows from invariance.

Suppose first that a pair maps to another pair at the same level. The prescribed type of the mark in the second base is the free extension of its type in the first base. For each bounded t in the first upper object, Lemma 7.2 therefore says that its joint type with the mark in the second base is the free extension of its old joint type. Choose bounded approximation lists in the old base computing this joint type and its mark marginal at the needed testing arities, and include the base entries of t . An isomorphism of the two bases matching these lists exists by bounded diagram homogeneity in $\mathcal{C}$. The approximation calculation and the testing bound show that it transports the whole joint type. The first part of the Lemma gives an isomorphism of the pairs taking the old tuple to its image.

The argument also proves preservation when two small pairs with a bounded match are mapped to a common larger level. Their joint types with the marks lift freely by the mixed-level form of Lemma 7.2. Matched small approximation lists can be matched in the two large bases by underlying homogeneity. Those lists compute the transported types, and the pair isomorphism test applies at the large level.

For reflection, let a large pair isomorphism match the two images. It transports the first small approximation list into the second large base. Inside the second small base, realize a copy of this list over the second small approximation list and the required base entries. This is a bounded realization problem: the diagram of the second list is unchanged under its mixed embedding, and mixed saturation in the underlying scheme realizes the joint diagram. Small-level homogeneity extends this match to an isomorphism of the small bases.

To check that the small isomorphism transports the types, test a parameter tuple of the size required by the uniform testing bound. Match that test upstairs over both approximation lists. Each list computes the free extension of its small type there, and the large pair isomorphism identifies the answers. Transport back to the small bases. The answers agree on all tests up to the bound, and Theorem 6.6 identifies the whole types. Apply the small pair isomorphism test. All lists used have bounds in the input arities; closure puts them below the small level. $\square$

8.1 A template before chain closure

We first construct a template whose individual values are marked pairs. The proof first transfers isolation for tuples with a common finite bound on the support of each entry, then removes that restriction by embedding into an all-dominated object. Neither step uses chain closure for marked pairs. That remaining assertion is proved only after the two-part testing bound is available.

Lemma 8.3 (The marked-pair template). *There is a fixed set-sized finitary term template F^b such that, on a further class of sufficiently closed eligible levels, its values on well-orders are objects of $\mathcal{C}^b$. Order embeddings give the componentwise arrows of Definition 8.1, also between such levels. Every marked pair at a level is isomorphic to a value of this template of that size.*

At this stage the following properties are available without using chain closure in $\mathcal{C}^b$: mixed downward Löwenheim–Skolem, preservation and reflection of bounded diagram comparisons, bounded diagram saturation, and the exhaustive-list criterion for being an arrow.

Proof. The samples and their language. Let F be the template of $\mathcal{C}$. At arbitrarily high closed input levels choose a pair

$$X = F(J) \leq Y, \quad Y \cong F(Z),$$

with Y all-dominated over $X + b$. Choose the presentations sufficiently rich for any prescribed set-sized order-saturation bound. We may take J, Z , and a further order W , to be padded dense orders. Padding means that the order has a well-ordered subset of its own cardinality. In addition, W has many disjoint convex substrips, each of the level size. Proposition 6.15, in its presentation-invariant form, puts Y over X into

$$L = F(J + W).$$

All these choices concern the already working scheme $\mathcal{C}$.

Expand a sample by sorts for the orders and charts, chart evaluation maps, the inclusions, and the mark. For each element of a chart choose a representing label and its finite list of positions. Representation tags record the label arity; coordinate functions have default values outside their arity. Include a sort of retraction indices and an evaluation function for a family

$$r_i : Y \longrightarrow X$$

from Corollary 6.16. Distinct indices are retained as distinct indices, even if two of their maps happen to be equal.

Finally include finitary LS witness functions for the underlying K -vertices. For each finite list in a vertex choose a small strong submodel containing it and the models already chosen for its proper sublists, and enumerate that model. Coherence gives strong inclusions between these small models. Perform this in all the finitely many vertex occurrences, including the required inclusion data. In the term closure of any generating set, these enumerated small submodels form a directed cover of each vertex intersection: every finite tuple has its witness model there, and a witness for a combined tuple contains the earlier witnesses. Thus the intersection is strong by the directed-union axioms. Designated entries keep the necessary sorts nonempty.

This is one fixed set-sized finitary expansion. It names neither the diagram table nor all the elements of a high sample. Its number of terms has a bound in the old template language and the fixed input data.

Homogeneous finite hulls. In each sample take a long ordinal-indexed sequence bundling an increasing sequence in each order sort and a sequence of distinct retraction indices. Color a

finite index set by the complete labelled atomic diagram, with negations and equalities, of its term hull and the inclusions among the hulls of its subsets. The palette has a fixed set bound. The finite-arity extraction used in Theorem 2.6 chooses compatible colors at all finite arities while retaining arbitrarily large homogeneous samples. Retain at the same time arbitrarily high prescribed order-saturation bounds. Treat the requested homogeneous size and the requested order saturation as one cardinal demand. If each possible next color had a bound on the demands it could meet, the supremum of those set-many bounds would contradict the availability of samples. Thus a single next color retains unbounded demands. Thinning the horizontal sequence does not change the saturation of its ambient chart orders.

Let H be a well-order of a sufficiently closed input size λ , and evaluate the extracted term data on H . Write the resulting chart orders as J_H, W_H, Z_H , and the underlying objects as X_H, Y_H, L_H . The evaluation and representation functions give both directions of generation:

$$X_H = F(J_H), \quad Y_H \cong F(Z_H), \quad L_H = F(J_H + W_H).$$

Representation equalities concern fixed finite tuples of terms and therefore hold in actual homogeneous samples. The increasing distinguished sequences pad the chart orders. Extending the old template to padded orders uses only the underlying scheme's chain and directed-union properties.

The vertex LS witnesses prove

$$X_H \longrightarrow Y_H \longrightarrow L_H$$

componentwise strong over $X_H \rightarrow L_H$. They also prove this for maps induced by order embeddings of H , including mixed sizes. Each witness model is an isomorphic copy of its sample model, and the full component map is a directed union of these strong maps. Since arrows of $\mathcal{C}$ are componentwise, these are underlying arrows. All-domination remains to be proved.

Bounded horizontal arity. Fix a bounded list $u = (b, c)$ from Y_H such that each individual entry has a term representation using at most q horizontal indices, for one common finite q . This is a bound per entry: the union of the supports of the whole bounded list may be infinite. Term depths need not be bounded. Coordinate functions and representations preserve finite support. Thus the positions representing u in $J_H + W_H$ use a fixed finite number of indices per position. The positions representing $r_i(u)$ in J_H , for the λ distinguished retraction indices, require only the additional index i . Comparing two such positions requires at most twice a fixed finite arity.

Take an actual sample homogeneous at these arities, with a horizontal index set containing an order copy of H . The specified positions match those representing an actual tuple $u_* \in Y$ and its images under the corresponding actual retractions. Choose the sample with order saturation above λ . Extend the partial matches to embeddings of the *entire* orders J_H, W_H into the respective sample orders. In a recursion of at most λ steps, each next position specifies an order pattern over at most λ earlier positions, and sample order saturation realizes it. The initial partial maps are order embeddings by the recorded comparisons. We are not extending a map of the entire expanded term hull.

The host-order embedding identifies u with u_* and embeds X_H into X . The mixed stationary prescription first shows that the mark in Y_H realizes p_{X_H} . We next prove the stronger assertion needed for isolation:

$$\text{tp}(u_*/X) \text{ is the free extension of } \text{tp}(u/X_H). \quad (7)$$

The images of u_* under the λ selected actual retractions belong to the embedded X_H , because their representing positions were included in the partial matching. Choose a test bound $\chi < \lambda$ above

the costs for the arity of u , and retain a set of κ selected retractions with $\chi + \aleph_0 < \kappa < \lambda$. Their images of u_* form a bounded list in X_H . For each $Z \subseteq X$ with $|Z| \leq \chi$, all but at most $|Z| + \aleph_0$ of these retractions fix Z pointwise. At every other index, diagram preservation identifies the diagram of its image of u_* with Z with the diagram of u_* with Z . Thus this image list computes the actual type over X with the majority estimate of Lemma 6.11. That Lemma gives (7). Only images of the tested tuple have been placed in X_H ; closure of the entire small upper object under these retractions is neither claimed nor needed here.

The actual sample pair is all-dominated. Apply Lemma 7.6 to (7); its fragment bound is below λ by closure. It follows that $\text{tp}(b, c/X_H)$ is isolated relative to b .

Removing the arity restriction. Let V be all-dominated over $X_H + b'$, with b' of the prescribed type. We construct an arrow

$$Y_H \longrightarrow V$$

over X_H , taking b to b' . Choose nested bounded lists exhausting Y_H in countably many steps. They may be chosen so that at each step every entry uses at most a fixed finite number of horizontal indices: increase a cardinal bound cofinally to λ and the finite index-arity bound at the same time. Every term has finite arity. Include also full enumerations of small vertex witness models for finite index sets in the corresponding bounded part of H . Their enumerated entries still have uniformly finite horizontal arity per entry, independently of term depth.

At each step the whole new list, the preceding list, and b satisfy the isolation conclusion already proved. The constructive primeness argument of Proposition 7.5 realizes the new list over the previous assignment and X_H . Its isolating fragment together with the preceding list is bounded; underlying saturation supplies the realization. Isolation gives equality of the full types over X_H . The included vertex witness models give strongness of the union map. Every bounded tuple of its image in V has isolated joint type with the mark. Consequently every bounded tuple of Y_H , with no arity restriction, has this property. Thus Y_H is all-dominated.

The remaining pre-closure properties. Any marked pair at the level has a base isomorphic to X_H , and Lemma 8.2 extends this to a pair isomorphism. The same Lemma gives orbit preservation and mixed reflection. For a set of at most ν entries in a larger pair, take its horizontal supports and a well-ordered subpresentation of size ν containing them. Its value is a marked pair by the individual template assertion just proved; coherence gives mixed downward Löwenheim–Skolem.

Naturalness on well-orders follows by putting any fixed bounded patterns into a common well-order enlargement and using orbit preservation. Mixed reflection makes these comparisons independent of the eligible level. Here are the remaining diagram details, without assuming the union clause in Definition 5.2. Define comparison of bounded term patterns by their pair isomorphism orbits at any eligible level. The preservation and reflection just proved make this independent of the level. At each fixed arity, the labels and order patterns belong to a fixed set, giving a uniform orbit bound. A mixed arrow factors, by mixed downward Löwenheim–Skolem and coherence, through a subpresentation of the target at its source level. Same-level orbit preservation in this factor and the level-independent pattern comparison give mixed diagram preservation. For saturation, represent a possible joint diagram in a larger well-ordered pair presentation. Its support has size below the receiving level, so that well-ordered support pattern embeds in a well-order at the receiving level. Evaluating F^b there realizes the joint diagram. The old subtuple has the orbit of the prescribed actual old tuple, so an isomorphism of the pairs carries the new realization over that tuple. An exhaustive diagram match is componentwise strong by its small vertex enumerations and respects the inclusions and mark, so it is a pair arrow. No union-object assertion for $\mathcal{C}^b$ was used. $\square$

8.2 A testing bound beside an old small pair

The following form of testing will be used when extending a partial map. It separates the use of the pair template from pair-chain closure, which is still to be proved.

Lemma 8.4 (Testing beside an isolated small upper object). *Let $\nu < \mu$ be sufficiently closed levels where Lemma 8.3 holds. Suppose*

$$A \leq U, \quad A \leq C, \quad C \leq T, T'$$

are underlying arrows, A, U are at level ν , and C, T, T' are at level μ . Suppose (A, U, b) is a small marked pair inside T , its specified image (A, U', b') is inside T' , and the map $U \cong U'$ fixes A . Suppose further that b/C and b'/C are the prescribed free extensions and corresponding bounded tuples of U, U' , jointly with the marks, have equal types over C .

For each $\sigma < \nu$ there is a bound $k(\sigma) < \nu$, uniform in these objects, with the following property. If $v \in T$ and $v' \in T'$ have length at most σ , their comparisons on joint tests from U and C of total size at most $k(\sigma)$ determine their comparisons on all such tests of total size $< \nu$. Tests in U' are read by the specified map.

Proof. Present (A, U, b) as $F^b(H)$ on a well-order of size ν . Given a questioned bounded tuple from C , include it over A in a level- ν subobject $V \leq C$. Append a universal strip to H , retaining size ν , to obtain H' . The base presentation of the pair template is a natural term presentation into the already working scheme $\mathcal{C}$. The presentation-invariant universal-strip result embeds V over A into $A_{H'}$. Corollary 6.7 then embeds $A_{H'}$ into C over this copy of V .

The joint bounded diagrams of U and $A_{H'}$ inside T are the diagrams of the natural layout

$$U_H \cup A_{H'} \quad \text{inside } U_{H'}. \tag{8}$$

Indeed, each bounded tuple of U , together with b , is isolated over A , whereas the marginal of b over $A_{H'}$ is the prescribed free extension. Lemma 7.2 determines its joint type. The argument applies both to (8) and to the specified image inside T' . It includes equalities between the two parts of the layout; an additional actual overlap is not ignored.

For a bounded candidate tuple in either ambient, each bounded test over the layout has a local realization in $U_{H'}$, by mixed diagram saturation in $\mathcal{C}$. Apply Lemma 6.4 to lift its comparisons to a placed comparison over (8). The layout has two well-ordered horizontal segments with designated term parts. Proposition 6.3 gives a bound depending on the candidate arity and the fixed template, not on H, H' or V .

If the two candidates disagree on the questioned test, their placed comparisons disagree and yield a witness within that bound. Its second-part parameters lie in $A_{H'}$, already embedded into C ; its first-part parameters lie in the original U . This is the required small witness. Close ν under the testing and placement costs for this fixed layout.

All padded-order repetitions here concern the horizontal top presentation as a functor into $\mathcal{C}$. Its extension to padded orders uses chain closure of $\mathcal{C}$, not of $\mathcal{C}^b$. $\square$

8.3 Chain closure with all bounded tuple lengths

Lemma 8.5 (Chain closure). *At further sufficiently closed levels, $\mathcal{C}^b$ has unions of increasing chains of total size equal to the level. The union is a marked pair. The common-upper-bound clause holds as well, including into an eligible larger level.*

Proof. Fix a level μ closed under the bounds of Lemma 8.4, with cofinally many smaller levels at which Lemma 8.3 and the required underlying results hold. We do not require pair-chain closure at the smaller levels.

Write the chain, replacing arrows by inclusions, as

$$(C_i, T_i, b) \quad (i < \alpha),$$

and its component unions as C, T . The underlying scheme gives $C \leq T$ and its union maps. The actual marginal of the finite mark over C is p_C : the marginals at proper stages are the prescribed free extensions, and Proposition 6.13 gives stationary continuity for this finite tuple.

Induct on the chain length. The shorter-length assertion allows intermediate limit unions to be inserted. Restricting cofinally, we may consider a continuous chain indexed by a regular cardinal δ . If a bounded tuple $d \in T$ is contained in some T_i , its type with b is isolated over C_i ; Lemma 7.2 gives isolation over C .

Suppose d is not contained in a proper stage. Its entries occur at cofinally many indices, so

$$\delta \leq |d| < \mu.$$

Choose an eligible $\nu < \mu$ strictly above $|d| + \delta + \aleph_0$, closed under the arity, isolation, approximation, and two-part testing bounds to be used. Small pair objects below will come from Lemma 8.3 or from explicit descent; pair-chain closure at ν is not an induction hypothesis. Choose

$$B \geq C \quad \text{all-dominated over } C + b', \quad b' \models p_C.$$

It suffices to find an image d' in B with $\text{tp}(b, d/C) = \text{tp}(b', d'/C)$, because the type on the right is isolated relative to b' . We obtain this image by building small pairs inside the given chain and compatible maps from their upper objects into B . The maps must preserve types over the entire union base C , including for tuples that span a limit of the small pairs.

Specifically, we construct continuous systems of underlying level- ν objects $A_i \leq U_i$ inside $C_i \leq T_i$, for $i < \delta$, and compatible arrows $f_i : U_i \rightarrow B$, satisfying:

1. $b \in U_i$, $f_i(b) = b'$, and f_i fixes A_i ;
2. (A_i, U_i, b) is a marked pair;
3. for every tuple u from U_i of length $< \nu$,

$$\text{tp}(b, u/C) = \text{tp}(b', f_i(u)/C); \tag{9}$$

4. every entry of d is eventually included.

Equation (9) is computed in the actual underlying ambients T, B .

Retractions prepared for the limits. For every nonzero limit $j \leq \delta$, put $C_\delta = C$, and choose in advance families of ν approximate retractions

$$r_\xi^j : T \rightarrow C_j, \quad s_\xi^j : B \rightarrow C_j \quad (\xi < \nu). \tag{10}$$

They exist in the underlying scheme since $C_j \leq T, B$; each old parameter of C_j is fixed outside a finite set of indices in each family. At a limit we require

$$r_\xi^j[U_j] \cup s_\xi^j[f_j[U_j]] \subseteq A_j \quad (\xi < \nu). \quad (11)$$

We describe the scheduling. Once an entry x and its image $f_i(x)$ have been chosen at stage $i < j$, each value in (10) belongs to C_j , hence to some C_k with $k < j$ by continuity. Assign its inclusion in the small base to a successor stage k' with $i < k' < j$ and $k \leq k'$. Such a stage exists because j is a limit. A value is never requested before it is available. New entries and images create at most $\nu \cdot \nu \cdot |\delta| = \nu$ requests at a stage. The union of the requests assigned to any one stage still has size at most ν : there are $< \nu$ stages, each of size ν . Underlying mixed downward Löwenheim–Skolem can therefore include all of them. This also schedules $j = \delta$, whose families were chosen in advance.

Small objects at a nonlimit stage. At an initial or successor stage i , include the preceding small base in the new base and the preceding small top in the new top, the requests now due, and the designated entries of d available in T_i . Choose ν approximate retractions $T_i \rightarrow C_i$. Alternate for ω steps the following operations: choose a level- ν subobject of C_i containing the current base, all due base entries, and all the chosen retraction images of the current top; then choose a level- ν subobject of T_i containing that base, the current top, and required top entries. Each request has size at most ν . Underlying coherence gives increasing arrows and the base-to-top arrows. Take underlying unions to obtain $A_i \leq U_i$.

The marginal of b over A_i is p_{A_i} , by restriction of the mixed prescription from C_i . Lemma 6.17 says that every $< \nu$ tuple from U_i , jointly with b , has actual type over C_i freely extended from A_i . Its type over C_i is isolated relative to b , since the given large pair is all-dominated. The uniform fragment bounds are below ν . Lemma 7.6 proves that (A_i, U_i, b) is a small marked pair. Only underlying unions were used in obtaining it.

Extending the map at a nonlimit stage. Let U^-, A^-, f^- be the preceding small top, base, and map, when present. The old map fixes every element of $U^- \cap C$, not just A^- : apply (9) with that element as a parameter. Thus fixing A_i is compatible with the old map on their overlap.

Enumerate U_i by nested bounded lists in countably many steps, including the mark and whole smaller strong vertex submodels. Place these lists in B , retaining the old map, and require equality on all joint tests of size $< \nu$ from U^- and C . For a current source list v , let $\chi_0 < \nu$ be the bound of Lemma 8.4. With no preceding stage, use whole-base testing instead.

For every old test u from U^- of length at most χ_0 , the tuple (b, u, v) lies in T_i . Its type over C_i is isolated, and its actual mark marginal over C is free. Lemma 7.2 gives an isolated type over C . Choose an isolating fragment

$$Z_u \subseteq C, \quad |Z_u| \leq h(|b, u, v|) < \nu.$$

Include the empty old test. The union of the fragments has size at most $\nu^{\chi_0} \cdot h(|b| + \chi_0 + |v|)$. This may exceed ν , but it is less than μ , since μ is strong limit and $\nu < \mu$.

The placement parameters are $f^-[U^-]$, the fragments, b' , previous assignments in the current exhaustion, and the required fixed entries of A_i . Their total size is $< \mu$. Choose a test size $\chi < \nu$ large enough for χ_0 , the current list, the fragment bounds, all relevant equalities, and the compact-placement costs. Specify joint diagrams on all parameter tests of size at most χ , using the actual source diagrams.

Every such *simultaneous* local request is possible in B . A query of size at most χ may meet many of the fragments Z_u , but all those entries together form one tuple from C of size at most χ .

Combine that tuple, its old entries from U^- , fixed base entries, and previous assignments into one parameter list. For the first placement, (9) matches its old entries over C ; if there is no old stage, the prescribed mark types give the initial match. At later placement steps, the maintained joint comparisons give the match also with previous assigned entries. The combined list has size $< \nu$. Underlying saturation in B realizes the diagram of the new source list over these parameters.

The compact-placement cost depends on the current tuple arity, χ_0 , the uniform fragment bound, and the fixed labels, rather than on the size of the entire union of fragments. For an infinite $\sigma < \nu$ bounding the current tuple arity, choose $\chi < \nu$ above these costs with $\chi^\sigma = \chi$. Each parameter subset of size at most χ then has at most χ support profiles. The hit-and-separation construction tests only a final parameter set of size at most χ . Apply Lemma 6.4 to the coherent local requests in a well-ordered presentation of B . It gives a placement in an order enlargement. The whole parameter set has size $< \mu$, so underlying saturation realizes its joint diagram with that placement back inside B . In particular the realization agrees on every Z_u and all previously imposed equalities. Its mark marginal over C is p_C . Isolation gives the correct whole type over C for (b, u, v) when $|u| \leq \chi_0$. Lemma 8.4 upgrades this to all joint tests of size $< \nu$. Continue.

The union is an underlying arrow $f_i : U_i \rightarrow B$, by the included strong vertex enumerations. It fixes A_i and takes b to b' . A $< \nu$ tuple of U_i might span this countable exhaustion. For such a tuple, the completed arrow gives the correct small type over A_i . The small pair is all-dominated, and both actual mark marginals over C are the prescribed free extensions. Mixed Lemma 7.2 identifies their types over C . This proves (9) for every $< \nu$ tuple.

Proper limit stages. For a nonzero limit $j < \delta$, take underlying unions

$$A_j = \bigcup_{i < j} A_i, \quad U_j = \bigcup_{i < j} U_i, \quad f_j = \bigcup_{i < j} f_i.$$

They are underlying objects and an underlying arrow by the chain and mixed upper-bound clauses of $\mathcal{C}$. Their size is ν , since $|j| < \nu$. Scheduling gives (11). For every $< \nu$ tuple of U_j , apply Lemma 6.17 on each side: its actual type over C_j is the free extension of its type over the common A_j . These small types agree because f_j fixes A_j .

The given large pair (C_j, T_j, b) is all-dominated. Downward isolation transfer proves all-domination over $A_j + b$, without small-pair chain closure. The equality over C_j , isolation there, and prescribed free mark marginals over C then give equality over C by Lemma 7.2. All proper-stage invariants are established.

The final limit. Take underlying unions $A_\delta, U_\delta, f_\delta$. The families prepared for $j = \delta$, condition (11), and Lemma 6.17 give equality of the actual types over C from their common type over A_δ . No isolation assertion for the final source pair has been used. In particular,

$$\text{tp}(b, d/C) = \text{tp}(b', f_\delta(d)/C),$$

since $|d| < \nu$ and all its entries were included. The right-hand type is isolated by all-domination of B ; hence the left-hand type is isolated relative to b . This covers every bounded tuple not already contained in a proper stage, and proves all-domination of T over $C + b$.

The union maps are componentwise strong by the underlying scheme. If all stages lie in a common marked pair, the underlying upper-bound clause gives a componentwise strong union map preserving the data and mark. That is precisely a pair arrow. The argument includes a larger eligible common upper bound. $\square$

Theorem 8.6 (The marked-increment working package). *Let $\mathcal{C}$ be a working scheme with componentwise arrows and a stationary finite prescription as in Definition 8.1. Assume the eligible-subclass hypothesis of Convention 5.3. On an unbounded further class of sufficiently closed eligible levels, $\mathcal{C}^b$ is again a working scheme, with componentwise arrows, fixed template F^b , and mixed arrows between these levels. The output class retains the eligible-subclass hypothesis.*

The added language and bound recipes at this step have set-sized bounds in the input template and prescription. Thus the construction can be performed at each fixed finite depth without class-sized parameters or a language size depending on the level.

Proof. Lemma 8.2 gives the isomorphism test, orbit preservation, and mixed reflection. Lemma 8.3 gives individual template values and their mixed arrows, mixed downward Löwenheim–Skolem, natural bounded diagrams and saturation, and the exhaustive-list arrow criterion. Lemma 8.5 supplies the missing union-object and upper-bound assertions. Composition and coherence are componentwise.

The extraction added only finitary chart, representation, retraction, and LS-witness symbols with a fixed set bound. Further bounds were testing and placement costs for the old template and the fixed two-part layout of the new template, isolation and approximation costs, and a refinement permitting cofinally many smaller template-valid levels. These are fixed-parameter operations in the closure bookkeeping. No step requires small-pair chain closure before it is proved. $\square$

9 Finite iteration and exact boundary maps

We work with schemes whose arrows are componentwise strong system embeddings, preserving the specified distinguished data. All levels in this section are eligible strong limit cardinals of countable cofinality. The additional closure requirements below are imposed for each fixed finite iteration depth. A common choice for all finite depths will be made separately in Theorem 13.5. Unboundedness after the refinements, including the supply of smaller completion levels for exact lifting, uses the eligible-subclass hypothesis of Convention 5.3. In the finite marked application, Section 13 verifies this hypothesis by maintaining containment of G_j for some finite j , in the hierarchy previewed in Section 5.

Completion and uniqueness for finite systems are central in multidimensional independence. Shelah developed dimension-raising existence and uniqueness methods in [She83, Sections 3–5]; compare the distinction between extension, uniqueness, and strong uniqueness in [SV24, Definition 8.16]. Here amalgamated completion and extension onto the prescribed target are proved separately.

9.1 Forgetting components and recovering types

Iteration requires more than forgetting vertices of an object: a type prescribed in the root must determine a unique type over the entire system. The following correspondence supplies that requirement and shows that it is compatible with free lifting.

Proposition 9.1 (Type correspondence under projection). *Let $P : \mathcal{C} \longrightarrow \mathcal{D}$ forget some components of systems. Suppose that both schemes satisfy the working package, that P takes objects and arrows to objects and arrows, including mixed arrows, and that the following diagram comparison holds: for bounded tuples in the retained sorts, equality of their $\mathcal{C}$ -diagrams is equivalent to equality of their $\mathcal{D}$ -diagrams after projection. The comparisons are made at arities bounded at both levels when levels differ.*

At sufficiently closed levels, restriction induces a bijection on types of bounded tuples in any fixed list of retained sorts, given by

$$\text{tp}_{\mathcal{C}}(v/X) \mapsto \text{tp}_{\mathcal{D}}(v/PX).$$

The parameters on the left are all of X , including its forgotten components. The bijection commutes with restriction and isomorphisms, and with free lifting along arrows, including mixed arrows when the tuple arities and the required testing bounds lie below the smaller level. It also takes witnessed free products to their projected witnessed free products.

Proof. First consider surjectivity at a level μ . Present $X = F(I)$ on a well-order, and let p be a projected type over PX . For a bounded list a of parameters from PX , saturation in $\mathcal{D}$ realizes the prescribed joint diagram of (v, a) in PX itself. Such a realization has term representations in the retained sorts of $F(I)$. Thus every one of the simultaneous local requests used in compact placement has an actual placement in the chart of X .

Here we prescribe projected diagrams only; diagrams involving the forgotten sorts have yet to be chosen. The proof of Lemma 6.4 applies to this form of request as well. To verify this explicitly, fix a sufficiently large bounded cardinal $\chi = \chi^{|v|+\aleph_0}$, larger than the label and testing bounds. A profile records the labels of v , its ordered supports, and their comparisons with the chosen representations of the projected parameters under consideration. There are at most χ profiles over at most χ parameters. If a continuation of a profile fails a prescribed projected diagram, record a witnessing projected test and add it to the next parameter set. The comparison of continuations with no new event uses equality of natural $\mathcal{C}$ -diagrams on retained sorts; the hypothesis on P then gives equality of the requested $\mathcal{D}$ -diagrams. Consequently the same new-hit or new-separation argument as in Lemma 6.4 rules out failure at every step. The local possibility used at the final step is the simultaneous request on the entire final bounded parameter set, which is realizable in PX by saturation.

The resulting placement in an enlargement of I satisfies the projected requests up to χ . Taking χ beyond the testing bound in $\mathcal{D}$ makes its projected type equal to p . The placement itself defines a $\mathcal{C}$ -type over all of X , which proves surjectivity. In particular, no diagram on a forgotten parameter was assumed to have been prescribed in advance.

Next let $q = \text{tp}_{\mathcal{C}}(v/X)$. Its approximation lists from Lemma 6.9 consist of tuples in the same retained sorts as v . A query whose parameters lie in the projection sorts can be moved over such an approximation list using the working package. The projection comparison identifies precisely the equality tests used in this computation. The majority answer and its exception estimate therefore compute the projected free lift. Increasing the approximation length or the test arity gives the same answers by the compatibility part of the approximation construction. Thus free lifts project to free lifts. This reasoning uses bounds below the originating level and consequently applies also to mixed arrows. Projecting the witnessing chain of a free product now gives its claimed projected witnessing chain.

For injectivity, suppose $q_0 \neq q_1$ have the same projection. Choose a bounded tuple f from X witnessing their inequality: the diagrams of (v, f) for the two types have distinct values δ_0 and δ_1 . Choose a bounded infinite χ such that

$$\chi^{|f|+\aleph_0} = \chi,$$

and such that χ bounds $|v|$ and the label and orbit-counting costs. All constructions below fit below μ : in particular $2^\chi < \mu$, and a tuple of χ many copies of v remains bounded at μ .

For each $\eta \in {}^\chi 2$, construct a genuine witnessed free sequence $(v_i^\eta : i < \chi)$ over X with successive marginal $q_{\eta(i)}$. Successor steps use free extension and amalgamation; limit steps use the chain axiom. The uniqueness of whole types of witnessed chains in Lemma 7.12 applies also to these bounded tuple entries. Their projected marginal types are all the same, so their projected sequences have the same whole type over PX . In particular their parameter-free projected diagrams agree. By the hypothesis on P , their whole $\mathcal{C}$ -diagrams agree as well.

Fix one such sequence $a = (a_i : i < \chi)$ in a level- μ object. Bounded diagram homogeneity gives, for every η , an isomorphism from the object containing v^η onto this fixed object, carrying v_i^η to a_i for every i . Write f_η for the image of f . Preservation of diagrams gives

$$D_{\mathcal{C}}(a_i, f_\eta) = \delta_{\eta(i)}.$$

If $\eta \neq \eta'$, a coordinate where they differ shows that f_η and $f_{\eta'}$ have different orbits over the entries of a . There would be 2^χ such orbits of $|f|$ -tuples over at most χ parameters. Lemma 6.1 bounds their number by χ , a contradiction.

Restriction and transport commute with projection directly. Surjectivity and injectivity, together with the free-lift calculation, give all the stated compatibilities. In particular a projected extension of a type is free exactly when its corresponding full extension is free: the free full extension has the required projection, and injectivity makes it the only possible full lift. $\square$

Corollary 9.2 (Lifting an extension of a projection). *Under the hypotheses of Proposition 9.1, let X and Z be at the same eligible level, and let $PX \rightarrow Z$ be an arrow. There are a $\mathcal{C}$ -extension $X \rightarrow X^*$ and a $\mathcal{D}$ -arrow $Z \rightarrow PX^*$ whose composite on PX is the projection of $X \rightarrow X^*$.*

Proof. Identify the given base with its image. Choose nested bounded lists d_i exhausting Z for $i < \omega$. In each vertex include enumerations of increasing smaller strong submodels exhausting that vertex. We construct extensions $X \rightarrow X_i$ and assignments of d_i into PX_i that extend the previous assignments and have the correct type over PX .

Suppose the assignment of d_i is given. Amalgamation in $\mathcal{D}$, with identification of the already matching tuple, puts Z and PX_i in a common extension, agreeing on PX and on d_i . The type there of d_{i+1} over PX_i lifts by Proposition 9.1 to a type over X_i . Realize it in a $\mathcal{C}$ -extension X_{i+1} . The part corresponding to d_i is already a tuple of parameters from PX_i , so its equality diagram forces the new assignment to extend the old one. At the first step use the same argument with the empty tuple.

Take the union of the X_i . The assigned lists match all their bounded diagrams, and the included vertex submodel enumerations give componentwise strongness by the working-package map criterion. Thus their union is the required arrow from Z . $\square$

9.2 The finite systems and their faces

Theorem 9.3 (Finite marked systems). *Start with the single-model scheme, allowing fixed bounded named data of a fixed diagram. Let r be a prescribed, definable, isomorphism-invariant stationary type of a finite tuple in its root sort. Stationarity here includes free lifting under mixed arrows at eligible levels. For each finite n , the n -fold marked increment construction gives a working scheme C_n at sufficiently closed levels. Its objects are systems indexed by the subsets of $\{1, \dots, n\}$, with the prescribed marks and componentwise strong arrows.*

For $s \subseteq \{1, \dots, n\}$, retaining the axes in s in their induced order gives a projection

$$P_s : \mathcal{C}_n \longrightarrow \mathcal{C}_{|s|}.$$

Every such projection takes objects and arrows to objects and arrows, including mixed arrows, and preserves and reflects diagram equality on bounded tuples in its retained sorts. It consequently has the exact type correspondence of Proposition 9.1. The type in the root sort lying over r exists uniquely over every whole system and lifts freely.

These conclusions include the projection to the base root and each single-axis projection. In a single-axis projection the upper object is all-dominated over its lower object together with the prescribed mark; forgetting other axes does not remove that assertion.

Proof. We induct on the number of axes. At depth zero the working package and the prescription r are the given ones. Suppose the assertion is known for $\mathcal{C}_n$. By the exact type correspondence for the root projection, r determines a unique type over every $\mathcal{C}_n$ -object. Projection commutes with free lifting, so this prescription is stationary, including under mixed arrows. Theorem 8.6 therefore supplies the working scheme $\mathcal{C}_{n+1} = (\mathcal{C}_n)^b$.

First forget its last axis. The result is its lower object X . If two bounded tuples in the lower objects have equal $\mathcal{C}_n$ -diagrams, a lower-object isomorphism matches them. It transports the prescribed type of b , and hence extends to an isomorphism of the marked pairs by Lemma 8.2. This proves diagram reflection; preservation follows by restriction. Objects and arrows plainly project to their lower objects and arrows.

Next apply an earlier projection $P : \mathcal{C}_n \rightarrow \mathcal{C}_k$ to both layers of $(X \leq Y, b)$. The marginal of b over PX is the required type, by Proposition 9.1. We must verify all-domination of PY over $PX + b$. Suppose it fails for a bounded tuple a from PY . The nonisolation alternative of Proposition 7.3 gives an extension of

$$\text{tp}_{\mathcal{C}_k}(b, a/PX)$$

to an enlarged parameter object $V \geq PX$ that is nonfree, although its b -marginal is free. By Corollary 9.2, put V inside the projection of an extension X' of X . Freely extend the chosen joint type to PX' . It remains nonfree from PX , since otherwise its restriction to V would have been free. Its b -marginal remains free from PX by transitivity.

Lift this type to X' by the exact type correspondence. Its restriction to X is the original type of (b, a) , because their projections agree and the correspondence is injective. Its b -marginal is the free lift of the original one, again by injectivity and compatibility with free lifting. Its joint type is nonfree: a free joint type would project to a free one. This contradicts the relative isolation of $\text{tp}_{\mathcal{C}_n}(b, a/X)$ and Lemma 7.2. The projected pair is thus all-dominated as required. Componentwise projection of an arrow preserves the prescribed mark and gives an arrow of these pairs.

It remains to reflect pair diagrams. Suppose bounded retained tuples in two projected pairs have equal diagrams. The projected pair isomorphism test gives an isomorphism of their projected bases transporting their types jointly with b ; any indicated base entries are included in this match. Choose in the first base sufficiently long approximation lists for this joint type, at arities exceeding the full and projected testing bounds. Their entries are in retained sorts. Carry these lists, together with the indicated base entries, by the projected base isomorphism. The inductive reflection hypothesis and bounded homogeneity give an isomorphism of the full bases making precisely these bounded matches.

The transported approximation lists compute the same projected joint type over the second projected base. Exact type correspondence therefore says that the full base isomorphism transports the entire joint type with b . The full pair isomorphism test now extends it to an isomorphism of marked pairs matching the original tuple. Preservation is immediate from projection of an isomorphism. The same calculation uses free lifting and the mixed preservation and reflection in Lemma 8.2 when comparing eligible levels; all the tuple and approximation bounds are below the smaller one. Composing elementary axis deletions proves the assertion for every P_s . This completes the finite induction. $\square$

Definition 9.4 (Boundary). For an n -axis object J , its boundary ∂J consists of all the proper-axis projections $P_s J$, with their specified inclusions. For $n = 0$ the boundary is empty. A boundary map $h : \partial J \rightarrow \partial K$ is a compatible family of arrows $h_s : P_s J \rightarrow P_s K$ for proper s . A boundary isomorphism is such a family of isomorphisms.

Lists in a boundary are lists of occurrences in its vertex sorts. The diagram of a joint list records the specified inclusion maps and all resulting equalities. In particular, entries supplied from different faces can coincide in a containing vertex. Compatibility on the named subfaces alone is not asserted to settle these extra equalities; their preservation will be part of the next argument.

9.3 Completing a boundary into an extension

Lemma 9.5 (Joint diagrams on the boundary). *At the levels used for $\mathcal{C}_n$, every boundary map between two n -axis objects at the same level preserves all bounded $\mathcal{C}_n$ -diagrams on their boundary unions. This includes joint diagrams of entries from different faces and all equality information on their actual overlaps.*

Theorem 9.6 (Amalgamated boundary completion). *For each finite n , at sufficiently closed levels the following holds. If J and K are $\mathcal{C}_n$ -objects at the same level and $h : \partial J \rightarrow \partial K$ is a boundary map, there are arrows*

$$K \longrightarrow K^*, \quad J \longrightarrow K^*$$

in $\mathcal{C}_n$ such that the second restricts to the composite of h with the first on every proper projection. The extension is in the same scheme and at the same level. No assertion about completion between distinct levels is needed here.

Proof of Lemma 9.5 and Theorem 9.6. We prove the two assertions together by induction on n . For $n = 0$ the diagram assertion is empty, and joint embedding gives completion. At a positive dimension, after proving the diagram assertion we shall derive completion by a placement argument. Thus only lower-dimensional completion will be used in proving the next diagram assertion.

For $n = 1$, its only proper projection is its base. Theorem 9.3 identifies the full-scheme diagrams on that base with its projected diagrams. The given base arrow preserves the latter, proving the assertion.

Let $n = m + 1$ with $m \geq 1$, and write the source and target as

$$(X^- \leq Y^-, b^-), \quad (X^+ \leq Y^+, b^+)$$

over $\mathcal{C}_m$. A boundary map gives a full base arrow $e : X^- \rightarrow X^+$ and a boundary map $h : \partial Y^- \rightarrow \partial Y^+$ agreeing with e on ∂X^- ; the upper faces also specify $h(b^-) = b^+$. Let v be a bounded tuple

from ∂Y^- , enlarged to include b^- . We first claim that its type over X^- in $\mathcal{C}_m$ equals the restriction along e of the type of $h(v)$.

Suppose instead that a bounded tuple f from X^- witnesses a difference. Put

$$\delta_0 = D_{\mathcal{C}_m}(f, v), \quad \delta_1 = D_{\mathcal{C}_m}(e(f), h(v)), \quad \delta_0 \neq \delta_1.$$

Choose an infinite bounded χ satisfying $\chi^{|f|+\aleph_0} = \chi$, bounding $|v|$ and all the label and orbit-counting costs. Let ϵ be the least cardinal with $2^\epsilon > \chi$. Cantor's Theorem gives $\epsilon \leq \chi$, and minimality gives $2^{|i|} \leq \chi$ for every $i < \epsilon$. Consequently

$$|^{<\epsilon}2| \leq \chi, \quad 2^\epsilon \leq 2^\chi < \mu,$$

where μ is the common level. Every accumulator and branch world below will have size μ , and the total number of amalgamations remains below μ .

We construct branch worlds $W_s \in \mathcal{C}_m$ for $s \in {}^{<\epsilon}2$, carrying tuples f_s , with coherent full arrows along each branch. Their proper boundaries have compatible arrows

$$q_s : \partial W_s \longrightarrow \partial A$$

into an increasing common accumulator $A \in \mathcal{C}_m$. The accumulator may be enlarged at each operation; existing boundary arrows are always composed with its enlargement arrow. The invariant requires coherence of these boundary arrows with the branch arrows. It does not require full maps of all the branch worlds into the accumulator to be retained.

Start with $W_\emptyset = X^-$, its tuple f , and the identity boundary map into $A = X^-$. At a node s , diagram preservation along the branch shows that $D(f_s) = D(f)$. Bounded homogeneity supplies an isomorphism

$$k_s : X^- \xrightarrow{\cong} W_s, \quad k_s(f) = f_s.$$

There is no coherence requirement on the choices of k_s . Choose the minus child to be a copy of Y^- attached over W_s by k_s^{-1} , and the plus child to be a copy of Y^+ attached by ek_s^{-1} . These choices give the full branch arrows. In each child, f denotes the image of f_s ; the corresponding upper boundary lists have diagrams δ_0 and δ_1 with it. Transporting h gives a boundary arrow

$$h_s : \partial W_{s0} \longrightarrow \partial W_{s1}$$

over ∂W_s , matching the two copies of v .

Apply the already proved completion theorem in dimension m to q_s . After enlarging the accumulator, this gives a full map $W_s \rightarrow A_0$ extending q_s . Ordinary amalgamation in $\mathcal{C}_m$ then puts the plus child in an enlargement A_1 over this full map. Let $G_1 : W_{s1} \rightarrow A_1$ be the resulting arrow. Retain its boundary restriction as q_{s1} , and define

$$q_{s0} = q_{s1}h_s.$$

Both restrict to the preceding q_s , after enlargement, and the two boundary copies of v have one common image v_s in A_1 . The temporary full maps to A_0 and A_1 are no longer part of the invariant. In particular the minus child has not been required to send its copy of f to the temporary image of f_s in the accumulator.

Process the nodes of each level successively. At a limit height take the union of the worlds along each branch and the union of the accumulators constructed so far. Their sizes are still μ . The boundary arrows into the union accumulator exist by the chain and common-upper-bound

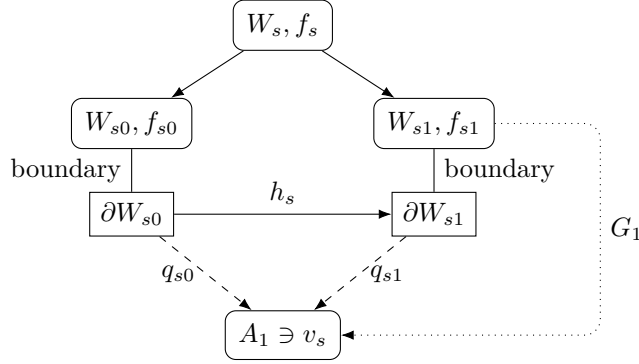


Figure 2: A return-tree split. Solid arrows between worlds are retained along their separate branches. Dashed boundary maps persist in the common accumulator and agree on the test v_s . Each dashed arrow denotes a compatible family of maps into the proper projections of A_1 . The dotted full map is used to choose the plus boundary map and need not be retained at later splits. Full maps of all leaf worlds are installed only after the splitting tests have been fixed.

clauses in every proper projection. Full branch arrows and the carried tuples also pass to the unions. This establishes the invariant through height ϵ .

At that height, apply lower-dimensional completion to each leaf boundary map, successively enlarging the accumulator. This supplies a full map of each W_η , for $\eta \in {}^\epsilon 2$, extending its retained boundary map. There are fewer than μ leaves, so the final union is still at level μ . If two leaves first differ after a node s , their images of f have diagrams δ_0 and δ_1 over the common tuple v_s . Indeed the corresponding copy of v was in that child's boundary, remained there along that branch, and its boundary image was never changed except by composition with an accumulator arrow. Thus the final images of f occupy more than χ orbits over the entries of $(v_s : s \in {}^{<\epsilon} 2)$. This parameter list has size at most χ . The orbit bound of Lemma 6.1 gives the contradiction. The claimed equality of types of v follows.

To pass from this type equality to the full pair diagrams, retain an arbitrary bounded tuple a from the lower face X^- . The marginal of b^+ over X^+ is the prescribed free lift from $e(X^-)$. Relative isolation of the source type of (b^-, v) and Lemma 7.2 therefore identify the actual type of $(b^+, h(v))$ over X^+ with the free lift of that source type.

Choose sufficiently long bounded approximation lists in X^- computing the source type. Their images under e compute its free lift over X^+ by Theorem 6.10. Base homogeneity gives an isomorphism $X^- \cong X^+$ matching a and these lists with their images under e , since e preserves their joint bounded diagram. The transport assertion for the approximation lists says that this isomorphism carries the whole source type to the type of $(b^+, h(v))$. Lemma 8.2 now extends it to an isomorphism of the marked pairs matching (a, v) with $(e(a), h(v))$. Every bounded boundary list is of this form. Thus its full $\mathcal{C}_n$ -diagram is preserved, including cross-face equalities.

We finish the induction by deriving completion from this diagram equality. Present the source J on a well-order. Its boundary union is a natural collection of the designated term sorts in that presentation. Choose nested bounded lists d_i exhausting J , including increasing strong submodel enumerations in every vertex. We construct extensions K_i of K and compatible assignments of d_i into K_i satisfying

$$D_J(d_i, u) = D_{K_i}(d'_i, h(u)) \quad \text{for every bounded boundary list } u. \quad (\text{C})$$

Here and subsequently h is composed with the current extension arrow. At the start the assertion

for the empty list is exactly the boundary diagram equality just proved.

For the next list prescribe the source joint diagrams on the parameter set consisting of the boundary images and the previous assignment. Each bounded local request, including the whole previous assigned list, is possible by saturation: the parameter diagram agrees by (C). This also checks repetitions and equalities between old entries and boundary entries, so the requests define one coherent prescription. Apply compact placement in a well-ordered chart of K_i , keeping these previous entries fixed. Work at one bounded cardinal large enough for the profile count, the new tuple, and the testing bound for the source boundary layout with the previous tuple included. The resulting placement lies in a level- μ extension K_{i+1} .

To check (C) beyond that initial testing cardinal, pull the candidate diagrams back along h to the natural boundary layout of J . Their local possibility there follows from the boundary diagram equality and saturation. Lemma 6.4 compactifies these requests over this source layout, and Proposition 6.3 says that agreement at the chosen testing bound determines all bounded comparisons. The actual images of the boundary inside K_i have not been assumed to form a natural layout. All uses of that hypothesis are on the source layout after pullback.

Continue through ω steps and take the union of the K_i . The working-package criterion, with the included strong submodel enumerations, makes the union assignment an arrow $J \rightarrow K^*$. Whenever a boundary entry is first included in d_i , its equality test against that boundary entry forces its assigned image to be h of that entry. The union arrow therefore extends the entire boundary map. The map from K is the chain inclusion. This proves completion and closes the induction. $\square$

9.4 A bounded support for boundary tests

We have extended boundary maps into common extensions. Exact lifting must instead land onto the given target. The next lemma supplies bounded data that control a tuple's boundary comparisons at a fixed bounded test arity. Together with uniform testing, this permits back-and-forth to remain inside the two given systems.

Lemma 9.7 (Fixed parameters for a boundary test function). *Fix $n < \omega$. At further sufficiently closed levels, let $J \in \mathcal{C}_n$, let z be a bounded tuple of J , and let σ be a bounded infinite test arity. There is a bounded list a in ∂J such that every boundary automorphism α fixing a pointwise satisfies*

$$D_J(z, u) = D_J(z, \alpha(u)) \quad \text{for every boundary tuple } u \text{ of length at most } \sigma. \quad (\text{F})$$

A boundary automorphism means a compatible family of isomorphisms of all proper projections; a full automorphism of J is not assumed.

Proof. The assertion is immediate for the empty boundary. Suppose it fails otherwise. Choose a bounded infinite χ with $\chi^{|z|+\aleph_0} = \chi$, bounding σ and the orbit-counting and label costs. Let ϵ be the least cardinal with $2^\epsilon > \chi$. Choose a smaller eligible level ν such that

$$|z| + \sigma + 2^\chi < \nu < \mu,$$

where μ is the level of J , with all required testing bounds below ν and with Theorem 9.6 available at ν . The further closure requirement on the level of J ensures this choice. It is a finite refinement for this fixed n .

Fix a well-ordered presentation $J = F_n(I)$. Construct continuous subpresentations J_i of size ν for $i \leq \epsilon$, beginning with z in J_0 . At step i , the boundary of J_i is a bounded parameter set in J . By the assumed failure, a boundary automorphism α_i of J fixes it pointwise and changes the value of the function at some tuple u_i of length at most σ . Thus

$$D_J(z, u_i) \neq D_J(z, \alpha_i(u_i)). \quad (\text{S})$$

We choose J_{i+1} containing both tests so that α_i restricts to an automorphism of its boundary.

Here is the support closure for this choice. Begin with the positions for J_i , z , u_i , and $\alpha_i(u_i)$, padding to size ν . For every element of every proper projected subpresentation currently generated, add positions for its image and its inverse image under the corresponding face map of α_i . There are finitely many faces and ν elements in their union, and each representation uses finitely many positions. Repeat this operation countably many times. The resulting position set still has size ν . Every element generated at its union appeared at a finite stage, so both its image and inverse image are generated in the union. Hence all restricted face maps are onto and remain compatible with the specified inclusions. They give the asserted boundary automorphism of J_{i+1} . At a limit use the union of the position sets. Mixed diagram preservation transfers (S) to J_{i+1} , since $|z| + \sigma < \nu$.

Now build a binary tree of boundary embeddings

$$q_s : \partial J_i \longrightarrow \partial A, \quad s \in {}^i 2,$$

where A is an increasing accumulator of size ν . Start with $A = J_0$ and the identity map on ∂J_0 . At a successor node first apply completion to q_s on J_i and then amalgamate J_{i+1} over that full copy of J_i . This yields an arrow $G_s : J_{i+1} \rightarrow A'$ whose boundary extends q_s . Define the two next boundary embeddings by

$$q_{s0} = \partial G_s, \quad q_{s1} = (\partial G_s) \alpha_i^{-1}.$$

They restrict to the same old q_s , because α_i fixes ∂J_i pointwise. The common parameter test $G_s(u_i)$ is the image of u_i on branch $s0$ and of $\alpha_i(u_i)$ on branch $s1$. Only these boundary embeddings are retained for later steps; the full map G_s is temporary.

At limits the continuity of the J_i and the common-upper-bound clauses give the boundary maps into the union accumulator. At height ϵ , complete the boundary maps to full maps of J_ϵ , one leaf at a time. The full maps at the leaves carry z . If two leaves first separate at $s \in {}^i 2$, their images of z have the two distinct diagrams from (S) over the one retained test $G_s(u_i)$. This follows from preservation by their final full maps and from coherence of their boundary maps with the chosen maps at the split.

There are at most χ nonterminal nodes, hence at most χ parameters in all the splitting tests. There are $2^\epsilon > \chi$ distinguishable leaf images of z . All leaf completions and all unions remain at level ν , since $2^\chi < \nu$. Lemma 6.1 again contradicts this number of orbits over the tests. Thus the bounded fixing list a exists. $\square$

Theorem 9.8 (Exact boundary lifting). *For each finite n , at further sufficiently closed levels every boundary isomorphism $h : \partial J \xrightarrow{\cong} \partial K$ between $\mathcal{C}_n$ -objects at the same level extends to an isomorphism $J \cong K$ onto the given target. All the prescribed marks and bounded named data are preserved.*

Proof. We build a back-and-forth sequence of matches $a \mapsto a'$ of bounded tuples, maintaining

$$D_J(a, u) = D_K(a', h(u)) \quad \text{for every bounded boundary tuple } u. \quad (\text{BF})$$

For the empty tuple this is Lemma 9.5. Suppose that a match satisfying (BF) has been chosen, and let z be any bounded enlargement of a in J .

Take a bounded infinite σ at least the uniform testing bound for a tuple of length $|z|$ over the natural boundary layout of a $\mathcal{C}_n$ -object. Choose fixing data c for z, σ in ∂J by Lemma 9.7. The maintained equality with the particular boundary list c gives

$$D_J(a, c) = D_K(a', h(c)).$$

Working-package homogeneity supplies a full isomorphism $H : J \cong K$ carrying a to a' and c to $h(c)$. Put $\alpha = (\partial H)^{-1}h$. This is a boundary automorphism of J fixing c pointwise. Therefore, for every boundary tuple u of length at most σ ,

$$\begin{aligned} D_K(H(z), h(u)) &= D_J(z, \alpha(u)) \\ &= D_J(z, u). \end{aligned} \tag{BT}$$

We justify explicitly the passage from these tests to all bounded boundary tests. By Theorem 9.6, embed J into a same-level extension K^* of K by an arrow j extending h . Both $j(z)$ and $H(z)$ are now tuples in K^* , and (BT) says that their diagrams agree over boundary tests from K of size at most σ . The boundary of K , in a well-ordered chart of K , is a natural collection of term sorts. The types of these two tuples over K have order placements by the one-dimensional calculus. Proposition 6.3 restricted to this natural boundary layout therefore gives equality over every bounded boundary test. Preservation by j then shows

$$D_K(H(z), h(u)) = D_J(z, u)$$

for all bounded u . Thus $z \mapsto H(z)$ extends the previous match and satisfies (BF).

Apply this extension step alternately to J, K with h , and to K, J with h^{-1} . Their level has countable cofinality, so countably many bounded enlargements can exhaust both systems. In each vertex choose increasing smaller strong submodels whose enumerations are included in these tuples. The union assignment is onto in every component and is an arrow by the working-package map criterion; the reverse construction gives its inverse arrow. If an entry lies on the boundary, its equality test with that boundary entry in (BF) forces its image to be exactly the image under h . The resulting isomorphism therefore extends all of h , as required. $\square$

10 A minimal type and its geometry

We continue with the fixed AEC K that is categorical in unboundedly many cardinals. Throughout this Section, a level is eligible for the single-model working scheme and for the refinements of the calculus that are explicitly invoked. The first level chosen below is denoted by λ_0 . The level used for all finite marked constructions will be a later cardinal λ ; these two choices serve different purposes.

Lemma 10.1 (Preservation of nonalgebraicity). *Let p be a nonalgebraic singleton type over an object A of a working scheme. Its free lift over an extension B is nonalgebraic. The same assertion holds for a mixed arrow whenever the input arity and the testing bounds lie below the smaller level.*

Proof. Suppose that the free lift were realized by $c \in B$. Include equality with c among its tests. To evaluate that test, the free-extension construction moves c into A , over enough of the chosen approximations for p . Its value is then the value of equality with an element of A . Every such

equality has value false for p , whereas equality with its alleged realization c has value true. This contradicts the evaluation rule in Theorem 6.10. The mixed-level construction uses the same rule with the bounds taken below the input level. $\square$

Proposition 10.2 (A minimal type). *There is a sufficiently closed single-model level λ_0 , a model $M_0 \in K$ of size λ_0 , and a nonalgebraic singleton type p_0 over M_0 such that, over every $M_0 \leq_K A \in K$ of size λ_0 , there is exactly one nonalgebraic extension of p_0 . This extension is its free lift.*

Proof. Choose λ_0 with the single-model working package, the stationary calculus, the all-dominated hull construction, and the singleton type bound of Theorem 4.11(3). Thus there are at most λ_0 singleton types over a model of that size. We may require at the outset strict closure under all the fixed bounds used below for finite tuples and their bounded tests.

There is a nonalgebraic singleton type at this level. For example, extend an order presentation by a fresh skeleton point. That point is outside the old model: if it were a labeled term on finitely many old indices, adding a second distinct skeleton point with the same order pattern over those indices would make both equal that term, contrary to the distinctness of the skeleton.

Suppose that no nonalgebraic singleton type at this level is minimal in the stated sense. A nonalgebraic type always has a nonalgebraic free extension by Lemma 10.1. Consequently every such type has two distinct nonalgebraic extensions over some model of size λ_0 .

Let ϵ be the least infinite cardinal with $2^\epsilon > \lambda_0$. Then $\epsilon \leq \lambda_0$ and $2^{[i]} \leq \lambda_0$ for $i < \epsilon$. Construct a continuous chain $(A_i : i \leq \epsilon)$ of models of size λ_0 and types p_η over A_i , for $\eta \in {}^i 2$, as follows. At a successor step, split every current p_η over a model extending A_i . There are at most λ_0 such models. Successive amalgamation over A_i , followed by unions, puts all of them into one next base A_{i+1} of size λ_0 . Freely extend each of the two split types to that base. Their restrictions to the splitting model are different, so they remain different, and Lemma 10.1 keeps them nonalgebraic.

Along each branch retain a witness model and the same distinguished realization. The required successor map exists by amalgamation with identification of realizations having the same type over the preceding base. At a limit take the unions of the common bases and, separately on each branch, of its coherent witness models. The realization is outside every base stage, so it is outside the union base. All these unions have size λ_0 because the construction has length at most λ_0 . At the final stage distinct branches have different types over A_ϵ , witnessed at their first splitting stage. This gives more than λ_0 singleton types over A_ϵ , contrary to the type bound. A minimal p_0 therefore exists. Its unique nonalgebraic extensions are free by existence of free lifts. $\square$

The passage from a minimal type to closure and dimension follows the geometric pattern of first-order categoricity developed through strongly minimal sets; see Baldwin and Lachlan [BL71, Sections 1–2]. Here the required containment, exchange, and change-of-base properties are established directly for the AEC and its locally constructed types.

10.1 Finite configurations over the chosen model

Definition 10.3. For $M_0 \leq_K M$ with $\|M\| \geq \lambda_0$, let $P(M)$ be the set of realizations of p_0 in M . More explicitly, $a \in P(M)$ means that in a strong section $N \leq_K M$ of size λ_0 containing $M_0 \cup \{a\}$, the type $\text{tp}(a/M_0)$ is p_0 . A finite ordered list of distinct points in $P(M)$ is independent if its type over M_0 , computed in such a section, is the product of the corresponding copies of p_0 .

These definitions do not depend on the section. Two sections lie in a third section of size λ_0 , by the Löwenheim–Skolem axiom and coherence. The embeddings into that section preserve their types over M_0 . Products are permutation invariant, so independence is a property of the underlying finite set. Strong embeddings over M_0 preserve and reflect both membership in P and finite independence.

We give a precise meaning to the containment test used in constructing closure. Fix a finite configuration $F \cup \{x\} \subseteq P(M)$ and a size- λ_0 section N containing it and M_0 . Say that F *forces* x if, whenever $f : N \rightarrow W$ is a strong embedding fixing M_0 , with W of size λ_0 , and $M_0 \leq_K D \leq_K W$ has size λ_0 and contains $f[F]$, one has $f(x) \in D$. Thus the test permits a common extension of the finite configuration and the proposed model; it does not presume that intersections of strong models are strong models.

The test is invariant under the type of the entire finite configuration over M_0 . Indeed, an identification of two copies of that type is witnessed in an amalgam. Any further configuration witnessing failure of forced containment can be amalgamated with that identification over the first witness model. Its avoiding submodel and the omitted point remain disjoint at that point because the maps are injective. This transports the failure to the second copy. The reverse transport is identical with the copies interchanged.

Lemma 10.4 (The containment test). *For a finite independent set $S \subseteq P(M)$ and $x \in P(M)$, S fails to force x if and only if $S \cup \{x\}$ is independent with $x \notin S$.*

Proof. First suppose that a coupling contains a model D over M_0 and S but omits x . The type of x over D is a nonalgebraic extension of p_0 . By Proposition 10.2 it is the free lift of p_0 . Restricting its parameters to S shows, by Proposition 6.12, that x has product type with S over M_0 .

Conversely suppose that S, x has product type. Use Theorem 7.9 to obtain a model all-dominated over $M_0 + S$, and amalgamate it with the actual finite configuration, identifying the copy of S . The domination conclusion of Proposition 7.5 makes x free over this model from M_0 . It is outside the model by Lemma 10.1. The same amalgamation can retain any further specified finite entries of the original configuration. This is a coupling witnessing that S does not force x . $\square$

Theorem 10.5 (The geometry of the minimal type). *On $P(M)$, for every $M_0 \leq_K M$ of size at least λ_0 , the preceding finite independence is the independence relation of a finitary pregeometry. Its closure, denoted by cl_M , has the following properties.*

1. *For finite $F \subseteq P(M)$ and any maximal independent $S \subseteq F$, the set $\text{cl}_M(F)$ consists exactly of those points that do not independently extend S . Equivalently, $\text{cl}_M(F)$ is the set of points forced by F .*
2. *Strong embeddings over M_0 preserve and reflect finite closure. If $M_0 \leq_K A \leq_K M$, then $P(A)$ is relatively closed in $P(M)$.*
3. *For finite F , $|\text{cl}_M(F)| \leq \lambda_0$. For every infinite $F \subseteq P(M)$,*

$$|\text{cl}_M(F)| \leq |F| + \lambda_0.$$

In particular bases exist, all bases have the same cardinality, and each point of the span of a basis has finite basis support.

Proof. Choose a maximal independent subset S of a finite F . Every $y \in F \setminus S$ fails to independently extend S , so Lemma 10.4 says that every model containing S in the actual configuration must contain all of F . A point that fails to independently extend S is therefore forced by F . If x independently extends S , the avoiding dominated model from Lemma 10.4 can be adjoined while retaining F, x . It contains F by the preceding observation and omits x . Thus F does not force x . This proves the first assertion and its independence of the choice of S .

Define closure on arbitrary subsets by taking the union of closures of finite subsets. Extensivity and monotonicity hold directly from the containment test. To verify idempotence on finite configurations, suppose that finitely many y_j are in $\text{cl}_M(F)$ and that x independently extends a maximal independent $S \subseteq F$. The avoiding model over S can be coupled with the whole finite configuration $F, (y_j), x$. It contains F and every y_j , because all these points are forced by S , and it omits x . Hence the y_j cannot force x . Finite character now yields idempotence for all subsets.

For exchange, it suffices by finite character to take finite F and points x, y with

$$y \in \text{cl}_M(F \cup \{x\}) \setminus \text{cl}_M(F).$$

If $x \in \text{cl}_M(F)$, idempotence would imply $y \in \text{cl}_M(F)$, so $x \notin \text{cl}_M(F)$. Let S be a maximal independent subset of F . Then S, x and S, y are independent. Since every member of F is already forced by S , the set $S \cup \{x\}$ is maximal independent in $F \cup \{x\}$. The displayed membership says that S, x, y is dependent. Permutation invariance of products makes S, y, x dependent as well. Applying the first assertion with $S \cup \{y\}$ gives $x \in \text{cl}_M(F \cup \{y\})$. This is exchange.

Preservation and reflection follow from the invariant finite-type definition and the containment test. For relative closedness, take finite $F \subseteq P(A)$ and $x \in \text{cl}_M(F)$. Choose $D \leq_K A$ of size λ_0 containing $M_0 \cup F$, and then choose a section of M containing $D \cup \{x\}$. Forced containment gives $x \in D \subseteq A$. The finite closure of F lies in this same D , proving the finite cardinal bound. There are $|F|$ finite subsets of an infinite F , so finite character proves the displayed bound.

For completeness, a maximal independent subset of a set can be constructed by well-ordering it and retaining a point precisely when it is outside the span of the points retained earlier. Finite character makes the resulting set independent, and maximality makes it spanning. If an independent set is spanned by a finite set of n points, it has at most n points: successively insert its points into a spanning list, using exchange to remove one of the original points whenever a new independent point is inserted. After n insertions no further independent point is in the span. This also proves equality of finite basis sizes.

For infinite bases H, J of the same closed set, choose for each $j \in J$ a finite support $S_j \subseteq H$. Then $\text{cl}(J) \subseteq \text{cl}(\bigcup_{j \in J} S_j)$. Independence of H implies $H \subseteq \bigcup_{j \in J} S_j$, because a member of H cannot belong to the span of the other members. Hence $|H| \leq |J|$; interchanging the bases gives the reverse inequality. The finite argument excludes the possibility of one basis finite and the other infinite. Finally, for a closed C , the operator $U \mapsto \text{cl}(C \cup U) \setminus C$ on $P(M) \setminus C$ satisfies the same closure and exchange axioms. Finite character follows by discarding all but the finitely many entries of U in a dependence witness. Applying the preceding constructions to this operator gives relative bases and their finite supports. $\square$

10.2 Pointed levels and change of base

Choose an enumeration $\bar{m}_0$ of M_0 and place it as a strong model in a sufficiently large well-ordered presentation of K . Such a placement exists by the order construction and categoricity at λ_0 . Fix the stabilized diagram Δ of this entire enumeration. The initial supporting positions of this placement will be held fixed in later order presentations.

Proposition 10.6 (Pointed working levels). *At all sufficiently closed levels μ above the bounds for $|\bar{m}_0| = \lambda_0$, models with a distinguished strong copy of M_0 whose enumeration has diagram Δ form a categorical working scheme. Arrows preserve the indicated copy pointwise. For a pointed model B at such a level, p_0 has a unique nonalgebraic extension p^B over B ; it is its free lift. This family lifts freely along pointed arrows, including the permitted mixed arrows. Naming further bounded data of a fixed diagram within the base does not change type computation over the whole base.*

Proof. The enumeration is a bounded tuple at μ . Bounded-diagram homogeneity in Proposition 4.3 carries it between any two instances of Δ by an isomorphism of the entire models. The pointed version of the working package follows by reserving its supporting positions and including the enumeration in every relevant diagram, as in Proposition 5.7.

Existence of p^B is mixed free extension of p_0 , and Lemma 10.1 gives nonalgebraicity. To prove uniqueness, take two nonalgebraic extensions over B . A distinguishing test for singleton types may be chosen with a fixed bound below λ_0 . For any such parameter set $Z \subseteq B$, choose $M_0 \leq_K A \leq_K B$ of size λ_0 containing Z . Both types restrict to nonalgebraic extensions of p_0 over A and are equal there by Proposition 10.2. Their diagrams on Z are consequently equal. The bounded testing theorem gives equality over B . Uniqueness and transitivity of free extension prove the assertion for pointed arrows.

All distinguished data here already belong to the parameter model. An amalgam over that model fixes them, and an identification of types over that model respects them. Thus adding them as constants does not change this type computation. $\square$

Proposition 10.7 (Change of base for the geometry). *Let B be a pointed model at a sufficiently closed level much larger than λ_0 . A finite list of points in an extension of B is independent over the closed set $P(B)$ in the geometry of Theorem 10.5 if and only if its type over B is the product of the corresponding copies of p^B .*

Proof. A product can be witnessed by successive free realizations in models containing all preceding entries. Each realization is outside that model by Lemma 10.1. Its point set is relatively closed, so the realization is outside the geometric span of $P(B)$ and the earlier entries. This proves one implication.

For the converse let x be a finite list independent over $P(B)$. Fix a parameter test $Z \subseteq B$ large enough to be used in the uniform comparison of the type of x over B , but with $|Z| < \lambda_0$. Choose a regular infinite $\theta < \lambda_0$ with $|Z| < \theta$. We may and do choose λ_0 closed enough that the comparisons and local-character bounds used here fit below it.

Inside B choose lists x_i , $i < \theta$, so that every $D^K(x_i, Z)$ equals $D^K(x, Z)$ and all the x_i together are independent over M_0 . To make the successor choice, include in the bounded match the entire enumeration of M_0 , Z , and all earlier copies. The original x , being independent over $P(B)$, is independent from those earlier copies. Saturation realizes its joint diagram inside B . Diagram equality is witnessed by an isomorphism, so it preserves every finite independence comparison over M_0 . The total parameters at every choice are bounded at the level of B .

Take a strong section $N \leq_K B$ of size λ_0 containing M_0 , Z , and all the copies. Their finite sublists have product type over M_0 . The independent-list and finite-domination conclusions of Theorem 7.13 give a continuous witness chain $(A_i : i \leq \theta)$ inside N , starting at M_0 , in which x_i is free over A_i from M_0 and is included in A_{i+1} . One may keep the chain inside N : at each step primeness embeds the required all-dominated increment into N over its already placed base and the specified next list. At limits finite domination keeps the next finite list free, and unions remain strong in N .

Since $\text{cf}(\theta) > |Z|$, local character applies to $\text{tp}(Z/A_\theta)$. It is free from some A_i . For a later $j < \theta$, its restriction to A_{j+1} is free over A_j , so Z and x_j have product over A_j . The type of x_j over A_j is free from M_0 . The base-change identity in Proposition 6.12 therefore says that x_j has product with Z over M_0 . In particular $D^K(x_j, Z)$ is determined by p_0 and $\text{tp}(Z/M_0)$.

A known product of the p^B has this same joint diagram with Z : products of free lifts are free lifts of products, and one may restrict the resulting type to M_0 together with Z . Since $D^K(x, Z) = D^K(x_j, Z)$, the two candidate types over B agree on every test of the uniform comparison size. Bounded testing makes their full types equal. This proves the converse. $\square$

10.3 Eliminating proper extensions with unchanged geometry

The obstruction considered here is related to Vaughtian pairs: proper extensions of the same cardinality with no new realizations of a fixed nonalgebraic type; compare [GV06], where the type is based on a limit model. Here finite marked systems and exact boundary lifting eliminate the obstruction in the pointed setting.

The next use of the marked construction requires a type prescription that can be chosen from a fixed set before choosing the common level. We record the uniformity explicitly.

Lemma 10.8 (Anchoring a singleton type). *There is a fixed cardinal β , depending on the single-model template, its singleton comparison bound, and the fixed point, with the following property. If A is a sufficiently closed pointed model and r is a nonalgebraic singleton type over A , a list $c \subseteq A$ of length at most β can be named so that:*

1. *the joint stabilized diagram of $c, \bar{m}_0$, together with a fixed majority prescription, defines a unique nonalgebraic singleton type r^B over every sufficiently closed model B having that named diagram;*
2. *the prescription at A is r , and the family r^B lifts freely along arrows preserving the names, including across eligible levels;*
3. *the prescription occurs at arbitrarily large eligible single-model levels.*

The possible descriptors obtained this way form a set, with their length bound independent of $\|A\|$.

Proof. Fix a sufficiently large uniform testing arity τ for singleton types, including the bound needed to apply compact placement to those tests. The approximation construction for Theorem 6.10 gives a list $c = (c_i : i < \beta')$ of approximations in A that computes r by majority on all these tests. Its length and all ancillary approximation data can be bounded by one cardinal β , independently of the level. Choose the length larger than the exception bound for the tests; increase β also to include the fixed point data. The prescription is the value attained by all but the permitted number of entries in $D^K(c_i, z)$, for each test z of arity at most τ .

Freely lift the actual r to arbitrarily large eligible models containing A . Query matching over the fixed approximation list shows that these lifts compute exactly the same majority prescription.

They are nonalgebraic by Lemma 10.1. For another sufficiently closed pointed B with the same named diagram, embed B into such a larger instance, respecting the names. This is possible by bounded-diagram homogeneity and mixed placement. Restrict the lifted type to that copy of B . It is nonalgebraic, since its realization is outside the larger parameter model, and query matching makes its bounded answers the prescribed ones.

Any two types having these answers are equal by the singleton comparison bound. Thus the construction is independent of the larger instance and the chosen embedding. A free lift along an arrow preserving the names has the same prescribed answers: the approximation evaluation is transported by matching each query over the named list inside the old base. Uniqueness therefore identifies it with the prescribed receiving type. At A the answers are those of r by the choice of c .

The list lengths have a fixed set bound. Their stabilized diagrams, the fixed testing arity, and the majority rule range over a set of descriptors. Only descriptors realized at arbitrarily high eligible single-model levels are needed; the preceding construction proves that the descriptor extracted from an actual r has this property. The named diagram itself can be witnessed at one fixed eligible level large enough for arity β , by saturation for diagrams from larger levels. Proposition 5.7 therefore applies with a lower threshold depending on β and the original working scheme, rather than on $\|A\|$. In particular, extracting the names from a model of size λ does not require moving above λ in order to use the pointed working scheme at that level. $\square$

Theorem 10.9 (No proper pair with unchanged points at one level). *There is a sufficiently large cardinal $\lambda > \lambda_0$ such that there is no proper strong inclusion $A <_K B$ of pointed models of size λ with $P(A) = P(B)$.*

Proof. Take the set of anchored descriptors from Lemma 10.8. Choose λ at a common level for all finite marked constructions for these descriptors, using Theorem 13.5. Require also all the relevant single-model and pointed bounds, and $\lambda > \lambda_0 + \beta$. This is a choice uniform in the possible descriptor. In particular the choice does not depend on a pair that might later be assumed to exist.

Suppose toward a contradiction that $A <_K B$ is such a pair, and choose $a \in B \setminus A$. Apply Lemma 10.8 to $r = \text{tp}(a/A)$ and name its list c . We work in the pointed scheme also naming this list, with its fixed joint diagram. Write $\mathcal{C}_n$ for its n -fold marked iteration with the root prescription r , as in Theorem 9.3.

Every one-step all-dominated increment of this scheme has no new P -points. Indeed, pointed categoricity carries its base, including the named list c , onto A . Transporting the witness $A <_K B$ back gives an extension with a realization of the required r and unchanged P . Primeness embeds the all-dominated increment into that extension over its base and mark. Since strong maps preserve and reflect P and fix the base, the increment has exactly the base points.

It follows by induction that the P -points in the top of every $\mathcal{C}_n$ object are precisely those in its root. For the induction step, its lower and upper layers are $\mathcal{C}_{n-1}$ objects and hence each has the points of its own root. Projection onto the new axis is a valid one-step marked increment by Theorem 9.3. The two roots therefore have the same points, proving the assertion for the entire object.

We next construct coherent finite systems. For each $n < \omega$ choose an object $J_n \in \mathcal{C}_n$. For every $s \subseteq \{1, \dots, n\}$ choose an isomorphism

$$e_{n,s} : J_{|s|} \longrightarrow P_s J_n$$

such that these maps commute with all further face projections, using the increasing identifications of finite index sets. Here the full face map is the identity. To make these choices, induct first on n

and then on the size of a proper subset s . For a proper subset u of the axes of $J_{|s|}$, let t be the corresponding subset of s . The boundary map for $e_{n,s}$ on this face is $e_{n,t}e_{|s|,u}^{-1}$, with the indicated index identifications. The already chosen maps make these boundary prescriptions compatible. Theorem 9.8 extends them to the desired isomorphism. At dimension zero it is simply pointed categoricity. This constructs every $e_{n,s}$ with the asserted commutation.

For the finite subsets s of any ordinal I , take a tagged copy of the top vertex of $J_{|s|}$. If $s \subseteq t$, use the top component of $e_{|t|,u}$ followed by the inclusion of that face into the top, where u records the positions of s in the increasing list of t . The preceding commutation gives a directed system of strong maps. All its roots are copies of the same actual J_0 , and no vertex has more P -points than that root.

The marks corresponding to different indices of I are distinct in this directed system. It suffices to inspect a two-axis object. Its new mark lies in the root of its upper one-axis layer, whereas the old mark is the nonalgebraic mark of that layer and is outside its root. Face isomorphisms preserve the marks, so the distinction persists under all transition maps.

Take I of cardinality $\mu > \lambda$, where μ is a sufficiently closed single-model pointed level. The ordinary directed colimit in the AEC exists: a coherent diagram of injective maps is realized as a diagram of inclusions in its direct limit, and the directed-union axioms apply. There are μ finite subsets of I , each contributing at most λ elements, and there are μ distinct marks. Thus the colimit N has size exactly μ . Every element of N comes from one finite vertex. Preservation and reflection of P give

$$P(N) = P(J_0), \quad |P(N)| \leq \lambda.$$

The enumeration of its distinguished M_0 retains the fixed stabilized diagram Δ under the strong colimit embeddings.

On the other hand, there is a pointed model of size μ with more than λ points. Start with any pointed model of that size. Successively realize a nonalgebraic free extension of p_0 outside the current model, for λ^+ steps. The receiving models and their unions may all be kept of size μ . The new realizations are distinct and belong to P , so the union has at least λ^+ points. Pointed categoricity at μ would make this model isomorphic to N over M_0 , a contradiction.

Only the stationary prescription r was lifted to higher levels. The assumed witness with unchanged P was used solely at λ . The large colimit used the full AEC directed-union axioms, rather than a same-level closure assertion for a marked scheme. $\square$

Corollary 10.10 (No-growth and rank on the entire tail). *Increase the threshold λ if necessary to include the bounds for the fixed point diagram. In arbitrary cardinalities, if $A \leq_K B$ are pointed models with $\|A\| \geq \lambda$ and $P(A) = P(B)$, then $A = B$. Moreover, for every pointed model M with $\|M\| > \lambda$,*

$$|P(M)| = \text{rk}(P(M)) = \|M\|.$$

The point sets of strong submodels are relatively closed, and relative bases and finite basis supports are available as in Theorem 10.5; no eligibility condition is imposed on these arbitrary cardinalities.

Proof. Suppose that $a \in B \setminus A$ and $P(A) = P(B)$. Construct increasing strong submodels $A_i \leq_K A$ and $B_i \leq_K B$ of size λ , for $i < \omega$, all containing the distinguished copy of M_0 , so that

$$A_i \leq_K B_i, \quad a \in B_i, \quad P(B_i) \subseteq A_{i+1}.$$

For the successor choice, $P(B_i)$ has size at most λ and is contained in A , so the Löwenheim–Skolem axiom places it, together with A_i , in the next A_{i+1} . Choose B_{i+1} to contain $A_{i+1} \cup B_i \cup \{a\}$.

Coherence makes all the required inclusions strong. The two unions $A_\omega \leq_K B_\omega$ have size λ . The point diagram restricts correctly to them, and

$$P(B_\omega) = \bigcup_{i < \omega} P(B_i) \subseteq A_\omega.$$

Thus their point sets are equal, while $a \in B_\omega \setminus A_\omega$. This contradicts Theorem 10.9.

Now let $\|M\| = \kappa > \lambda$. If $|P(M)| < \kappa$, choose a strong submodel containing $M_0 \cup P(M)$ of size at most $|P(M)| + \lambda_0$. Enlarge it inside M , if necessary, to a strong submodel of size at least λ and still less than κ . It has the same P as M , contrary to the assertion just proved. Hence $|P(M)| = \kappa$. If H is a basis of $P(M)$, the closure bound gives

$$\kappa = |P(M)| = |\text{cl}_M(H)| \leq |H| + \lambda_0.$$

Since $\kappa > \lambda \geq \lambda_0$, it follows that $|H| = \kappa$. The final assertions were established in Theorem 10.5 independently of eligibility. $\square$

11 Geometric systems and classification by bases

Throughout this section the distinguished model M_0 , its fixed pointed pattern, and the pregeometry P are those of Proposition 10.6 and Theorem 10.5. All model sizes exceed the threshold of Corollary 10.10. Thus a strong inclusion of pointed models with equal P -sets is equality. At an eligible uniform level μ , all vertices have size μ . A statement about unions of objects at that level concerns chains whose total size is μ , as in the working conditions of Definition 5.2.

We first prove structural and presentation assertions without assuming categoricity of systems. The classification argument then proceeds by the number of axes. Its first assertion extends a root isomorphism and prescribed bijections of the other basis pieces; this gives categoricity for the current shape. That categoricity supplies the working package used to prove the second assertion: prescribed bijections of the basis pieces added to a fixed system extend to an isomorphism over that system. By splitting off the last axis, the second assertion gives the first at the next dimension. We establish the conditional package before carrying out this paired induction.

11.1 Shapes, relative bases, and arrows

Definition 11.1 (Admissible shapes and geometric systems). Let A be a finite linearly ordered set of axes. An admissible family $L \subseteq \mathcal{P}(A)$ of *active labels* is defined recursively. On no axes the family is $\{\emptyset\}$. If k is a new last axis, choose an upward closed subset Q of the previously constructed family L , and put

$$L' = L \cup \{s \cup \{k\} : s \in Q\}.$$

Here upward closed means that $s \in Q$, $s \subseteq t$, and $t \in L$ imply $t \in Q$.

A geometric system of shape L is a family $(X_t : t \subseteq A)$ of pointed models in a common top model, with $X_t \leq_K X_u$ when $t \subseteq u$, satisfying the following condition. There is a partition $(H_s : s \in L)$ of a basis of $P(X_A)$ such that $H_s \subseteq P(X_s)$ and

$$P(X_t) = \text{cl}_{P(X_A)} \left(\bigcup_{s \in L, s \subseteq t} H_s \right) \quad (t \subseteq A). \quad (12)$$

At uniform level μ we require $|H_s| = \mu$ for every $s \in L$. The resulting scheme is denoted $\mathcal{G}_L$. The basis partition witnesses the definition; it is not named in the system structure.

For a vertex t , write

$$D_t(X) = \text{cl}_{P(X_A)} \left(\bigcup_{r \subseteq t} P(X_r) \right).$$

For $t = \emptyset$ this is the closure of the empty set in the pregeometry. An arrow $X \rightarrow Y$ is a commuting family of pointed strong embeddings for which a witnessing partition for X extends to a witnessing partition for Y . Equivalently, after identifying X with its image, each finite tuple from $P(X_t)$ independent over $D_t(X)$ remains independent over $D_t(Y)$.

The equivalence in the last sentence, including its independence from the initially chosen partition, will be proved below. All closures and independence assertions are computed in a common containing pointed model. Theorem 10.5 makes this convention independent of the chosen common strong extension. The notation $\text{cl}_{P(X_A)}$ means the closure cl_{X_A} defined there; its subscript emphasizes the pregeometry in which it is computed. We omit that subscript when the ambient model is clear.

Lemma 11.2 (Combinatorics of shapes). *An admissible family contains $\emptyset$ and is closed under unions. Its restriction to any subset of the axes is admissible, with the inherited order. Consequently every $t \subseteq A$ has a largest active subset*

$$a_L(t) = \bigcup \{s \in L : s \subseteq t\}.$$

In a geometric system, $X_t = X_{a_L(t)}$ whenever t is inactive.

Proof. The union assertion follows by induction on the axes. Unions of old labels are old. If a union involves $s \cup \{k\}$ with $s \in Q$, its old part is an active label containing s , and therefore belongs to Q . This also handles the union of two labels containing k . Restriction to a set of axes replaces L and Q , at every retained step, by their restrictions. The restricted Q is upward closed in the restricted L . Steps on discarded axes make no contribution. This proves the restriction assertion.

The displayed union is an active label, since the family is finite and contains $\emptyset$. The collections of active labels below t and below $a_L(t)$ agree. Equation (12) therefore gives equality of their P -sets. The inclusion $X_{a_L(t)} \leq_K X_t$ and Corollary 10.10 give equality of the models. $\square$

Remark 11.3. Union closure alone would not imply the recursive last-axis description. For example, $\{\emptyset, \{1, 2\}\}$ is union closed, but its two layers on splitting off axis 2 have different active families. We use the recursive definition, including the upward closed set Q , when identifying a system with an arrow between systems on fewer axes.

Lemma 11.4 (Replacement and extension of relative bases). *Suppose X is a geometric system. For each active s , choose any basis B_s of $P(X_s)$ over $D_s(X)$. Then the B_s form a witnessing partition for X .*

A componentwise strong system embedding is an arrow precisely when it preserves the relative independence in Definition 11.1. For an arrow, every witnessing partition of its domain extends to one of its codomain.

Proof. Fix a witnessing partition $(H_s : s \in L)$, and enumerate L in an order extending inclusion. For active s ,

$$D_s(X) = \text{cl} \left(\bigcup_{u \in L, u \subsetneq s} H_u \right).$$

When the piece at s is replaced, previously processed labels not below s are independent, together with the original piece at s , over this lower span: they are parts of a single independent set. Replacing H_s by any basis of the same flat over its lower span preserves both independence and the span of the union. Explicitly, let C be the current pieces below s . Then $\text{cl}(C \cup B_s) = \text{cl}(C \cup H_s) = P(X_s)$, and $C \cup B_s$ is independent. The remaining current pieces were independent over $\text{cl}(C \cup H_s)$, since the full current partition was independent. They remain independent over the equal flat $\text{cl}(C \cup B_s)$. Thus the new full partition is independent and has the same span. Applying this replacement at the finitely many active labels gives the first assertion and Equation (12) for the new partition.

If pieces extend along an embedding, any finite independence calculation in a vertex over its lower span is the same calculation using the corresponding pieces in the larger independent set. Relative independence is therefore preserved.

Conversely, let (B_s) be an arbitrary witnessing partition of the domain, viewed as a subsystem of Y . Preservation implies that B_s is independent over $D_s(Y)$. Extend it to a relative basis $B_s \cup C_s$ of $P(Y_s)$ over $D_s(Y)$. The first assertion, applied to Y , says that these choices assemble into a witnessing partition of Y . This proves both the equivalence and the final assertion. $\square$

Lemma 11.5 (Structural properties and the last-axis description). *Arrows compose and satisfy coherence. Same-level chains of total size the level have unions in $\mathcal{G}_L$, with the common-upper-bound clause also for arrows into a larger eligible level.*

Restriction to any subset of axes preserves systems and arrows. If $L' = L \cup \{s \cup \{k\} : s \in Q\}$, splitting off the last axis identifies a system of shape L' with an arrow $X \rightarrow Y$ in $\mathcal{G}_L$ such that the complementary piece at s has size μ for $s \in Q$ and is empty for $s \notin Q$.

Proof. Composition follows from the relative independence criterion. For coherence, suppose $X \subseteq Y$, and both embed by arrows in Z . Independence over $D_t(X)$ implies independence over $D_t(Z)$, and hence over its subset $D_t(Y)$. Componentwise coherence is the AEC coherence axiom.

For a continuous chain, extend witnessing pieces successively using Lemma 11.4, and take their unions at limit stages. Every finite independent sublist and every finite dependence witness occurs at a stage. The resulting union pieces are independent and span every vertex of the componentwise union. Each contains μ elements from the first stage and has size at most μ , so all pieces still have size μ . For a chain which is not initially continuous, insert its earlier unions: the preceding construction applies recursively, and the common-upper-bound argument in the next paragraph supplies its connecting arrows.

If each stage maps by an arrow into Z , a finite tuple independent over the lower span of the union is independent over the lower span at any stage containing it. Its image is independent over $D_t(Z)$. The componentwise AEC common-upper-bound axiom supplies the strong maps, so the union map is an arrow. This argument also permits a larger level for Z .

On restricting to axes $B \subseteq A$, retain the pieces whose labels are contained in B . They are a basis of $P(X_B)$ and satisfy the required span identities. For a retained vertex all its proper subsets are still retained; its relative independence test is therefore exactly the old test.

Finally let Z have shape L' , and put $X_s = Z_s$ and $Y_s = Z_{s \cup \{k\}}$. A partition for Y is obtained by combining H_s with $H_{s \cup \{k\}}$ when $s \in Q$, and using H_s alone otherwise. These choices extend the pieces of X , and their complements have the asserted sizes. Conversely, extend pieces of X to pieces of Y along such an arrow, give the old pieces labels s and their complements labels $s \cup \{k\}$, and use X and Y as the two layers. The span identities give a system of shape L' . Inactive vertices cause no exception, by Lemma 11.2. $\square$

For later use, preservation of relative independence also preserves finite rank calculations modulo the lower span. For finite tuples a, c in a vertex, the condition that a be independent over $D_t(X) \cup c$ is determined by the ranks, modulo $D_t(X)$, of sublists of a, c . These ranks are determined by which sublists are independent. Thus arrows preserve this condition as well.

11.2 Order presentations and their rank

Proposition 11.6 (Tiled presentations). *Let L be admissible and let μ be sufficiently large for the pointed pregeometry and the no-proper-pair conclusion. The order template E gives a presentation in $\mathcal{G}_L$ as follows. Choose a fixed initial order carrying the whole enumeration of M_0 . For each active label t , choose a tile T_t , preceded by a buffer B_t , and concatenate these finitely many pairs after the initial order, in any fixed order. Every tile and buffer is a dense linear order without endpoints of size μ . In vertex s retain the initial order, all buffers, and just the tiles with labels $t \subseteq s$.*

These presentations have the following properties.

1. *Each active relative piece has rank μ .*
2. *Order embeddings of the corresponding tiles and buffers, fixing the initial pointed data, induce arrows, including between different eligible levels.*
3. *A presentation has an extension by an arrow gaining μ independent complementary coordinates at every active label simultaneously.*
4. *Tiles may be chosen with μ pairwise disjoint convex substrips, each of size μ . Further insertions of such substrips preserve the preceding assertions and the naturalness of the underlying K -diagrams.*

No categoricity assumption about $\mathcal{G}_L$ is used here.

Proof. Let I_s be the indicated suborder and set $X_s = E(I_s)$, using the literal embeddings supplied by Theorem 2.6. All I_s have size μ , since they contain all buffers. The models have size μ and are strongly included according to their labels. The initial order was chosen to contain the supports of the fixed enumeration of M_0 . Its unchanged term pattern carries the same pointed model in every vertex, and coherence gives the required strong inclusion of M_0 .

Choose a relative basis H_s of $P(X_s)$ over the union of its proper lower P -sets for every active s . We first prove that the indexed family of these choices is independent; this also proves that the pieces are disjoint. Suppose there is a finite circuit of labeled entries, allowing a two-entry equality dependence when two differently labeled entries coincide. Choose a maximal label s represented among its entries. Retain the finite term supports of all entries with label s , and indeed retain every support position in a tile available at s . In each unavailable tile replace all support positions by distinct positions in its preceding buffer, beyond all the finitely many retained positions in that buffer. Such positions exist because the buffer has no greatest point and is dense. No retained position lies in that unavailable tile. Consequently this replacement preserves the order and equality pattern of all circuit supports and fixes the whole initial pointed order.

The labels and this order pattern preserve the pointed finite geometry comparisons. One can formulate this either using the natural K -diagrams or by taking the term model on the fixed initial order and the finite supports and applying its two pointed strong embeddings into the original presentation. In the latter formulation Theorem 10.5 preserves and reflects the independence relation.

An entry with label $t \neq s$ in the circuit has no label properly containing s , by maximality. Thus $s \not\subseteq t$. After the replacement its term uses only common buffers, initial data, and tiles with labels contained in $s \cap t$. Its image belongs to $P(X_{s \cap t}) \subseteq D_s(X)$. The entries with label s have not moved. The preserved circuit makes a sublist of H_s dependent over $D_s(X)$, a contradiction. The union is therefore independent. It spans every vertex by induction on the labels: its relative basis spans that vertex over lower vertices, which have already been spanned. If a vertex is inactive, the order retained there is exactly the order retained at its largest active subset. This proves the decomposition without first assuming the system axioms.

We give the fresh-skeleton observation needed for the rank assertion. If $J \subseteq I$ and $i \in I \setminus J$, the skeleton entry at i does not belong to $E(J)$. Otherwise it equals a labeled term on a finite tuple of positions from J . Enlarge the order by a second position i' having the same order pattern as i relative to that finite tuple. Literal pattern preservation would make both distinguished skeleton entries equal to that term. This contradicts their required distinctness. The enlargement need only duplicate the cut relative to the finite tuple, not the cut relative to all of J .

For an active t , omit T_t from I_t . The resulting order still has size μ , contains the pointed data and all proper lower vertices, and presents a strong submodel. Enumerate μ points of T_t and adjoin them one at a time. The fresh-skeleton observation makes each successor inclusion proper. All these models have size μ , because the buffers were retained. By Corollary 10.10, each successor adds a point of P . Choose one such point at each successor. Relative closedness of P in strong submodels makes these choices independent over the initial omitted-tile model. There are μ choices, so the rank over the proper lower vertices is at least μ . It is at most μ by the size of X_t . This proves the first assertion, including the root case $t = \emptyset$.

Next consider componentwise order embeddings into enlarged tiles and buffers, identifying the source with its image. Suppose a finite tuple a in $P(X_s)$, independent over $D_s(X)$, becomes dependent over $D_s(Y)$. Dependence has a finite witness consisting of points from proper lower vertices of Y . Fix the finitely many old positions supporting a . Move every other position needed for the witness into its corresponding old tile or buffer, preserving its order relative to the fixed positions. Density and absence of endpoints of the old order suffice for each finite pattern; it is not necessary that the old order be dense in the enlarged order. The initial pointed order remains fixed. Each witness remains in its designated proper lower vertex, now in X , and its finite dependence diagram is preserved. This contradicts independence of a over $D_s(X)$. Lemma 11.4 therefore gives an arrow. The proof did not require equal source and target sizes.

To obtain simultaneous gains, enlarge all tiles with μ fresh points and, if desired, enlarge the buffers. For a particular active t , begin the strict-growth argument over the order containing the old t -tile, all enlarged proper-lower tiles, all enlarged buffers, and the initial data. This base contains X_t and all proper lower vertices of Y . Adjoining the fresh points of the t -tile yields μ independent points over that larger base. Hence the complementary rank at t is μ . The preceding arrow assertion allows any old pieces to extend, and Lemma 11.4 combines the relative choices into one partition. This proves the simultaneous assertion.

Finally a lexicographic sum, indexed by a linear order of size μ , of dense endpoint-free orders of size μ is still dense and endpoint-free and has size μ . Its summands are the desired convex substrips. In particular, for a presentation indexed by a well-order H , one can use $H \times \mathbb{Q}$ and reserve intervals of H of order type and cardinality μ . Taking H of order type $\mu \cdot \mu$ supplies μ disjoint such intervals. Additional summands or strips do not change any of the preceding arguments. Underlying K -diagrams remain natural because the entire system was obtained by designated term subcollections of the single template E . $\square$

11.3 Obtaining the working package from categoricity

Proposition 11.7 (Conditional working package). *Fix an admissible shape L . Suppose that $\mathcal{G}_L$ is categorical at every level in an unbounded class of eligible levels closed sufficiently for the preceding constructions. Assume also the eligible-subclass hypothesis of Convention 5.3. On an unbounded further class of sufficiently closed levels, it satisfies the full working package of Definition 5.2, with its actual system isomorphisms and its arrows from Definition 11.1. The resulting diagrams, saturation, and calculus include mixed arrows between eligible levels, and the output class retains the eligible-subclass hypothesis.*

The closure requirements are uniform in the finite shape and the fixed pointed template. They use only the smaller and larger input levels specified by the general working-package construction.

Proof. The structural conditions are Lemma 11.5. Proposition 11.6 supplies term presentations and mixed presentation arrows before categoricity. By the assumed categoricity, every object at an input level is isomorphic to its presentation.

We check the extra relative-basis condition throughout the reversal argument. In each tile and buffer use the order $H(\delta)$ from Theorem 3.5. For a tuple of length θ , record its term labels, the fixed initial positions it uses, and, separately in each of the finitely many variable orders, the order and realized-cut record of its support positions. If $\theta < \text{cf}(\delta)$, equal records are carried to one another by order automorphisms of the respective tiles and buffers. Their product fixes the initial data and induces an automorphism of the whole system. The set of possible records has a bound depending only on θ , the finite shape, and the template. Hence the system orbits at that arity have a uniform set bound.

Number these orbits and record the arrows between them, together with the partial map from support records to orbit names. This is again a bounded palette. If reversal fails unboundedly often, choose repeated profiles at eligible levels $a < \mu$ exactly as in Theorem 3.11, with $a^\theta \leq \mu$ and the required large cofinalities for the orders. Apply the ordinal-position shrinking of Proposition 3.9 simultaneously to the finitely many tiles and buffers. Given at most a parameters in the large system, the resulting small presentation contains them, has size a in every tile and buffer, and computes the support records of all its θ -tuples correctly. Its inclusion is an actual arrow by Proposition 11.6. Thus it is an aligned small geometric system, not merely a componentwise subsystem.

For completeness, the contradiction uses two orbit names p, r with $p \rightarrow r$ and $r \not\rightarrow p$. Repeatedly choose aligned small copies, each containing the previous one, and process their at most a^θ tuples in the large level. Whenever a tuple can still be sent to p , send it onward to r . From that point it cannot reach p . Coherence makes the successive small copies a chain of arrows. After θ^+ rounds every θ -tuple of their union occurred at a proper round. Both unions exist by Lemma 11.5: the small union has total size a , and the large construction has total size μ . Alignment says that none of the tuples of the small union can reach p . Categoricity of that union at a says that its orbit p is realized, a contradiction. This proves eventual reversal at each fixed arity.

Mixed Löwenheim–Skolem closure also has the required form. In a presentation, close the supports of a prescribed small set under finitely many intermediate and exterior positions in every tile and buffer, repeat countably many times, and pad each order to the chosen smaller eligible size. The resulting dense endpoint-free suborders give a small presentation containing the set. The initial pointed order is retained in full. Its inclusion is an arrow by Proposition 11.6. Transport through a presentation of an arbitrary large object gives mixed containment; coherence supplies comparisons between such subobjects.

We now apply the first route of Proposition 5.8. Its structural hypothesis is Lemma 11.5;

its presentation and mixed downward-closure hypotheses are Proposition 11.6 and the preceding paragraph. Categoricity is the input assumption, and the required arity-by-arity reversal and uniform orbit bounds were proved above. Thus the proposition applies to the actual system arrows, including their relative-basis condition.

In particular, the canonical comparison at a smaller input level is computed in a small geometric subobject. Two such sections are contained in a common small section by mixed downward closure, and coherence gives its arrows, so the comparison is well defined. The projections between comparisons at two input levels are onto because the smaller system is a tiled presentation and extends by the mixed order arrows of Proposition 11.6. The bounded palette therefore stabilizes, exactly as in the first route of Proposition 5.8.

Choose receiving levels which remain eligible, are strictly closed under the resulting arity bounds, and have sufficiently large smaller input levels below them for every bounded request. These are the closure refinements described in Section 5.2, supplied simultaneously in Theorem 13.5; closure under the next input level is not being used to infer eligibility by itself. At such a level, countable alternating matching in smaller input-level sections gives actual system isomorphisms on equal bounded diagrams, and realizes possible bounded joint diagrams over their given sub tuples. Thus diagrams, homogeneity, and saturation have the system interpretation required here, rather than merely their componentwise K interpretation. Order embeddings and order amalgamation make the diagrams natural on term tuples, so Proposition 5.9 also applies.

It remains to check the additional exhaustion condition, which Proposition 5.8 explicitly leaves as a separate obligation. A finite tuple independent modulo the proper lower vertices stays so when its target diagram agrees with a tuple placed along an actual presentation arrow: the comparison is realized by a system isomorphism, and such an isomorphism preserves relative independence. Every tuple in an exhaustive sequence is handled at a stage containing it. The finite relative-independence criterion in Lemma 11.4 then supplies the additional arrow condition. Componentwise strongness is the conclusion of the same proposition's exhaustion argument. This completes the working-package verification.

All refinements used here are the bounds for the original reversal, alignment, diagram stabilization, and saturation arguments, applied to finitely many orders and the fixed initial data. None introduces an amalgamation assumption on the original AEC; the amalgamation used below is the conclusion of the working package and Theorem 6.6. $\square$

11.4 Fresh coordinates and the two induction assertions

Fix pieces $(H_s^X : s \in L)$ in a system X . For an arrow $X \rightarrow Y$, extend them to pieces

$$H_s^Y = H_s^X \cup C_s^Y.$$

The sets C_s^Y are its *complementary pieces*. Their sizes may be any cardinals at most μ , including zero, even though all the total pieces have size μ .

Definition 11.8 (Joint freshness). A list of coordinates in an extension Y of X , with individual active labels, is jointly fresh over X if it is a sublist of the complementary pieces for some such extension of the pieces of X . At a single label s , the corresponding condition is independence over

$$\text{cl}(P(X_s) \cup D_s(Y)).$$

For several labels these relative conditions assemble into joint freshness by Lemma 11.4. Coordinates are regarded as occurrences in their designated vertex sorts.

Freshness is preserved by arrows of extensions. Indeed every failure or success of finite independence over the displayed flat is determined by finite rank calculations using parameters from $P(X_s)$ and the lower span. The observation following Lemma 11.5 preserves those calculations. This also shows that freshness is an invariant of the type over X once the working package is available.

Theorem 11.9 (Classification with prescribed basis maps). *For each finite admissible shape L , at all sufficiently closed uniform levels μ , the following assertions hold.*

1. *If $X, Y \in \mathcal{G}_L$, an isomorphism $e_\emptyset : X_\emptyset \simeq Y_\emptyset$ and arbitrary bijections between the chosen nonroot pieces of X and Y extend to a system isomorphism $X \simeq Y$. In particular $\mathcal{G}_L$ is categorical at these levels.*
2. *Suppose $X \rightarrow Y$ and $X \rightarrow Z$ are arrows in $\mathcal{G}_L$. Choose pieces of X , extend them to pieces of Y and Z , and prescribe a bijection $C_s^Y \simeq C_s^Z$ for every active label s . Then there is a system isomorphism $Y \simeq_X Z$ realizing every prescribed bijection. The complementary cardinalities may be less than μ , and zero is allowed.*

The level requirements are uniform over the finitely many shapes at any fixed number of axes.

Proof. We prove the two assertions in order, by the number of axes. On no axes, the first assertion is exactly the assertion that a prescribed isomorphism of the root models is an isomorphism of the systems. Existence of such a root isomorphism, hence categoricity of the zero-axis scheme, is Proposition 10.6.

Suppose the second assertion is known for all shapes on fewer axes. For a shape with last-axis description L' , apply the first assertion for L to its lower layers and then the second assertion for L to its upper layers. The pieces added at labels in Q are exactly the pieces bearing labels $s \cup \{k\}$ in L' . Prescribe their given bijections, and prescribe the empty maps at the other labels. This produces the desired isomorphism of the two-layer systems by Lemma 11.5. If an auxiliary root basis has been chosen, replace the target root basis by its image under the prescribed root isomorphism; Lemma 11.4 permits that replacement without changing any prescribed nonroot piece. Thus the first assertion at the next number of axes follows from the preceding second assertion.

It remains to prove the second assertion for a fixed shape L whose first assertion has now been established. Refine the levels using Proposition 11.7. All the one-dimensional calculus and dominated-hull results are then available for $\mathcal{G}_L$. The rest of the proof is given in four claims.

Claim 11.10 (Uniqueness of finite fresh types). *For a fixed base X and a finite ordered list of active labels, all jointly fresh tuples with those labels have the same type over X .*

Proof. At zero axes this is the base-change calculation of Proposition 10.7. Types in the pointed scheme over a whole pointed base are the underlying K -types: an amalgam or type identification over that base already fixes the point. All objects at the level have full P -rank by Corollary 10.10.

At a positive number of axes suppose two finite jointly fresh tuples b^0, b^1 have different types over X . Bounded testing gives a bounded tuple f from X such that

$$D(f, b^0) \neq D(f, b^1).$$

Choose a cardinal $\chi < \mu$ large enough for the orbit bound at arity $|f|$, for the finite tuple lengths, and for the template labels, with $\chi^{|f| + \aleph_0} = \chi$. Let ϵ be the least cardinal for which $2^\epsilon > \chi$. Then $\epsilon \leq \chi$, there are at most χ nodes at all proper levels of the binary tree of height ϵ , and $2^\epsilon \leq 2^\chi < \mu$.

Build a tree of extensions of X at level μ . At a node V , the arrow from X preserves the diagram of f . Bounded-tuple homogeneity therefore supplies an isomorphism $X \simeq V$ fixing f . Transport the two witness extensions along this isomorphism to form the children of V . Their marked tuples are fresh over V , and their diagrams with f have the two specified different values. At limit nodes take unions along the branch. Extend chosen basis pieces at every step. Hence along every branch all marked coordinates, at their respective labels, lie in complementary pieces over the initial X . The total size along a branch is μ , so all unions are allowed. Notice that the return isomorphism fixing f uses diagram homogeneity, not categoricity alone.

At every leaf make a further rich extension by Proposition 11.6, so that the root has complementary rank μ over the original $X_\emptyset$. There is a common system U with the same property: present any system and make a rich extension, transporting it over X using categoricity. For each level $\eta < \epsilon$ reserve in U a finite tuple c_η of coordinates with the same labels as the marked tuple at that level. Choose all of these coordinates from complementary pieces over X . Their total number is at most χ .

For a leaf W , its root marked coordinates and those in U are independent over $P(X_\emptyset)$. Both full root complements have size μ , and the prescribed match uses fewer than μ coordinates. Extend that match to a bijection of root complements. The already proved second assertion for zero axes gives

$$W_\emptyset \simeq_{X_\emptyset} U_\emptyset$$

carrying all root marked coordinates correctly. At nonroot labels extend the prescribed matches of marked coordinates to bijections of the full pieces, whose sizes are μ . The first assertion for the present shape extends these maps and the root isomorphism to an isomorphism $W \simeq U$. The old nonroot pieces of X need not be fixed in this step.

Let f_W be the image of f in U . Two different branches first separate at a level η ; preservation of diagrams along their arrows gives different values of $D(f_W, c_\eta)$ after the two leaf isomorphisms. Thus the 2^ϵ images of f lie in different orbits over the fixed set of at most χ reserved coordinates. The parameter-orbit bound of Lemma 6.1, applied to the working scheme, is χ at these arities, a contradiction. $\square$

For each active s , let p_s^X denote the unique fresh singleton type given by the claim. Such types exist by the rich extensions of Proposition 11.6 and categoricity. Their realizations lie outside the base vertex, so they are nonalgebraic.

Claim 11.11 (Free lifts preserve freshness). *The free lift of p_s^X along an arrow $X \rightarrow V$ is fresh over V . Finite jointly fresh tuples have the product of their singleton types. Bounded jointly fresh lists have a unique whole type specified by their labels and admit all-dominated hulls.*

Proof. Let a realize the free lift in an extension Y of V . If a is not fresh, finite character gives a finite tuple $c \subseteq P(V_s)$ with

$$a \in \text{cl}(c \cup D_s(Y)).$$

This condition is invariant under system isomorphisms and is therefore determined by the bounded diagram of (a, c) in Y . Its witnesses in lower vertices need not lie in V ; the isomorphism-invariant condition itself is what the diagram records.

Put $m = |c| + 1$, and choose an m -tuple b jointly fresh over X , all at label s . Every singleton marginal is p_s^X by Claim 11.10. Take a sufficiently long bounded family of approximations, inside X , for $\text{tp}(b/X)$, as in Lemma 6.9. By the support-copy construction in that Lemma, every approximating tuple has the same diagram as b without parameters, and is consequently

independent modulo $D_s(X)$. In the free lift to V , each coordinate has the same marginal as a . For each coordinate, the majority estimate gives the diagram of (a, c) outside a uniformly bounded set of approximation indices. Choose the family longer than the union of these finitely many exceptional sets. One approximating tuple $(a_1, \dots, a_m)$ is independent modulo $D_s(X)$ and has, coordinate by coordinate, the required diagram with c in V .

Since $X \rightarrow V$ is an arrow, this tuple remains independent modulo $D_s(V)$. The comparison just obtained is an equality of $\mathcal{G}_L$ -diagrams,

$$D_{\mathcal{G}_L}(a, c; Y) = D_{\mathcal{G}_L}(a_i, c; V).$$

The established working package therefore gives a system isomorphism $F_i : Y \cong V$ taking a to a_i and fixing the displayed tuple c . It carries every lower vertex onto its counterpart and preserves the pointed closure, so $F_i[D_s(Y)] = D_s(V)$. Transporting the dependence witness gives $a_i \in \text{cl}(c \cup D_s(V))$ for every i . The latter flat has relative rank at most $|c|$, whereas the tuple has relative rank $m > |c|$. This contradiction proves that the free lift is fresh.

Realize a finite product successively by free lifts along a chain of extensions. The assertion just proved puts each new coordinate in a complementary piece over the preceding system. Extending pieces along the chain makes the whole tuple jointly fresh over X . Claim 11.10 then says that every jointly fresh finite tuple has this product type. A bounded jointly fresh list is consequently product-independent in the sense of Theorem 7.13. That theorem gives its unique whole type and its witnessing free chains; its all-dominated-hull conclusion gives existence over the list. $\square$

We now know the types of fresh coordinate lists and have all-dominated hulls over them. To extend a partially prescribed basis map, we also need to cover an arbitrary bounded tuple using only boundedly many new coordinates. The next claim provides this size control, including when the target basis has already been chosen.

Claim 11.12 (Bounded rank cost and chosen coordinates). *Let $X \rightarrow Y$ be an arrow. A finite tuple d of Y is contained in a subextension $X \rightarrow U \rightarrow Y$ whose complementary pieces have finite sizes. For an arbitrary bounded tuple d , their sizes can be bounded by $\max(|d|, \aleph_0)$.*

If specified pieces of Y extend those of X , the subextension containing a bounded d can be chosen with all its complementary pieces subsets of the specified complementary pieces of Y , still of bounded total size.

Proof. For finite d , take an all-dominated object over $X + d$ and embed it into Y by Proposition 7.5. Call its image U . Suppose one complementary piece of U is infinite. Choose a sequence $(a_i : i < \omega)$ from that piece. It is jointly fresh and hence product-independent over X . The witnessing-chain assertion of Theorem 7.13 retains this configuration in a chain

$$X = V_0 \longrightarrow V_1 \longrightarrow \dots \longrightarrow V_\omega,$$

where a_i is free over V_i from X and belongs to V_{i+1} . We may construct this chain inside the actual U . At each stage, finite domination makes the actual next coordinate a_i free over V_i from X , since its products with the preceding coordinates remain those of the original list. Choose an all-dominated increment over $V_i + a_i$ and use primeness to embed it into U , fixing this base and this coordinate. At limits take unions inside U , using the common-upper-bound clause. Thus $V_\omega \leq U$, and the tuple d and every a_i retain their actual positions in U . Equivalently, the witnessing-chain amalgamation may retain all of these data in a common extension.

The tuple d is finite, so Proposition 6.13 at the limit of cofinality ω makes the actual type $\text{tp}(d/V_\omega; U)$ free from some V_n . Hence d and a_n have product over V_n . The base-change rule of

Proposition 6.12, together with the freeness of a_n over V_n from X , gives that a_n and d have product over X . Domination of U over $X + d$ now makes $\text{tp}(a_n/U)$ free from X . This is impossible by Claim 11.11: the free lift of its fresh singleton marginal must be fresh over U , whereas $a_n \in P(U_s)$. Therefore every complementary piece of U is finite.

For a bounded list of cardinality τ , include its entries successively by this finite construction, with the preceding subextension as base, and take unions at limit stages. There are finitely many labels. At most τ finite increments therefore add at most $\kappa = \max(\tau, \aleph_0) < \mu$ coordinates in total. All objects continue to have size μ , and their unions have total size μ . This proves the bounded version.

For the final assertion fix the target partition $H_s^X \dot{\cup} C_s^Y$. Starting with a bounded-rank subextension containing d , repeat the following construction countably many times. Choose relative bases for the current subextension over X . Each gained basis element in vertex t has a finite support in the fixed target basis, consisting of old coordinates and coordinates of C_s^Y with $s \subseteq t$. Request all the latter supporting coordinates in their own vertices, and include them in the next subextension using the bounded construction just proved. At every step at most κ coordinates are requested and the total new rank remains at most κ .

In the union let C_s consist of the requested coordinates of label s . These sets are independent, together with the pieces of X , because they are subsets of the target partition. They span every union vertex over its old pieces: an element occurs at a stage, is spanned there by the gained pieces and the old pieces, and the finite target supports of those gained pieces were requested at the next stage. Conversely the closed P -set of each union vertex contains all the indicated coordinates below its label. Thus the span identities hold in both directions. The C_s are precisely complementary pieces of the union, have total size at most κ , and are subsets of the prescribed C_s^Y . $\square$

Claim 11.13 (Primeness over bounded complementary pieces). *If $X \rightarrow U$ has a bounded list b enumerating all its complementary pieces, then U is all-dominated over $X + b$. It embeds over X into any extension realizing a prescribed matching of these coordinates.*

Proof. The list b is product-independent by Claim 11.11. Take an all-dominated object over $X + b$ and embed it into U over these data. Its image V contains in each vertex the old P -set and all the complementary coordinates with labels below that vertex. Relative closedness and the span identity imply $P(V_t) = P(U_t)$ for every t . The no-proper-pair conclusion then gives $V_t = U_t$ for every vertex. Thus $V = U$, proving the domination assertion.

A prescribed matching of b into another extension gives the same whole type over X , by Claim 11.11. Primeness of the all-dominated object therefore gives the desired arrow fixing X and realizing that matching. $\square$

We finish the second assertion of the theorem. Identify the two bases with X , and keep the prescribed bijections of all complementary pieces. Choose countable exhaustions of the two targets by bounded lists; these exist because the eligible level μ has cofinality ω and there are only finitely many vertex sorts. Construct alternating isomorphisms between subextensions of the two targets. Initially use id_X . Maintain the invariant that the complementary pieces in each current subextension are subsets of the specified target pieces, and that the current isomorphism agrees with their prescribed bijections.

At a forth step, apply Claim 11.12 over the current subextension on the first side to include the next bounded list and to use only remaining specified coordinates. Its new complementary list is bounded. The corresponding coordinates on the second side are jointly fresh over its

current subextension. Claim 11.13, transported by the current isomorphism, embeds the enlarged subextension into the second target over that isomorphism and realizes their prescribed matching. Its image is the next subextension there. The span identities show that its complementary pieces are exactly the specified images of the new pieces, so the invariant is preserved. A back step applies the same construction with the roles interchanged.

The union of these countably many coherent isomorphisms covers both targets, since every list in both exhaustions is eventually included. It is therefore a system isomorphism over X . It realizes all prescribed basis bijections: every basis element is eventually in the domain, and the invariant determines its image. This proves the second assertion, with no lower bound on complementary cardinalities.

The induction is complete. At each fixed finite number of axes there are only finitely many active-label families. Taking the maximum of their finitely many bounds gives the stated uniformity. Simultaneous closure through all finite numbers of axes is addressed separately in Theorem 13.5. $\square$

12 Exact isomorphisms of geometric systems

We continue with the pointed pregeometry of Theorem 10.5, above the threshold of Corollary 10.10. A shape and its active labels have the meaning of Definition 11.1. Unless otherwise stated, every active basis piece in this section has cardinality equal to the uniform level μ . The bases remain auxiliary choices.

Definition 12.1 (A prescribed boundary isomorphism). Let a be a finite ordered set of axes, and let $\mathcal{X}^- = (X_s^- : s \subseteq a)$ and $\mathcal{X}^+ = (X_s^+ : s \subseteq a)$ be systems of the same shape. A prescribed boundary isomorphism is a family

$$h_s : X_s^- \longrightarrow X_s^+ \quad (s \subsetneq a)$$

of pointed isomorphisms such that $h_t \upharpoonright X_s^- = h_s$ whenever $s \subseteq t \subsetneq a$. This condition refers to the displayed inclusions. It does not add an assumption about intersections of two incomparable vertices inside the top.

A list on the boundary can consequently be specified by assigning a proper face to each of its occurrences. Before consistency on actual intersections has been proved, its prescribed image means the list obtained by applying the corresponding h_s to each occurrence. In particular, the assertion that two occurrences name the same element is one of the atomic comparisons that must be preserved.

Theorem 12.2 (Exact geometric boundary lifting). *For every finite allowed shape, at all sufficiently closed uniform levels, every prescribed boundary isomorphism between two systems of that shape extends to a pointed isomorphism of their tops. The top isomorphism, together with its prescribed restrictions, is an isomorphism of the systems.*

The refinements of the eligible levels needed for a fixed shape use only the one-dimensional bounds for its working scheme, the bounds for its tiling layouts, and the corresponding assertions for shapes on fewer axes. In particular these refinements have the uniformity specified in Section 13.

We prove the theorem by induction on the number of axes. At zero axes it is pointed categoricity. The induction at a positive number of axes has two parts: preserving the joint diagrams of the entire boundary, and realizing successive extensions of a partial match inside the given target. The first part supplies, among other things, the missing compatibility on actual intersections.

Lemma 12.3 (Joint boundary comparisons). *Assume exact geometric boundary lifting for all allowed shapes on fewer than n axes. At a sufficiently closed uniform level, a prescribed boundary isomorphism between two n -axis systems preserves the bounded K -diagram of every occurrence-tagged boundary list, computed in the respective top models.*

Proof. If the top label is inactive, let u be the largest active label. It is a proper subset of the axis set, and the top vertex is X_u . Every vertex is the vertex at its largest active lower label, which is contained in u . Thus the map at u already gives all the comparisons in question. We may assume the top label is active.

Write the axis set as $a \cup \{k\}$, with k last and $|a| = n - 1$. Splitting off k writes the two systems as

$$X^- \longrightarrow Y^-, \quad X^+ \longrightarrow Y^+$$

in the scheme $\mathcal{G}_L$ on a . Its categoricity is supplied by Theorem 11.9; after refining the levels, Proposition 11.7 supplies its working package. The arrows add rank μ at the labels in a fixed upward closed $Q \subseteq L$ and add no rank at the other labels. Call such a step a step of shape Q . The boundary prescription gives a full system isomorphism $e : X^- \simeq X^+$ and compatible isomorphisms

$$h_s : Y_s^- \simeq Y_s^+ \quad (s \subsetneq a)$$

whose restrictions to X_s^- agree with e .

If joint comparison fails, choose bounded lists f from X^- and d^- from the proper vertices of Y^- witnessing failure, and put $d^+ = h(d^-)$. All occurrences from the lower layer may be included in f ; all the remaining occurrences lie in these upper proper vertices. The comparisons of

$$(f, d^-) \text{ in } Y^- \quad \text{and} \quad (e(f), d^+) \text{ in } Y^+ \tag{13}$$

differ. The notation in (13) includes the indicated vertex occurrences. If there is no upper test d^- , the base isomorphism and the strong arrows already give equality, so this case cannot witness failure.

A system isomorphism matching these lists would in particular be a K -isomorphism of the tops matching them. Hence the working $\mathcal{G}_L$ -diagrams of the lists in (13) also differ. Choose an infinite cardinal $\sigma < \mu$ bounding the lengths of the lists. Enlarge a cardinal $\chi < \mu$ so that $\chi^\sigma = \chi$, it bounds the relevant labels and parameter-orbit constants, so that Lemma 6.1 gives at most χ orbits of σ -lists over χ parameters. Choose μ closed far enough that all the following tree costs are below it. Let ϵ be the least infinite cardinal with $2^\epsilon > \chi$. There are at most χ nodes below height ϵ , and $2^\epsilon < \mu$ leaves.

We construct a binary tree of branch worlds C_η in $\mathcal{G}_L$, for $\eta \in {}^{\leq \epsilon} 2$, and a single increasing reference chain $(R_i : i \leq \epsilon)$ in that scheme. All these objects have level μ . A branch world carries a copy f_η of f . For every node η of length i the invariant also includes a prescribed isomorphism

$$b_\eta : \partial C_\eta \simeq \partial R_i. \tag{14}$$

Along a branch, these boundary maps commute with the inclusions into later branch and reference worlds. At each splitting node, the two new copies of its upper test d have the same image in the next reference boundary. Those images will be retained at all later stages.

Start with $C_\emptyset = R_0 = X^-$ and the identity boundary map. Suppose the objects at level i have been constructed. Choose a reference step $R_i \rightarrow R_{i+1}$ of shape Q . Such a step exists: categoricity of $\mathcal{G}_L$ identifies R_i with X^- , and we transport the given extension $X^- \rightarrow Y^-$ along that identification.

For each $\eta \in {}^i2$, an isomorphism $X^- \simeq C_\eta$ can be chosen to take f to f_η . Indeed arrows along the branch preserve its bounded diagram, and equality of bounded diagrams in the working scheme is implemented by an isomorphism. Transport the minus and plus extensions along this return isomorphism, using e on the plus base. They give children $C_{\eta \smallfrown 0}$ and $C_{\eta \smallfrown 1}$ over C_η , with opposite comparisons to f_η and with the given identification between their proper upper vertices. On their lower proper vertices that identification is the identity of the current branch world.

It remains to extend (14) to the children; merely choosing arbitrary isomorphisms of the new worlds would not suffice. First do this for the plus child. Process all proper subsets $s \subsetneq a$ in increasing cardinality. Consider the restriction of the plus step to the axes $s \cup \{k\}$ and the corresponding restriction of $R_i \rightarrow R_{i+1}$. On the base vertex, the prescribed map is $(b_\eta)_s$. On every other proper vertex it is either a restriction of that map or one of the upper maps already constructed at a proper subset of s . These maps commute on all indicated subfaces by construction. The restriction has at most $n - 1$ axes, so the induction hypothesis lifts this boundary prescription onto its upper vertex. This supplies the map at s , whether that vertex is active or repeated from a smaller active one. The resulting family is the required plus boundary map. Compose it with the given minus-to-plus boundary identification to obtain the minus boundary map. Both extend b_η , and the two copies of the test have equal reference images.

At a limit $i \leq \epsilon$, take unions along each branch and in the reference chain. The lower-dimensional working scheme has the chain and common-upper-bound properties, and these unions have size μ . The coherent boundary maps have unions which are onto the reference boundary: each previous reference vertex was already the image of the corresponding branch vertex. Thus the invariant persists. No assertion that a limit inclusion is itself a step of shape Q is needed. At its next successor stage a fresh step of that shape is available over the union by categoricity.

At a leaf, apply the induction hypothesis once more, this time to the $(n - 1)$ -axis worlds C_η and R_ϵ , with the boundary prescription b_η . Obtain a full isomorphism $j_\eta : C_\eta \simeq R_\epsilon$. Let A be the list of all reference images of the splitting tests. Its length is at most χ , since there were at most χ splitting nodes and each test had length at most σ .

For two different leaves, use their first splitting node. The associated test has the same image in A on both branches, whereas its comparison with the images of f is opposite. Arrow preservation of diagrams carries this distinction to the leaves, and j_η preserves it in R_ϵ . Consequently the 2^ϵ lists $j_\eta(f_\eta)$ lie in different orbits over A . This contradicts the bound χ . Joint boundary comparison follows. $\square$

Remark 12.4. The proof just given applies to equalities between occurrences from incomparable faces. It therefore proves that the prescribed maps have a well-defined injective union on the actual union of the boundary vertices. Applying the same observation to the inverse family proves that this union is a bijection between the two boundary unions. Equality of actual intersections was not an input to the argument.

The boundary prescription now respects actual intersections and all bounded joint diagrams. The remaining task is to extend a bounded match inside the prescribed target. We first establish testing for its tiled boundary layout, then use a strip avoided by all boundary supports to realize the resulting comparison without enlarging that target.

Lemma 12.5 (Testing on rationally tiled parameter orders). *Fix the finite-support order template E and its natural stabilized K -diagrams. Fix a nonempty set-sized initial order J_0 , finitely many tile labels $i < m$, and a finite family $\mathcal{A}$ of subsets of m . These are fixed descriptor data. For*

well-orders H_i , put

$$J = J_0 + \sum_{i < m} (H_i \times \mathbb{Q}),$$

with lexicographic order on each product. For $A \in \mathcal{A}$, let J_A consist of J_0 and the tiles whose labels belong to A . Let B be the union of the designated term sets $E(J_A)$, with the term representations inherited from E . Occurrences may retain their designated-part tags.

There is a monotone cardinal bound $g_A(\sigma, \tau)$, depending also on E and J_0 , with the following property whenever $|J| = \mu$ and μ is a sufficiently closed level above the input cardinals, $|J_0|$, and this bound. Let c be a fixed tuple in $E(J)$ of length at most τ , with fixed term representations. Two placements of tuples of length at most σ in arbitrary order enlargements of J have the same comparison over B together with the fixed tuple c if they agree on every parameter query from B of size at most $g_A(\sigma, \tau)$, with c retained as fixed data. Here σ and τ may be increased to infinite cardinals. The bound is independent of the horizontal order lengths and of μ , and is uniform over a fixed set of descriptors of bounded hereditary size. The candidate placements need not be terms of a horizontal reindexing of the tiled template.

Proof. Amalgamate the two supporting order enlargements over J and discard all added positions not supporting either candidate. Let T be the union of the supporting positions of the two candidates and the chosen old supports of c . Put

$$\zeta = \max(\sigma, \tau, |J_0|, |E|, \aleph_0).$$

Thus $|T| \leq \zeta$. A fiber means the old convex copy $\{h\} \times \mathbb{Q}$ at a horizontal position in some H_i . Call an old fiber exceptional if it contains a position of T , or if some added position of $T \setminus J$ has old fiber positions strictly below and strictly above it. Each position of T hits or splits at most one old fiber: two distinct fibers are disjoint convex suborders. Hence at most ζ fibers are exceptional.

Freeze the entire initial order J_0 , every rational position of every exceptional fiber, and all positions of T . The resulting ordered set F has cardinality at most ζ , since every fiber is countable. Also freeze the following record: which positions of F are old, the tile label and rational coordinate of each old position in an exceptional fiber, the exceptional-fiber grouping, the labels and supporting lists of both candidates and c , and all order comparisons and equalities among these positions. The fixed copy of J_0 is retained literally. Up to renaming the other positions on a set of size at most ζ , these records form a set. In particular, arbitrary cuts made by added supports within a rational fiber are part of this bounded record; those supports have not been replaced by horizontal-template terms.

Every nonexceptional old fiber lies wholly in one strict cut of F : it contains no point of F , no candidate support splits it, and distinct exceptional fibers are convex and disjoint from it. For each tile label i and strict cut u of F , record the ordinal length $\alpha_{i,u}$ of the horizontal positions of nonexceptional fibers of tile i lying in that cut. These positions form an interval of the well-order H_i . There is a set of coordinates (i, u) ; the set of cuts can have cardinality as large as $2^{|F|}$, which still has a bound depending only on ζ . We do not assume that the supporting order T , or the set of its cuts, is discrete or well-ordered.

For a fixed frozen record, retain only ordinal vectors giving valid configurations of the displayed tiled form. The vector and frozen record determine the parameter order and both candidate support patterns up to the specified order identifications. If two valid vectors satisfy $\alpha_{i,u} \leq \beta_{i,u}$ at every coordinate, embed the shorter ordinal interval into the longer one at each coordinate, and send each full rational fiber by the identity on its rational coordinate. Together with the identity on F , these maps give an order embedding of the two configurations, fixing the candidate supports

and the old frozen data. Every old position stays in its tile, and J_0 stays fixed. Consequently a designated term remains a term of the same designated part $E(J_A)$, with the same occurrence tag. Naturality of the stabilized K -diagrams preserves each joint diagram comparison under this embedding.

Consider the definable class of valid vectors for which the two candidate comparisons differ on some parameter query. Diagrams on any particular set-sized query can be evaluated in a sufficiently large eligible padded presentation, as in Proposition 4.8. Thus this is a first-order definable class with the fixed record as a set parameter. Apply Lemma 4.7 to obtain a set of valid unequal vectors such that every valid unequal vector dominates one of them. No assertion that an arbitrary ordinal vector describes a valid horizontal well-order is needed. Coordinatewise domination between the two valid configurations gives the embedding just constructed, so the query witnessing inequality in the smaller configuration still witnesses inequality after that embedding.

The old parameter orders in this set of witnesses have a common cardinal bound. The number of designated finite-support terms on one such order is at most its cardinality plus $|E| + \aleph_0$. Bound this cardinal over the set of witnesses and then over the set of frozen records. An unequal comparison in any valid configuration therefore differs on a designated parameter query of at most that bounded cardinality, retaining c as fixed data. Increase the bound to dominate the input cardinals and take monotone majorants in σ and τ . This defines the required g_A independently of the level. Taking the same supremum over a fixed set of bounded descriptors proves the last uniformity assertion. $\square$

We next record the realization step, with its support calculation explicit. A permitted parameter layout means either a natural layout covered by Proposition 6.3 or a rationally tiled designated-part layout covered by Lemma 12.5. A bounded list of additional fixed parameters is allowed in either case. A testing bound for a permitted layout will always mean the bound supplied by the corresponding one of these two results.

Lemma 12.6 (Realization using an unoccupied strip). *Let $E(J)$ be a pointed K -presentation of cardinality μ at a sufficiently closed uniform level. Let $V \subseteq J$ be a convex substrip of cardinality μ . Let $B \subseteq E(J)$ be a permitted parameter layout, with chosen representations of all its entries using positions in $J \setminus V$. Include the fixed point data among these represented parameters.*

Suppose a bounded tuple z^ in a presentation $E(J^*)$, where $J \subseteq J^*$, has a placed comparison over B . Then a tuple z of $E(J)$ has the same comparison over every bounded list from B . Any specified bounded old entries of z^* are retained as actual entries of z , provided their chosen representations avoid V .*

Proof. Shrink J^* to J together with supports for z^* , so its cardinality is still μ . Include the old entries to be retained in the tuple under consideration and in the bounded fixed data of the layout. Choose a bounded test cardinal $\chi < \mu$ at least the tuple length and the testing bound for this permitted layout. Let T^* be the chosen supporting positions of z^* in J^* . For the invocation of Lemma 6.14, replace V by its convex hull in J^* . Since V is convex in J , this hull meets J in exactly V . Thus its old complement is still $J \setminus V$, and the side of this convex hull on the old complement is the side of the original strip. The support-closure lemma therefore has the required convex strip in the ambient order; the actual realization still uses $E(S \cup V)$. By Lemma 6.14, there is a bounded $S \subseteq J \setminus V$ containing the supports of the fixed old entries and the point such that the following holds. Given a list Q of at most χ positions in $J \setminus V$ and at most χ positions R in S , one can move Q to a list in S , preserving all order comparisons and equalities relative to R and T^* , as well as its side of V .

For completeness, the size requirement on this use of support closure is bounded independently of μ . If $\lambda < \mu$ bounds χ , the initial supports, and T^* , the witnesses can be chosen through χ^+ stages inside a set of size at most 2^λ . At each stage there are at most 2^λ realized requests over at most χ old positions, since $(2^\lambda)^\chi = 2^\lambda$. Every at-most- χ subset of the union occurs at an earlier stage. Strong-limit closure keeps the resulting set S below μ .

The subpresentation $E(S \cup V)$ is a strong submodel of $E(J)$ and has cardinality μ . The complete list of elements generated by S is bounded, since the number of labels and $|S|$ are below μ . Saturation for bounded K -diagrams in $E(S \cup V)$ therefore realizes the joint diagram of z^* with this entire list, producing z . The diagram on that parameter list is the same in both presentations by naturalness. Equalities to its entries preserve the specified old assignments.

Choose representations of z using positions $T \subseteq S \cup V$. Let q be a test of at most χ parameters from B , with its chosen supports $Q \subseteq J \setminus V$. Apply support closure with $R = T \cap S$; this set has size at most χ . It gives a moved parameter list q' whose terms use only S . The joint order pattern of the supports of (z, q) agrees with that of (z, q') : comparisons to $T \cap S$ are retained, and comparisons to $T \cap V$ are forced by the retained side of the convex strip. The pattern of (z^*, q) likewise agrees with that of (z^*, q') , because comparisons with T^* were retained. Naturality and the diagram chosen in $E(S \cup V)$ give

$$D^K(z, q) = D^K(z, q') = D^K(z^*, q') = D^K(z^*, q).$$

Here q' need not belong to B : all terms on S were included in the saturation request. This proves equality on tests of size at most χ . Both tuples have placed comparisons over the original permitted layout, so its testing bound gives equality on every bounded test, as asserted. $\square$

Proof of Theorem 12.2. Continue the induction on the number of axes. The zero-axis and inactive-top cases were addressed above. Suppose the top is active and exact lifting is available on fewer axes. Refine the uniform levels for Lemma 12.3, the working packages, and the indicated one-dimensional layout bounds. The compatible face maps have a bijective union $h : B^- \rightarrow B^+$ preserving all bounded joint K -diagrams in the tops.

Use Proposition 11.6 and geometric categoricity to present the two systems by tiled presentations. Choose each tile and buffer as $H_i \times \mathbb{Q}$, where H_i is a well-order of cardinality μ . In the top-only tile choose H_i of order type $\mu \cdot \mu$; its μ consecutive intervals of order type μ give pairwise disjoint convex substrips of cardinality μ . Every proper face is generated on the fixed point positions, all buffers, and its designated tiles, so its chosen representations omit the top-only tile. The finite family of these designated parts is exactly a parameter layout covered by Lemma 12.5. That lemma also allows the bounded old matched tuple as fixed data. Use its testing bound for this layout in the following placement and realization arguments.

Construct nested bounded tuple matches in alternating directions. The invariant for a match $a^- \mapsto a^+$ is

$$D^K(a^-, b) = D^K(a^+, h(b)) \quad \text{for every bounded occurrence-tagged } b \text{ in } B^-. \quad (15)$$

It includes the equality comparisons needed for a partial injection, and initially holds with the empty tuple by Lemma 12.3.

For a forward step, enlarge a^- to a requested bounded tuple v^- in the source top. Prescribe its joint comparisons with B^+ by transport along h , and prescribe the actual values a^+ for the old coordinates. Every bounded local request is possible in the target top. To see this, collect all its boundary parameters into one bounded list. The invariant matches that list jointly with a^- .

Bounded diagram homogeneity, or the equivalent saturation clause, then realizes the requested extension v^- over its target copy and the actual old tuple.

Apply Lemma 6.4 to the chosen boundary parameter set, keeping the old tuple fixed. The underlying E -order need not be well-ordered, so use its padded-order clause with the coherent locally possible prescriptions on every bounded subset. For each bounded desired test size, closure under the full profile-count recursion in that lemma supplies a placement in an enlargement of this E -order realizing all prescriptions through that size. Choose the test size at least the bound from Lemma 12.5, including the old tuple as fixed data. Any two such outputs agree on that bound, so the lemma identifies their full placed comparisons. Fix one output. For any bounded query choose another output at a test size including that query; their equality proves that the fixed output also has the prescribed value there. Thus a bounded tuple v^* has the required placed comparison on every bounded query, with the previous coordinate equalities included. These are actual placements in arbitrary enlargements of the E -order; no horizontal-term representation of their new supports is assumed.

All boundary supports avoid the top-only tile, and the old matched tuple has boundedly many supporting positions. Some convex substrip of cardinality μ in that tile avoids those positions as well. Apply Lemma 12.6 to this substrip. It realizes the placed comparison by a tuple v^+ inside the actual target, retaining a^+ and satisfying (15) for the larger tuple. This completes the forward step. Interchanging the two systems and using h^{-1} gives the backward step with the same invariant.

Since $\text{cf}(\mu) = \aleph_0$, choose countable bounded exhaustions of both tops and alternate these two steps so that both are covered. Include in the requested tuples the enumerations of smaller strong submodels used in the working-package map criterion. The union map is onto and preserves every atomic formula and its negation, because any finite tuple occurs in a stage. It is therefore a K -isomorphism of the tops; the same map criterion also supplies strongness directly. Equality tests in (15) show that its restriction to each proper face is the prescribed h_s . It is consequently an isomorphism of the whole geometric systems.

For a fixed finite shape the dimension induction used only its finitely many proper restrictions. The remaining refinements were the uniform arity bounds for working diagrams, the fixed tiling layouts, support closure, and the bounded tree costs. These are the finite recipes recorded in Section 13, establishing the last assertion of the theorem. $\square$

13 Uniform definitions and common closure levels

The arguments above use two different finite inductions. The first iterates marked increments, with a fixed prescription in the root sort. The second ranges over finite geometric shapes. In each application we need one cardinal at which every finite stage is available. We now give the definability and closure bookkeeping that permits this in ZFC. The fixed point data, and subsequently the threshold furnished by the marked-increment application, are fixed before performing the bookkeeping for the geometric application. Thus the two applications may use different closure functions.

The bookkeeping has three stages. First we express the finite systems and their required properties by fixed formulas at a given cardinal. Next we list the bound requests and the order in which each construction uses them. Only after those formulas are fixed do we choose a common majorant and construct the nested eligible subclasses introduced in the [preview in Section 5](#). Their absorption property will allow each finite induction to retain an unbounded supply of levels.

13.1 Set definitions at a given cardinal

Proposition 13.1 (Uniform finite-depth definitions). *Fix the set code for K from Proposition 2.3, its order template, and the fixed data used in either of the two applications. There are a set $\mathcal{D}$ of normalized finite descriptors and fixed first-order formulas of set theory with the following properties.*

1. *Given $d \in \mathcal{D}$ and an infinite cardinal μ , the formulas define sets of standardized objects and arrows. For marked descriptors these sets are obtained by a finite recursion whose length is an input; at eligible levels they give precisely the categories $\mathcal{C}_n$. For a geometric descriptor they give the systems $\mathcal{G}_L$ and their arrows.*
2. *The formulas define orbit comparison, comparison of realized types over a base, relative isolation, the prescribed root marginal, and the finite projection and boundary conditions. On levels with the requisite working package these definitions agree with their mathematical meanings in the preceding sections.*
3. *There is one unary adequacy formula, with a descriptor as parameter, which can assert any of the packages needed during the finite inductions: the individual working conditions, the chart clause using D_F , the projection and type conditions, amalgamated completion, exact boundary lifting, or the geometric categoricity and basis conditions. Its use does not require a formula for a previously chosen closure function.*
4. *All candidate term templates, named patterns, and finite descriptors needed in the two inductions can be chosen from $\mathcal{D}$. In particular, their codes have one set-sized bound independent of the finite depth and of the level cardinal.*

These are definitions at every cardinal; a definition by itself makes no claim that the cardinal is eligible.

Proof. We specify the set recursion and the formulas used in it. The small-model code for K defines membership and strong substructure by the directed union formulas in Proposition 2.3. These are fixed formulas with a set parameter. They are therefore available in all the following set constructions.

Standardized structures and tuples. For a fixed finite descriptor and cardinal μ , use a separate copy of μ for each vertex sort, and represent each inclusion by its map. Pointed data and marks are represented by constants in their indicated sorts. This convention allows a proper same-cardinality inclusion: it is an injective map between the two copies, rather than the identity map on a common universe. A finite diagram of such structures is specified by sets of finitary relations, function graphs, constant values, and maps between these carriers. The collection of all candidates is a set. An arbitrary diagram of the specified cardinalities has a copy in this collection, by choosing bijections in its finitely many vertex sorts.

The bounded tuples used below are functions with ordinal domain of cardinality less than μ , together with a vertex occurrence for every coordinate. It suffices to use domains that are cardinals below μ : other enumerations are obtained by reindexing. All such functions form a set. A fragment in a system is likewise an occurrence-tagged subset of its finitely many carriers. Equalities between occurrences are tested after their specified maps into a common vertex; they are not inferred from equality of sort tags.

Raw comparison relations. Suppose sets $\mathcal{O}$ and $\mathcal{A}$ of objects and arrows have already been constructed. Isomorphism is expressed by an arrow with a two-sided inverse in $\mathcal{A}$. Put

$$\text{Orb}_{\mathcal{O},\mathcal{A}}(Y, u; Z, v) \iff \exists h: Y \cong Z \ (h(u) = v).$$

Here, and in the formulas that follow, the quantified objects and arrows belong to the displayed sets. If $e: X \rightarrow Y$ and $f: X \rightarrow Z$ are arrows, put

$$\begin{aligned} \text{Eq}_{\mathcal{O},\mathcal{A}}(X; e, Y, u; f, Z, v) \\ \iff \exists W \exists g: Y \rightarrow W \exists h: Z \rightarrow W (ge = hf \text{ and } g(u) = h(v)). \end{aligned}$$

These are formulas about sets. At a level where amalgamation has been proved, the second relation is precisely equality of the realized types used in the one-dimensional calculus, as characterized in [Theorem 6.6](#). Until amalgamation has been established we use the displayed relation itself, without taking a quotient or claiming transitivity.

Here is the isolation test in full. For $e: X \rightarrow Y$ and bounded b, c in Y , the test is

$$\begin{aligned} \exists z \in X^{<\mu} \forall (f: X \rightarrow Z) \forall b', c' \ [\text{Eq}(X; e, Y, b; f, Z, b') \\ \wedge \text{Orb}(Y, (b, c, ez); Z, (b', c', fz))] \\ \implies \text{Eq}(X; e, Y, (b, c); f, Z, (b', c')). \end{aligned}$$

The tuples b', c' have the same lengths and occurrences as b, c . The universal quantifiers range over all extensions in $\mathcal{O}$, not just over the objects about to be defined. The displayed notation $X^{<\mu}$ includes the occurrence tags. By the diagram and amalgamation clauses of the working package, this is exactly isolation relative to b : z enumerates an isolating fragment. Requiring this formula for every bounded c in Y defines all-domination over $X + b$. In particular, this test uses only $\mathcal{O}$ and $\mathcal{A}$ of the preceding category.

Root prescriptions and free lifting. The base descriptor includes the fixed named pattern. A prescription for a root type is given by the approximation code already used in [Lemma 10.8](#): a named bounded list of approximating tuples, its joint underlying K -diagram with the point, and a test arity sufficient to determine the type. For any test at that arity, the requested comparison is the comparison given by all but the allowed number of exceptions in the list. The allowed number is $|z| + \aleph_0$ for a parameter test z , and the list is longer than this number. Checking this rule for every such z is a fixed formula using the underlying K -diagram predicate. Existence and uniqueness of a type satisfying it, and its free lifting, are the conclusions proved when the prescription is introduced; they are not part of the assertion that the formula is defined.

This root test is applied after forgetting to the root K -sort. The projection type correspondence in [Proposition 9.1](#) identifies it, at the levels under consideration, with the required type over the whole system. Thus the definition does not successively quote a new formula for a type at each marked depth.

For completeness, references to free lifting in an adequacy condition also have a fixed set interpretation. Let p be represented by $(A \rightarrow Y, u)$, and let q be represented by $(B \rightarrow Z, v)$ over an extension $j: A \rightarrow B$, with q restricting to p . This is a same-level test: A, B, Y, Z are objects at μ . Mixed lifting is derived below using approximation costs bounded at the source level. For an infinite $\sigma \geq |u|$, an approximation test on $I = (u_i)_{i < \sigma}$ requires the following. For each parameter tuple z of length at most σ in the base, and each corresponding subtuple of u and the u_i , at most

$|z| + \aleph_0$ indices have a joint diagram with z different from that of the realized type. The free-lift test is

$$\forall \sigma (|u| + \aleph_0 \leq \sigma < \mu) \quad \exists \theta (\sigma < \theta < \mu) \quad \exists I \in (A^{|u|})^\theta$$

such that I passes this test for p over A and jI passes it for q over B . Here σ, θ are cardinals, and all joint diagrams are computed in the indicated realizing objects. The approximation construction gives these sequences for a free lift. Conversely, compare a sequence witnessing this test with the sequence computing the free lift at the same arity. Move a parameter query from B into A over both sequences by bounded saturation. Each majority then computes the original type on the moved query; the two majorities agree because their sequences have lengths exceeding their exception bounds. This proves equality with the free lift on every bounded test, and hence equality of types by [Theorem 6.6](#). All the sequences, tuples, exception sets, and realized types in this test range over sets at the given cardinal. This supplies a formula without a satisfaction predicate for a category definition.

The marked recursion. Let $\mathcal{O}_0(\mu)$ be the standardized pointed K -models with the specified named pattern, and let $\mathcal{A}_0(\mu)$ be their strong maps preserving the names. Given $\mathcal{O}_i(\mu), \mathcal{A}_i(\mu)$, form their raw comparison and isolation relations as above, and set

$$\begin{aligned} \mathcal{O}_{i+1}(\mu) = \{ & (e: X \rightarrow Y, b) : X, Y \in \mathcal{O}_i(\mu), \quad e \in \mathcal{A}_i(\mu), \\ & b \text{ passes the prescribed root test,} \\ & \forall c \in Y^{<\mu} \quad (b, c)/X \text{ passes the relative isolation test} \}. \end{aligned}$$

An arrow from $(e: X \rightarrow Y, b)$ to $(e': X' \rightarrow Y', b')$ is a pair $(f: X \rightarrow X', g: Y \rightarrow Y')$ of arrows in $\mathcal{A}_i(\mu)$ with $ge = e'f$ and $g(b) = b'$. This defines $\mathcal{A}_{i+1}(\mu)$ by Separation. Reading a pair as two layers gives the next axis; the earlier axes, their maps, their marks, and the root projection are part of its finite diagram code.

The operator in these two definitions is fixed. A finite sequence

$$(\mathcal{O}_i(\mu), \mathcal{A}_i(\mu))_{i \leq n}$$

satisfying the initial condition and this operator is a set and is unique. Existence follows by finite recursion using Power Set and Separation at each step. Consequently the assertion that a candidate belongs to its last object or arrow set is one first-order formula with n as an input. It is not a sequence of truth definitions on the universe. Induction, using the prescribed-type correspondence and [Theorem 8.6](#), identifies this recursion with the intended $\mathcal{C}_n$ at eligible levels. Raw tests at other levels have no role in that identification.

Geometric descriptors. Such a descriptor contains a finite axis set and its active-label family. The shape condition is a finite calculation. Membership in P , and independence of a finite list in P , are tested in a strong λ_0 -section over M_0 , as in [Theorem 10.5](#). A section, an isomorphism of sections, and a finite product witness are set-coded data. The independence predicate so obtained is fixed, with M_0, p_0, λ_0 as parameters. For a set H in a vertex, independence means that every finite subset is independent; spanning means that every relevant point is in the closure of some finite subset of H . Thus the existence of pieces $(H_s)_{s \in L}$ forming the required partitioned basis is a first-order condition on subsets of the finitely many carriers. Their required cardinalities are expressed by bijections.

The arrow condition is componentwise strongness together with preservation of finite independence in a vertex modulo its proper lower vertices. A dependence modulo the lower vertices has

a finite witness there. This gives a fixed formula, equivalent to extendibility of basis pieces by [Definition 11.1](#). The bases are existential witnesses in these tests; they are not extra structure that an isomorphism must preserve. Inactive vertices and the equalities forced by their largest active lower label are included as part of the system condition. This constructs the geometric object and arrow sets without a recursion in formula complexity.

A set of template codes. Choose an infinite cardinal b_0 bounding the fixed vocabulary, template labels, point code, and the lengths and pattern codes allowed for the named root prescriptions. The latter have such a bound because the approximation list length is bounded in terms of the underlying K -testing arity, independently of the level where an actual pair is found. Order and equality patterns for lists of this fixed bounded length form a set.

At one marked-template extraction, the expansion consists of old chart labels and representation tags, finitely indexed coordinate functions, the finite system maps and mark information, a retraction-index sort with its evaluation map, and the vertex LS-witness functions. For a vocabulary and label bound b , these symbols have a cardinal bound depending only on b and the fixed data. There is no symbol for every element of a sample or every entry of a sample-sized diagram table: sample entries and retraction indices are elements of sorts. Complete atomic diagrams on finite lists of terms are subsets of the set of atomic term instances and their negations.

For definiteness, increase bounds generously and take

$$b_{m+1} = (2^{b_m})^+, \quad b_* = \sup_{m < \omega} b_m.$$

Induction bounds the languages, finite-term sets, and their possible diagram codes at each finite extraction by a corresponding b_m . Normalize symbol names by ordinals below these bounds. The union over $m < \omega$ of the resulting sets of candidate codes is a set containing every candidate template needed at finite depth. The explicit geometric tile templates obey the same bound. Finally take finite diagrams of these codes, together with finite axis data, root codes, and the flags described next. This is the promised set $\mathcal{D}$.

One adequacy formula. Use the following fixed menu of assertions. A descriptor can request any finite selection of them for any of its indicated lower schemes. Every selection, including a preliminary geometric categoricity or basis selection, requires a strong limit cardinal of countable cofinality above all its fixed parameter bounds.

1. For every well-order I of cardinality μ , the indicated term interpretation $F(I)$ is an object, every object at the level is isomorphic to $F(I)$, and substitution along every same-level order embedding gives an arrow. The term interpretation is its set of terms modulo the template equality relation. It suffices to quantify over orders on a fixed copy of μ , so this is a first-order test on sets. In particular, validity of only the presentation on the initial ordinal μ is not the presentation clause used here.
2. Same-level arrows preserve bounded tuple orbits, and comparison of bounded tuples in well-ordered charts is exactly the fixed comparison D_F of [Proposition 5.9](#).
3. A chain of objects whose union has the level cardinality has an object as its union. For this test it suffices to use a chain with at most μ indices: from any chain with a union of size μ , choose stages covering its elements in all vertex sorts. The resulting subchain has the same union and has order type below μ^+ .

4. The root prescription, the indicated projections on objects and arrows, diagram comparison on projection sorts, and the stated type correspondence over a base hold. The comparisons use the raw relations above, now together with the requested working conditions. The free-lift clause uses the approximation test just specified.
5. A compatible boundary arrow has an amalgamated completion, or a compatible boundary isomorphism extends to an isomorphism of the whole objects, as indicated by the flag. These assertions quantify over the finite families of maps on the proper axis subsets and the object and arrow sets at this cardinal.
6. For a geometric descriptor, the indicated categoricity, lifting with prescribed nonroot basis bijections and root map, or lifting over a fixed base with prescribed complementary basis bijections holds. The pieces and the bijections range over sets of subsets and functions of the standardized carriers.
7. A designated term interpretation T into an indicated lower scheme has the following properties. Its values on the orders used by the specified natural layout are objects of that lower scheme, its same-level order substitutions give lower-scheme arrows, and comparison of its bounded term lists agrees with the lower scheme's D_F . Require explicitly that this comparison is natural in the new interpretation: two instances with the same T -labels and joint horizontal support pattern have the same lower-scheme diagram, and the descriptor's support substitutions preserve the designated term parts. This is the comparison used for the natural-layout test; arbitrary old-chart representations without this horizontal naturality assertion do not suffice. Orders, their finitely many designated parts, term evaluations, and support patterns are set-coded at the level. This flag may be requested for the base and upper interpretations of a new marked pair template while omitting the new pair's chain and chart flags. It uses the already established lower-scheme comparison.

There are only finitely many forms in this menu. The geometric categoricity and basis item can be used before the diagram clause has been obtained for the current geometric shape. A descriptor records which assertions are currently required, and for which finitely many lower descriptors. Bounded quantification over that finite record yields one formula, denoted $\text{Adeq}(d, \mu)$. Its ingredients were all defined by set operations or by the fixed underlying K and D_F formulas. This proves all four assertions. $\square$

The definitions above describe objects and assertions at one cardinal; they do not themselves establish eligibility there. Before collecting the bounds, we check how the individual adequacy conditions supply the maps and comparisons between different eligible levels that the earlier constructions use.

Lemma 13.2 (The mixed-level interface). *For either application, suppose the relevant individual adequacy conditions of Proposition 13.1 hold on a class S of levels, above the fixed parameter bounds. When the full individual working conditions, including the chart clause, are requested, the mixed presentation, coherence, LS, diagram, saturation, and map conditions needed in the preceding arguments hold on S . A preliminary descriptor requests only the corresponding already established clauses: in particular, its natural-interpretation flag supplies comparisons in the indicated lower working scheme and makes no assertion of a new pair's chain closure or chart clause.*

Proof. Presentation arrows across levels are supplied by the construction of the templates: the vertex term-hull LS witnesses give componentwise strongness in the marked application, and the tile

specialization of [Proposition 11.6](#) gives the additional basis condition in the geometric application. Coherence is componentwise coherence, with the finite independence test for geometric arrows. Once a componentwise union is an object, its arrows and its common-upper-bound property, including into a larger object, follow from the AEC chain axioms and, for geometry, finite character of independence.

Suppose an object X at $\nu \in S$ maps to an object Y at a larger $\mu \in S$. Present Y on a well-order. The image of X uses at most ν positions: each of its ν entries has finite term support, and there are finitely many vertices. Include these positions in a well-ordered subpresentation of cardinality ν , also including the fixed named positions. Coherence gives an arrow from X to this subpresentation. This is mixed LS.

For a tuple bounded at ν , the latter arrow preserves its orbit by the same-level adequacy condition. The chart clause identifies that orbit with D_F . Inclusion of the subpresentation in the large chart leaves the labeled support pattern unchanged, so the same D_F test computes the diagram in Y . This proves preservation along arbitrary mixed arrows. A bounded well-ordered pattern from a larger chart can also be placed in an order of size ν . When its old tuple has the same diagram as a specified tuple in a ν -object, the same-level chart clause gives an isomorphism matching those tuples. Transporting the enlargement through that isomorphism proves the required mixed-level bounded saturation.

Root prescriptions pass along mixed arrows because the approximation lists give the same majority comparisons there. To compare the full system types after projection, apply the type correspondence at the receiving level to those equal root marginals. Finally, the criterion for maps obtained by exhausting bounded tuples is componentwise the strong-submodel criterion of the working package. In the geometric case its additional condition is a finite independence comparison, preserved by the matched system diagrams. These arguments verify the mixed clauses without inserting them as new proper-class parameters in the unary adequacy formula. $\square$

13.2 A fixed register of bound requests

We specify what is included when a closure requirement is invoked. Given a uniform unary formula $S(\mu)$ and a bound request with ordinal input α and descriptor d , write

$$B_q(d, \alpha; S) = \begin{cases} \text{the least ordinal satisfying the request,} & \text{if such an ordinal exists,} \\ 0, & \text{otherwise.} \end{cases}$$

In this notation q belongs to the finite register below. The notation does not permit an arbitrary formula as a value of q , or an arbitrary class as a parameter S .

Lemma 13.3 (Uniform bound register). *All the refinements in one application of the one-dimensional tools, one marked-increment step with its projection and boundary arguments, or one geometric categoricity, basis, or boundary step are expressible using a fixed finite register of first-order bound requests. Their parameters are finite descriptors, ordinal size inputs, and unary adequacy tests from Proposition 13.1. Any references within one such step to earlier refinements can be eliminated by a fixed finite number of substitutions, independently of the input depth. Every bound request needed on the levels where its preceding hypotheses have been established has an ordinal witness.*

Proof. Here is the register; its descriptions specify first-order tests, rather than codes for formulas of increasing depth.

Cardinal, order-pattern, and partition bounds. Include successors, power sets, and their bounded iterations. For example, profile growth through α^+ stages is bounded by the recursion $a_0 \geq \alpha$, $a_{\xi+1} = (2^{a_\xi})^+$, with suprema at limits, for $\xi \leq \alpha^+$. This is a set recursion with ordinal parameters. Cut filling and saturation of order patterns of size at most α use the same operations and the corresponding α^+ -stage construction. The finite-arity partition request is the least ordinal on which every coloring of k -element subsets by the given color set has a homogeneous subset of the prescribed size; k and the size bounds are descriptor or ordinal inputs. The partition argument used to construct the order templates proves existence. Small trees and their parameter accumulators use successors, powers, and the least ϵ with $2^\epsilon > \chi$. These operations cover the chain lengths and cardinal costs in the domination and return-tree arguments.

Placed comparison and syntax tests. For a specified permitted layout and tuple arity, the comparison request asks for a bound on the size of a parameter test witnessing inequality between two placed comparisons. More explicitly, quantify over an eligible level, its indicated order layout, the two candidate term lists in order enlargements, and the parameters generated by the designated parts. If their comparisons differ on some bounded parameter list, require that they differ on one of size at most the proposed bound. Inequality is evaluated in a larger eligible padded presentation, using isomorphism to a well-ordered chart, or equivalently by its D_F comparison where the chart clause has been obtained. All the objects and positions involved in a particular test are sets. Universal quantification over the level cardinal is an ordinary first-order quantifier.

The permitted layouts are the whole-model layout, designated vertex parts, and finitely segmented or tiled layouts with the prescribed order and colors. Their codes belong to the descriptor set. The ordinary and rationally tiled ordinal-vector arguments of [Proposition 6.3](#) and [Lemma 12.5](#) prove that this request has a uniform witness. The syntax request asks for a test size at which the fixed D_F comparison on patterns of a specified size is already determined. The set of such patterns and the set-structure back-and-forth construction in [Proposition 5.9](#) prove existence. Neither request mentions a truth predicate for $\mathcal{C}_n$.

For the rationally tiled layouts $\sum_i (H_i \times \mathbb{Q})$ used in the geometric application, the comparison request includes the bounded frozen data of [Lemma 12.5](#). That lemma supplies its witness also for candidates in arbitrary enlargements of the underlying E -order. The exceptional fibers, their frozen positions, and the set of cut coordinates have bounds in the input arities and fixed descriptor; no new dependence on the level or the finite depth is introduced.

Approximation, compactification, and isolation bounds. The approximation and compactification constructions take a comparison bound and a profile bound at the specified arity, increase the lengths to dominate the exception counts, and perform the bounded power-set iteration just described. Thus their sizes are fixed compositions of the first two requests and cardinal operations. The isolating-fragment bound is the size of the initial supports, the bounded initial parts of the candidate blocks, and the terms on those supports in the proof of [Proposition 7.3](#). These again use the same comparison and cardinal requests. The construction of all-dominated objects uses larger input ordinals in these requests, mixed LS, and the indicated successor-length recursions; it introduces no new form of predicate depending on the marked depth. The existence assertions used here are [Proposition 7.8](#) and [Theorems 7.9](#) and [7.13](#), with the mixed interface just verified.

Eventual orbit thresholds. The remaining request asks for the first threshold above which a specified reversal, orbit-preservation, or stabilization assertion holds at the eligible levels and the

indicated arity. The assertion itself uses the standardized object sets, arrows, and orbit relation defined above. Thus the threshold test has the form

$$\forall \mu (S(\mu) \wedge \mu > \beta \implies R(d, \alpha, \mu)),$$

or its two-level version, with R one of these fixed orbit assertions. The reversal and stabilization arguments prove the existence of the threshold when the current categoricity and the previous required conditions hold on the unbounded eligible class. The use for geometric shapes has exactly those hypotheses by [Proposition 11.7](#). Passing from these thresholds to diagram homogeneity also requires smaller eligible levels cofinal below the final level; that requirement is recorded separately as a limit-point refinement.

We check that this list accounts for a whole generic step. In a marked step, extraction uses partition bounds, the underlying comparison and isolation bounds, and arbitrarily large samples. It chooses a template from the set established in [Proposition 13.1](#); see [Lemma 8.3](#). After that choice, the two-part comparison of [Lemma 8.4](#), used in [Lemma 8.5](#), is the placed comparison request with this template as parameter. The downward isolation argument requires the same bounds and sufficiently closed smaller levels. Orbit comparison and saturation use the transport argument; the passage to the unary chart clause uses the syntax request. Projection, completion, and exact boundary lifting then use the ordinary comparison requests, the prior unary conditions, and the tree costs; exact lifting may require one further supply of cofinal smaller levels. These are respectively the constructions of [Proposition 9.1](#) and [Theorems 9.6](#) and [9.8](#).

For geometry, first use the current categoricity condition and prior basis conditions as unary inputs to the eventual orbit request, then require smaller eligible levels and the chart syntax bounds. Basis lifting and exact boundary lifting use the same one-dimensional and tiled-layout requests and tree costs, by [Theorems 11.9](#) and [12.2](#). A simultaneous projection calculation or a maximum over finitely many lower descriptors is expressed by bounded quantification over that finite descriptor. It does not require successively inserting their formulas into one another.

Within each of these generic proofs there are finitely many successive uses of closure and of cofinal smaller levels. Label those positions in the proof in their order. The predicate for a position is the initial adequacy predicate conjoined with the earlier stated threshold, closure, and limit-point conditions. Substituting these descriptions at the finitely many positions gives a fixed formula for each request. The number of positions is attached to the generic proof, not to the finite axis count; finite lower-scheme lists are handled by the bounded quantification just described. This produces the asserted finite register with S eliminated in favor of its uniform defining formula.

The eligible predicates in a marked step. The following substitution order also previews the common-level induction at the end of this section. References here to retaining G_j use the [advance notation in Section 5](#); the actual classes and their absorption property are constructed below, after the register is fixed. They supply levels for that induction, not parameters for the requests being defined here.

Let S_0 be the unary conjunction of the full individual working conditions for the input scheme and the projection and stationary-prescription conditions already established for the finitely many lower descriptors. The mixed interface supplies their mixed working conditions. First refine S_0 by the testing, approximation, isolation, and domination bounds of [Sections 6](#) and [7](#); call this class S_1 . On its sufficiently closed members the actual constructions give the samples used in [Lemma 8.3](#). The induction maintains that S_1 is unbounded by retaining a G_j ; unboundedness of an arbitrary class alone would not justify this step.

After selecting F^b , let $R(\mu)$ be the conjunction of $S_1(\mu)$, the individual presentation flag for F^b , and the natural-interpretation flags for its base and upper values in the old scheme. Evaluate the testing request for the layout $U_H \cup A_{H'}$ on this preliminary class R , using the old scheme's D_F . Lemma 8.3 supplies the individual validity and natural interpretations; Lemma 8.4 reduces the comparison to the old working scheme, where the ordinal-vector argument of Proposition 6.3 supplies its uniform bound.

Now refine R by that testing bound and its placement and arity costs; call the resulting class S_2 . Thus S_2 is the output of this refinement, not the eligible predicate in its own defining request. The preliminary predicate R uses only the earlier costs and the two specified kinds of flags. Neither the request nor its witness uses a D_{F^b} chart clause or chain closure for the pair scheme.

At members of $S_2 \cap \text{Lim}(S_2)$ closed under the same arity costs, Lemma 8.5 proves pair-chain closure, including bounded tuples spanning the chain. The actual pre-closure orbit argument and this chain assertion then give Theorem 8.6; only now impose the syntax bound for D_{F^b} and request its chart clause. Projection correspondence and amalgamated boundary completion follow after the fixed comparison and tree-cost refinements in Section 9. The fixed-parameter boundary argument and exact lifting may add one further limit-point refinement to obtain smaller levels where completion already holds. Each displayed eligible predicate is thus a finite conjunction of unary flags, previously listed bound closures, and limit-point conditions. Substitution follows this displayed order; no predicate for a previously chosen G_j or closure function is inserted into a bound request. More generally the auxiliary subclass T in the eligible-level convention is used only to ensure an unbounded supply of levels through these finite refinements. It is not an additional parameter of any request. In the present induction $T = G_j$ supplies this existence assertion, while each request still uses the unary adequacy predicate and the finitely expanded refinements just displayed.

The eligible predicates in a geometric step. The first assertion of Theorem 11.9 for the current shape follows from the previously proved assertions for its finitely many lower shapes. Start with the unary conjunction of this current categoricity assertion and the needed lower-shape assertions. The structural and tiled-presentation proofs give the actual arrows, coherence, union conditions, and mixed containment used by Proposition 11.7. Its finite-tile alignment and orbit calculation prove eventual reversal on this class. The first route in Proposition 5.8 then proves stabilization; impose those arity bounds and a limit-point refinement within the current eligible class. Actual diagram homogeneity is proved at the retained members before the D_F syntax bound is imposed.

The four subsequent claims in the proof of Theorem 11.9 use this established package, the preceding one-dimensional bounds, and finite tree costs. They prove the relative-basis assertion at the retained levels. For exact boundary lifting, include the already proved exact assertions for all proper restricted shapes in the initial unary conjunction. The tree and unoccupied-strip proofs of Theorem 12.2 add only the indicated layout, support-closure, and tree-cost refinements. Increasing the number of axes changes the finite list of lower descriptors and the ordinal size inputs; it does not add a new kind of eligible predicate or a new position in this generic substitution order.

Finally, the witness assertions are precisely the uniform bounds proved in the cited constructions with their stated hypotheses. An irrelevant descriptor, or a stage whose hypotheses fail, may give a request with no witness. Its value 0 makes a total definable ordinal function and has no mathematical force. At a used stage we invoke the preceding construction to establish existence before applying the bound. In particular, totalization supplies neither an adequate template nor a missing lifting or testing witness. $\square$

13.3 The common majorant and its limit refinements

We now construct the hierarchy used in the preceding previews. The register and its finite substitutions have been fixed independently of this construction; the common majorant bounds their values, while the limit refinements supply smaller eligible levels cofinally below a new level.

Choose a cardinal ϑ above the hereditary sizes of the normalized descriptor set and all fixed parameters. This is one cutoff, chosen before any finite-depth induction; it includes all candidate templates, not just those already selected. For $\alpha \in \text{Ord}$, take the supremum of the values of all the functions in the expanded finite register, for every descriptor and for all ordinal inputs at most α . Finite lists of size inputs are included by taking all their coordinates at most α . Each such collection is a set, and each request has been totalized, so Replacement bounds its values. We may therefore choose a fixed, monotone definable ordinal function Φ such that

$$\Phi(\alpha) > \vartheta, \quad \Phi(\alpha) > \alpha, \quad B_q(d, \vec{\beta}) < \Phi(\alpha) \quad (\max \vec{\beta} \leq \alpha).$$

Include the initial single-model requirements in this finite register. The formulas defining its members have already been fixed. No partial truth construction, and no bound on a variable formula's complexity, is being used here.

For a definable class A of ordinals write

$$\text{Lim}(A) = \{\lambda : \sup(A \cap \lambda) = \lambda > 0\}, \quad \text{Cl}(f) = \{\lambda : \forall \alpha < \lambda \ f(\alpha) < \lambda\}.$$

Multiple ordinal inputs can be replaced by their maximum in the closure condition. Define

$$\begin{aligned} G_0 &= \{\mu : \mu \text{ is a strong limit cardinal, } \text{cf}(\mu) = \omega, \ \mu \in \text{Cl}(\Phi)\}, \\ G_{j+1} &= G_j \cap \text{Lim}(G_j) \quad (j < \omega), \quad G_\omega = \bigcap_{j < \omega} G_j. \end{aligned}$$

Because $\Phi(0) > \vartheta$, every member of every G_j exceeds the common parameter cutoff. This detail cannot be replaced by discarding an arbitrary initial segment after each new template choice.

The hierarchy itself has a uniform definition. For an ordinal λ , form $A_0 = G_0 \cap (\lambda + 1)$ by Separation and then, by finite recursion on subsets of $\lambda + 1$, put

$$A_{i+1} = \{\beta \in A_i : \beta > 0 \text{ and } \sup(A_i \cap \beta) = \beta\}.$$

The assertion $\lambda \in A_j$ is one formula with j, λ as inputs, and is equivalent to $\lambda \in G_j$. Quantifying over $j \in \omega$ defines G_ω . In particular the countable constructions below can choose least witnesses using this formula and Replacement.

Lemma 13.4 (Closure and absorption). *Each G_j is unbounded and closed under suprema of strictly increasing countable sequences of its members. The class G_ω is unbounded. If a current eligible class S contains G_j , and a finite sequence of refinements consists of the bound closures and thresholds in the fixed register and restrictions to limit points of earlier eligible classes, then the resulting class contains G_{j+m} for some finite m . Here every used bound must have a witness under its stated hypotheses.*

Proof. Given any starting ordinal, choose strictly increasing infinite cardinals $(\kappa_n)_{n < \omega}$ above it with

$$\kappa_{n+1} > \max(2^{\kappa_n}, \sup\{\Phi(\alpha) : \alpha < \kappa_n\}).$$

The supremum κ of this sequence is a cardinal of cofinality ω . For every cardinal $\lambda < \kappa$, some κ_n exceeds λ , and then $2^\lambda \leq 2^{\kappa_n} < \kappa$. For every $\alpha < \kappa$, the same choice gives $\Phi(\alpha) < \kappa_{n+1} < \kappa$.

Thus $\kappa \in G_0$. If instead the κ_n are an increasing sequence in G_0 , their supremum is again a countable-cofinality strong limit cardinal and is strictly Φ -closed by these same two tests. This proves the two assertions for G_0 .

Suppose they hold for G_j . Above any starting ordinal choose a strictly increasing countable sequence in G_j . Its supremum belongs to G_j by closure and to $\text{Lim}(G_j)$ because the sequence is cofinal there. This proves unboundedness of G_{j+1} . A countable increasing sequence in G_{j+1} has its supremum in G_j , and its own members witness that this supremum belongs to $\text{Lim}(G_j)$. Hence G_{j+1} has the asserted closure as well.

Choose $\lambda_n \in G_n$, strictly increasing and above an arbitrary prescribed ordinal. For each fixed j , its tail lies in G_j because the classes are nested. The supremum belongs to G_j by its countable closure. It consequently belongs to G_ω , proving unboundedness.

For absorption, maintain that the current eligible class contains a particular G_k . A request value on inputs below a member $\mu \in G_k$ is less than $\Phi(\alpha)$ for a suitable $\alpha < \mu$, and hence less than μ . Thus a bound-closure refinement retains all of G_k . A constant threshold is included as a request with input 0 and has the same property. If the next refinement demands cofinally many members of the current class below the new level, retain G_{k+1} : its members are limit points of G_k and thus of the current class. If several earlier eligible classes are mentioned, the maintained index can first be increased to the maximum of their finitely many indices. Induction through the finite refinement list proves the claim, with m bounded by its number of limit-point refinements. $\square$

Theorem 13.5 (Common levels for all finite depths). *With the fixed data of either application and the constructions proved in the preceding sections, there is an unbounded class of countable-cofinality strong limit cardinals at which all its finite-depth conclusions hold simultaneously. In the marked application this includes the working packages, projection and type correspondences, amalgamated completion, and exact boundary lifting for every C_n , uniformly over the root prescription codes realized at arbitrarily large eligible single-model levels. In the geometric application it includes categoricity, the working packages, basis lifting, and exact boundary lifting for every finite shape. One such common level may be chosen above any specified set of fixed cardinal thresholds.*

Proof. Use the fixed register and Φ just constructed. The initial single-model results hold on G_0 after the initial requirements have been included. Suppose the assertions required as input to a finite induction step hold throughout G_j . Take S to be the class defined by their unary adequacy formula, with the current descriptor. It contains G_j . Lemma 13.2 supplies the mixed conditions for that step. Lemma 13.3 identifies its bound requests and gives the existence of each relevant witness at its place in the proof. Lemma 13.4 then retains some $G_{j'}$, with $j' < \omega$, through all its refinements.

When a marked step extracts a new template, its code lies in the fixed descriptor set by Proposition 13.1. The remaining requests with that code as parameter were already majorized by Φ . Thus making the choice introduces no new closure function or cutoff. For a given finite shape, the finitely many previously required lower shapes can first be brought to one G_j by taking the largest of their indices. The same observation applies to the finite projection lists in the marked argument. The two ordinary finite inductions therefore give, for each finite descriptor, all its required conclusions on G_j for some finite j .

In the marked application only root codes consistent at arbitrarily large eligible single-model levels are used. A code obtained from an actual pair at a sufficiently closed starting cardinal has that property: freely lift its type with the named approximation data to arbitrarily large single-model levels. The majority rule is preserved there, and restriction from such a level gives

the same stationary prescription at every sufficiently closed smaller single-model level with that named pattern. This is the prescription argument in Lemma 10.8, prior to the use of the finite marked systems. It requires no persistence of a no-growth statement about P at larger levels. The fixed code set and majorant include every such possible input code, so the common-level construction can precede the choice of the actual pair.

Every member of G_ω belongs to G_j for every finite j . It consequently satisfies the conclusion for every finite descriptor used in the application, although the index j and the selected template may depend on that descriptor. Lemma 13.4 makes G_ω unbounded, so choose a member above the desired thresholds. All constructions in this argument are ordinary set recursions, Separation, Replacement, and finite induction with uniform formulas. This completes the simultaneous closure justification in ZFC. $\square$

14 Transfer to all larger cardinals

The common levels of Theorem 13.5 provide one cardinal at which all finite geometric shapes have exact boundary lifting. This section uses that single cardinal as the base of an induction through the cardinals. The levels supplied by the closure construction themselves need not contain a tail.

The trade between one extra dimension and an increase in cardinality goes back to Shelah's multidimensional existence and uniqueness methods [She83, Theorems 5.1–5.2]; compare their later formulation in [SV24, Lemma 11.6]. The argument below is given for the exact boundary maps and cardinal assignments used here.

Fix a common uniform level γ from Theorem 13.5, for all the geometric shapes and their exact boundary assertions. Choose it strictly above the no-growth threshold λ of Corollary 10.10, and above all fixed point and single-model thresholds. In particular $\gamma > \lambda_0$, the cardinality of the distinguished model M_0 . In every pointed model of cardinality at least γ , model cardinality equals the cardinality of a basis of its point geometry.

Definition 14.1 (Prescribed piece cardinalities). Let L be an allowed family on a finite axis set a . A cardinal assignment on L is a family of cardinals

$$(\kappa_s : s \in L), \quad \gamma \leq \kappa_s \leq \kappa_t \quad (s \subseteq t \text{ in } L).$$

A system with this assignment satisfies the span and inclusion conditions of Definition 11.1, with a witnessing basis partition $(H_s : s \in L)$ such that $|H_s| = \kappa_s$. The bases are not named. A prescribed boundary isomorphism has the meaning of Definition 12.1.

For $t \subseteq a$ let $a_L(t)$ be its largest active subset. The span identities and monotonicity imply

$$\text{rk } P(X_t) = \max\{\kappa_s : s \in L, s \subseteq t\} = \kappa_{a_L(t)}. \quad (16)$$

All these ranks are at least γ , so Corollary 10.10 shows that the same formula gives $\|X_t\|$. It also shows that an inactive vertex equals $X_{a_L(t)}$. Lemma 11.4 applies without a uniform-cardinality requirement: its proof only uses the finite shape and the basis axioms of the pregeometry.

Lemma 14.2 (Choosing subcubes with prescribed supports). *Suppose two systems $\mathcal{X}^-$ and $\mathcal{X}^+$ have the same cardinal assignment, and prescribed isomorphisms h_t on their proper faces. Suppose their top is active and put*

$$\kappa = \max_{s \in L} \kappa_s, \quad Q = \{s \in L : \kappa_s = \kappa\}.$$

One can choose witnessing global basis partitions (H_s^-) and (H_s^+) such that

$$h_s[H_s^-] = H_s^+ \quad (s \in L, s \subsetneq a). \quad (17)$$

Fix such partitions. Let χ be a cardinal with $\gamma \leq \chi < \kappa$ which bounds all $\kappa_s < \kappa$. Given at most χ requested elements in the finitely many vertices on both sides, there are subsystems $\mathcal{U}^\pm$ inside $\mathcal{X}^\pm$ with the following properties:

1. Each requested element belongs to its indicated vertex. Every vertex whose size is strictly below κ is retained in full.
2. Their witnessing pieces are subsets $I_s^\pm \subseteq H_s^\pm$. They equal $H_s^\pm$ when $s \notin Q$, and have cardinality χ when $s \in Q$.
3. On every proper face, $h_t[U_t^-] = U_t^+$.

If compatible subsystems have already been chosen whose witnessing pieces are subsets of these fixed partitions $(H_s^\pm)$, and whose total vertex sizes are at most χ , their entire vertices can be included among the requests. Moreover, at every $s \in Q$ one can require exactly χ additional basis coordinates beyond those old selected pieces, on both sides. All resulting inclusions preserve the relative independence condition for system arrows.

Proof. Choose the pieces on the minus side. For each proper active s , the isomorphism h_s takes its lower span onto the corresponding lower span on the plus side, since it agrees with the prescribed maps on every lower vertex. Thus $h_s[H_s^-]$ is a relative basis at s . By Lemma 11.4, these relative bases assemble in the plus system. Complete them by a relative basis at its top. Their cardinalities have the required values because relative rank is invariant under isomorphism. This gives the claimed partitions.

We construct the desired subsystems by a countable closure process, on both sides and at all the finitely many vertices. Begin with the requested sets, the fixed point, and any specified old subsystems. At every growing active label $s \in Q$, also request χ basis coordinates. In the extension version choose these outside the old selected piece. There are enough coordinates since $|H_s^\pm| = \kappa$ and the old piece has size at most $\chi < \kappa$. At a proper label the choices may first be made on the minus side and transported by h_s ; these are still fresh on the plus side. At the top make the choices on both sides separately. Retain all nongrowing vertices in full.

At each finite stage choose, by the Löwenheim–Skolem axiom, strong submodels of the ambient vertices containing all requests so far. Their sizes are at most χ : the LS bound, point, retained smaller vertices, and requested data all fit within that cardinal. At the next stage make the following further requests.

1. For each inclusion of labels $s \subseteq t$, put all elements of the currently chosen model at s into the requests at t .
2. For every point obtained in a chosen vertex, choose a finite support for it from the union of the global basis pieces below that vertex. Request each supporting coordinate in the vertex of its own active label.
3. At each proper face, request the image under h_t of every element currently included on the minus side, and the inverse image of every element currently included on the plus side.

Also retain all previous requests and chosen models. These operations make at most χ requests at each stage. Each ambient point set has the given basis partition, so the supports in the second operation exist and lie in the indicated vertices. For a retained smaller vertex all supporting labels

are retained as well: if $s \subseteq t$ and $\kappa_t < \kappa$, monotonicity gives $\kappa_s < \kappa$. Thus no operation requires enlarging a retained vertex beyond its given ambient model.

Take the unions of the chosen models through the countable process. Coherence makes all successive and vertex inclusions strong, and the AEC chain axioms give strong limit models $U_t^\pm \leq_K X_t^\pm$. On a proper face the image and inverse-image requests give exactly $h_t[U_t^-] = U_t^+$. This construction used the maps one face at a time; it did not assume that their union on possible extra intersections was already a function.

Put $I_s^\pm = H_s^\pm \cap P(U_s^\pm)$. Every point of $P(U_t^\pm)$ occurs in a finite-stage model, and its supporting coordinates were subsequently placed at their own labels. Conversely, the point set of $U_t^\pm$ is relatively closed in the ambient model and contains all selected coordinates below t . Therefore

$$P(U_t^\pm) = \text{cl} \left(\bigcup_{s \in L, s \subseteq t} I_s^\pm \right).$$

The selected pieces are subsets of the single global independent basis and hence are independent together. They supply precisely the required decomposition. Nongrowing pieces and vertices were retained in full. At a growing label at least χ coordinates were requested and its model has size at most χ , so its piece has size exactly χ . In the extension version, at least χ fresh coordinates were requested; at most χ coordinates altogether were added. The additional piece therefore has cardinality exactly χ .

Finally, the selected subsets increase when old subsystems are included. Being subsets of the same global basis proves the relative independence requirement for their inclusions, using the criterion of Lemma 11.4. $\square$

The subcube lemma makes compatible smaller systems available, but an arbitrary isomorphism between the next two subcubes need not extend the previous one. The cardinal induction resolves this by adding an axis: the old full isomorphism becomes part of a boundary prescription at a strictly smaller maximum cardinal. This is why the induction must cover all finite shapes simultaneously.

Theorem 14.3 (Exact lifting at every larger cardinal). *For every finite allowed shape and every monotone assignment $(\kappa_s : s \in L)$ of cardinals at least γ , any prescribed boundary isomorphism between two systems with that assignment extends to an isomorphism of their tops.*

Proof. We induct on the cardinal $\kappa = \max_{s \in L} \kappa_s$, proving the assertion simultaneously for all finite shapes and all assignments with that maximum. For $\kappa = \gamma$ all pieces have size γ , so Theorem 12.2 applies at the chosen common level. Suppose $\kappa > \gamma$ and the assertion is known at every smaller maximum.

As before, an inactive top equals a proper active vertex and its prescribed map already gives the conclusion. Assume the top is active. Keep all smaller pieces and their vertices in full. There are only finitely many of them, so their cardinalities have a common bound below κ . Choose the global basis partitions of Lemma 14.2.

Let $\delta = \text{cf}(\kappa)$. Choose an increasing sequence $(\alpha_i : i < \delta)$ of ordinals below κ , cofinal in κ , and fix enumerations of every growing vertex on each side by κ . We construct continuous chains of subsystems

$$(\mathcal{U}_i^- : i < \delta), \quad (\mathcal{U}_i^+ : i < \delta),$$

and compatible full system isomorphisms $f_i : \mathcal{U}_i^- \simeq \mathcal{U}_i^+$. Every proper-face restriction of f_i is the given h_t on that subface. Their pieces are increasing subsets of the chosen global partitions.

Nongrowing vertices are retained in full; all growing pieces at stage i have the same cardinality $\rho_i < \kappa$, at least γ .

For the initial stage, choose $\chi < \kappa$ bounding γ , the retained pieces, and the sizes of the requested initial enumeration segments. Apply Lemma 14.2 on both sides. This gives systems with the same piece sizes and maximum below κ , whose proper-face maps are the restrictions of the given prescription. The induction hypothesis at that maximum supplies f_0 .

Suppose $\mathcal{U}_i^\pm$ and f_i are given. Choose a cardinal $\chi < \kappa$ which bounds γ , all retained piece sizes, all old vertex sizes, and the next requested enumeration segments. The size checks justifying this choice at every proper stage are given below. Apply the extension part of Lemma 14.2. It produces $\mathcal{U}_{i+1}^\pm$ containing the old systems and the requested segments, with exactly χ new independent coordinates at every label in

$$Q = \{s \in L : \kappa_s = \kappa\},$$

and none at the other labels. The set Q is upward closed: if $s \in Q$ and $s \subseteq t \in L$, then $\kappa \leq \kappa_t \leq \kappa$, so $t \in Q$.

Add a new last axis k . Regard $\mathcal{U}_i^\pm \rightarrow \mathcal{U}_{i+1}^\pm$ as the two layers of systems with active labels

$$L' = L \cup \{s \cup \{k\} : s \in Q\}.$$

The old pieces have their previous cardinalities and the additional pieces all have cardinality χ . These are valid decompositions: the old and added coordinate sets are disjoint subsets of the fixed global basis, and their unions span the new vertices. The cardinal assignment on L' is monotone. Comparisons among old labels use the old monotonicity, comparisons among new labels use equality to χ , and an old label below a new one has size at most χ by its choice. No new label is contained in an old one. All assigned cardinals are at least γ and the maximum is still strictly below κ .

Prescribe the boundary maps of these enlarged shapes by f_i on the entire old cube and by the given h_t on every enlarged proper face of the new cube. They commute on all indicated subfaces: on an old proper face the two prescriptions already agree by the invariant for f_i . The cardinal induction hypothesis applies to this shape as well, since it is simultaneous in the number of axes and its maximum decreased. It yields an isomorphism of the new tops extending f_i and all the enlarged proper-face maps. This is the required f_{i+1} .

At a nonzero limit $i < \delta$, take componentwise unions and put $f_i = \bigcup_{j < i} f_j$. The maps are coherent, so f_i is an onto isomorphism between these unions, with its required face restrictions. The component unions are strong submodels of the actual ambient vertices by the AEC common-upper-bound axiom. Union pieces are still independent by finite character. Every point in a union vertex occurs at some stage and has a finite support in that stage's selected pieces, so the union pieces span. These are thus systems of the required shape, and inclusions continue to preserve relative independence.

Here are the cardinal calculations at proper stages. If $\kappa = \tau^+$ is a successor cardinal, choose $\chi = \tau$ at the initial and every growth stage. Every proper stage index has cardinality at most τ , and every union at such a stage has size at most $\tau \cdot \tau = \tau$. The enumeration segments below κ also have size at most τ . Hence every stated choice and every proper-stage induction application has maximum strictly below κ .

If κ is a regular limit cardinal, there are fewer than κ earlier stages at every $i < \delta = \kappa$. Regularity bounds their sizes and their union by a cardinal below κ . Enlarge the next χ to cover that bound, the finitely many retained sizes, and the next enumeration segments. It can still be chosen below the limit cardinal κ , and may be increased cofinally through κ .

If κ is singular, at a proper stage $i < \delta$ fewer than $\text{cf}(\kappa)$ earlier cardinals have been used. Their supremum is therefore below κ . Both that supremum and $|i|$ have a common cardinal bound below κ , as do the finitely many retained sizes and the next enumeration segments. Their maximum bounds the union and can be used in choosing χ . In particular this argument applies when $\text{cf}(\kappa) = \aleph_0$; the final countable union has size κ , while every proper stage remains below it.

The variable pieces have equal sizes on the two sides and at all growing labels. This is immediate initially and at successors, where the same additional cardinal χ is used everywhere. At a limit their sizes are the same cardinal sum of the initial piece and the previously added disjoint pieces. The preceding bounds keep that sum below κ at a proper stage. Monotonicity also persists, since the smaller retained pieces were bounded by the initial variable size.

Finally take the unions over $i < \delta$. The requested enumeration segments are cofinal, so these unions are the two original systems. The union of the f_i is an onto isomorphism of the tops with every prescribed restriction. The final union is allowed to have cardinality κ ; the induction hypothesis was used only at proper stages, where the maximum was smaller. This completes the cardinal induction. $\square$

Corollary 14.4 (Pointed categoricity on a tail). *For every cardinal $\kappa \geq \gamma$, any two models of cardinality κ carrying the fixed point pattern are pointedly isomorphic.*

Proof. On zero axes the only active piece is a basis of the whole point geometry. Its size is the model cardinality by Corollary 10.10. Apply Theorem 14.3 with the one-element cardinal assignment κ . The boundary is empty. $\square$

Theorem 14.5 (Categoricity on a tail). *If an AEC K is categorical in unboundedly many cardinals, then there is a cardinal θ such that K is categorical in every cardinal $\kappa \geq \theta$. Here categoricity includes the existence of a model at each such cardinal.*

Proof. All preceding constructions were made for this fixed AEC under the hypothesis of unbounded categoricity. Choose γ as above. Fix a sufficiently large categorical cardinal $\rho \geq \gamma$ at which a pointed presentation with the chosen diagram of M_0 exists and the stabilized K -diagrams on its enumeration are preserved by strong maps into larger models. Such a ρ is available from the single-model working levels and Proposition 10.6. We may take $\theta = \rho$.

Let M be any model of cardinality $\kappa \geq \rho$. Choose a subset of its universe of cardinality ρ . The LS axiom gives a strong submodel $N \leq_K M$ containing that subset and of size at most $\rho + \text{LS}(K) = \rho$. Thus $\|N\| = \rho$ exactly. Categoricity in ρ identifies N with the pointed presentation just chosen. Transport its copy of M_0 into N and then into M . The fixed diagram on its whole enumeration is preserved by the strong inclusion, so M has the required point pattern. Notice that this uses categoricity at the fixed smaller cardinal ρ , not a categoricity assertion at κ .

Consequently any two models of cardinality $\kappa \geq \theta$ can be given the chosen point pattern. Corollary 14.4 identifies them pointedly, and forgetting that point gives a K -isomorphism. This proves uniqueness.

For existence, take a linear order I of cardinality κ . The order presentation of Theorem 2.6 is a member of K and has size exactly κ : its distinct skeleton gives the lower bound, and its at-most- $\text{LS}(K)$ labels on finite lists give the upper bound $\kappa + \text{LS}(K) = \kappa$. Hence there is exactly one isomorphism type, rather than merely at most one, at every cardinal in the asserted tail. $\square$

Conclusion of the uniform theorem

Proof of Theorems 1.3 and 1.2. Theorem 14.5 proves the qualitative assertion of Theorem 1.3. Apply the set-coding reduction of Proposition 2.3 at a fixed infinite LS bound λ . For each of its set-many AEC codes, either the categoricity spectrum is bounded, or the qualitative assertion gives categoricity on a tail. In the bounded case choose a strict upper bound for every categorical cardinal of that code. In the unbounded case choose a threshold for its categorical tail. Replacement over the set of codes gives a common cardinal bound $\mu(\lambda) \geq \lambda$ above all these choices.

If an AEC with LS number at most λ is categorical at any cardinal $\kappa \geq \mu(\lambda)$, its code cannot be one of the bounded-spectrum cases. Its categorical tail therefore contains every $\kappa' \geq \mu(\lambda)$. This is exactly Theorem 1.2, with the same threshold in its premise and conclusion. No regularity, successor, amalgamation, or no-maximal-model hypothesis has been added to the statement. $\square$

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